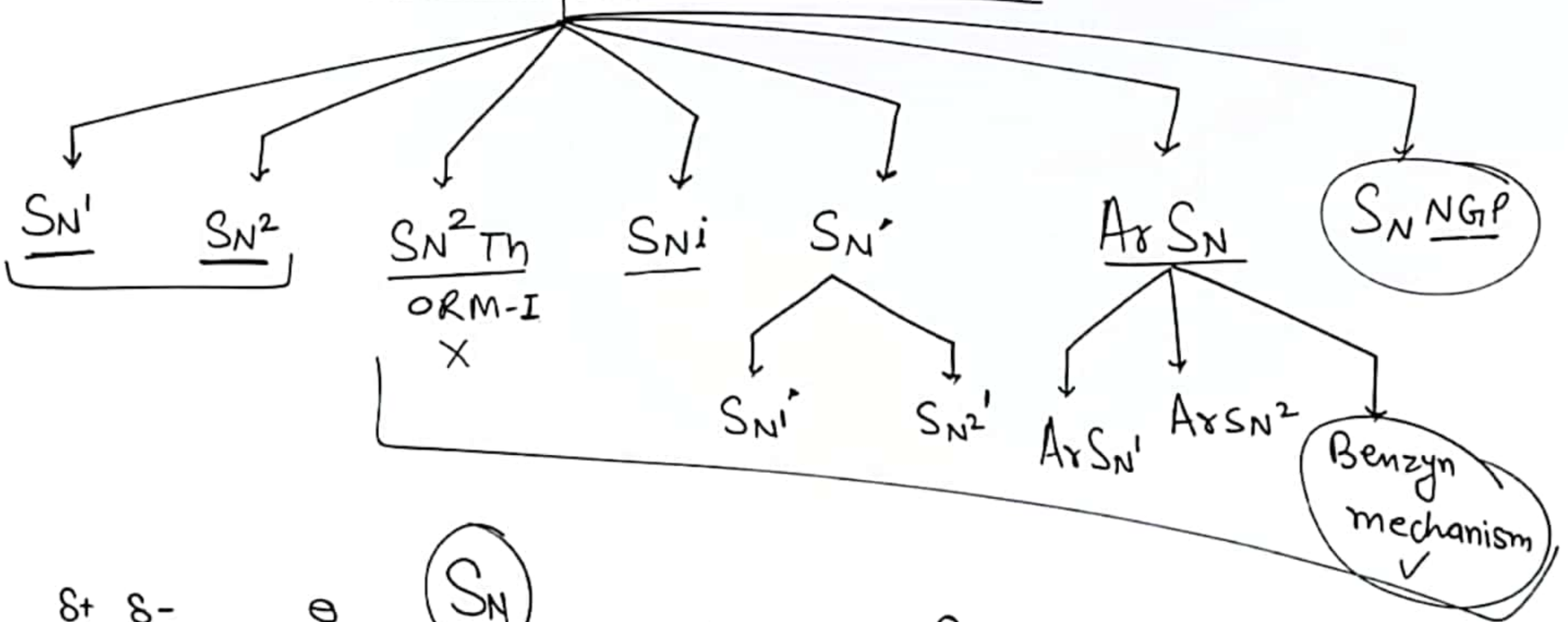
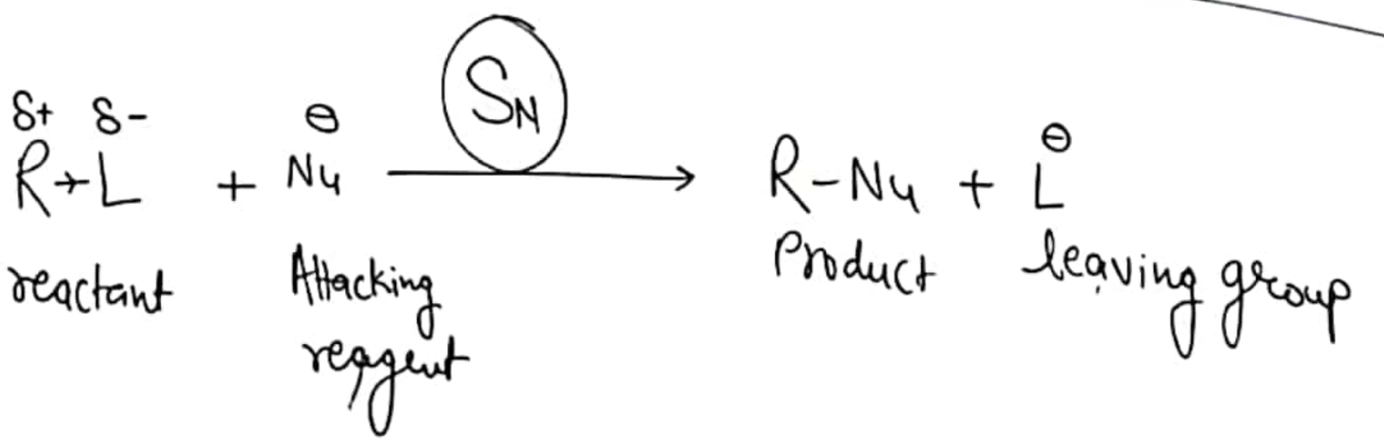
ORGANIC REACTION MECHANISM - III

ORM-III

Nucleophilic Substitution Reactions (S_N)

Alkyl halide
Alcohol
ether





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ORGANIC REACTION MECHANISM – III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

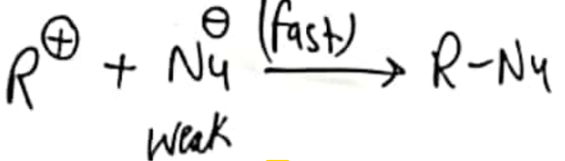
① Two step mechanism

② Mechanism

Step.1



Step.2

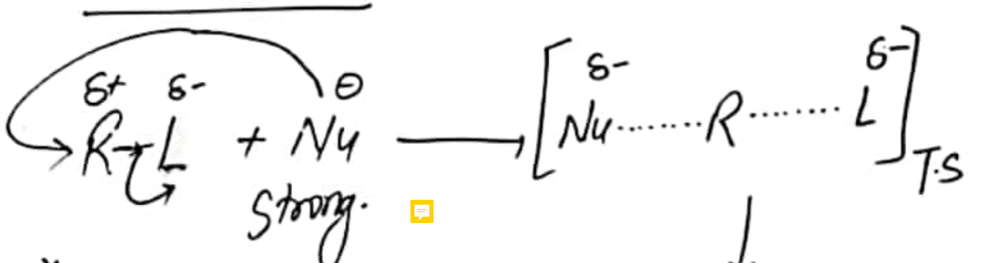


S_N2 (Bimolecular Nu. Sub. Rx^n

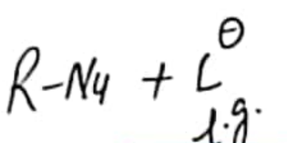
① Single step concerted mechanism

Where bond formation and bond breaking take place simultaneously (T.S.)

(2) Mechanism







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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

(3) Kinetics of S_N1

⇒ Rate of S_N1 reaction $\propto [R-L]^1 [Nu^{\ominus}]^0$

⇒ 1st order reaction.

(4) Nucleophile :-

Weak nucleophiles are favourable for S_N1 reaction.

⇒ Generally neutral nucleophiles, like $H_2\ddot{O}$, $R-\ddot{O}H$, H_2S, CH_3COOH etc

S_N2 (Bimolecular Nu. Sub. Rx^n

(3) Kinetics for S_N2

⇒ Rate of $S_N2 \propto [R-X]^1 [Nu^-]^1$

⇒ 2nd order reaction

(4) Nucleophiles

Strong nucleophiles are favourable for S_N2 Rx^n

⇒ Generally anionic nucleophile

$H^{\ominus}$, $X^{\ominus}$, $R^{\ominus}$, $NH_2^{\ominus}$, $OH^{\ominus}$, $OR^{\ominus}$, $SH^{\ominus}$, $SR^{\ominus}$, $R-COO^{\ominus}$, $CN^{\ominus}$, NH_3, $R-NH_2$, $R-NH-R$, $\ddot{P}R_3$ etc

some neutral nucleophiles are also strong Nu-

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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. Rx^n)

(5) Leaving group

⇒ reactivity for S_N1 ∝ leaving ability

⇒ R—I > R-Br > R-Cl > R-F (Reactivity)

↑

No S_N1 Rx^n

(6) Solvent:

Polar protic solvents are favourable for S_{N1}

S_N2 (Bimolecular Nu. Sub. Rx^n

(5) Leaving group.

⇒ same

⇒ R-I > R-Br > R-Cl > R-F (Reactivity)

↑

No S_N2 Rx^n

(6) Solvents:

⇒ Polar aprotic solvent - -

nucleophilicity increases in presence of polar aprotic solvents

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ORGANIC REACTION MECHANISM - III

ORM-III

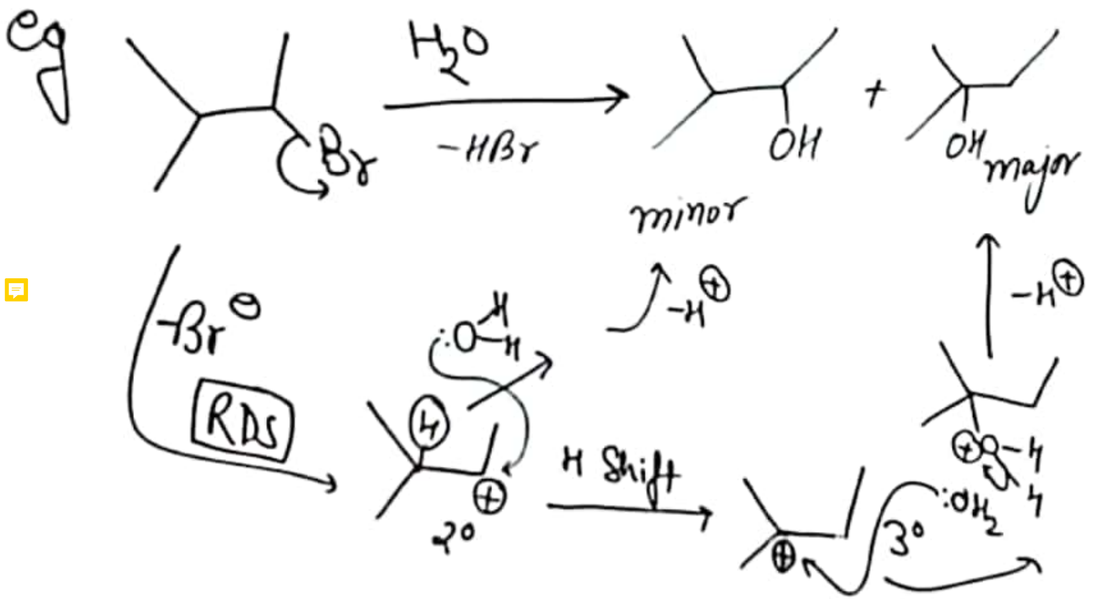
S_N1 (Unimolecular Nu. Sub. Rx^n)

(7) Rearrangement:-

Intermediate = free carbocation.

↓

Rearrangement in carbocation is possible

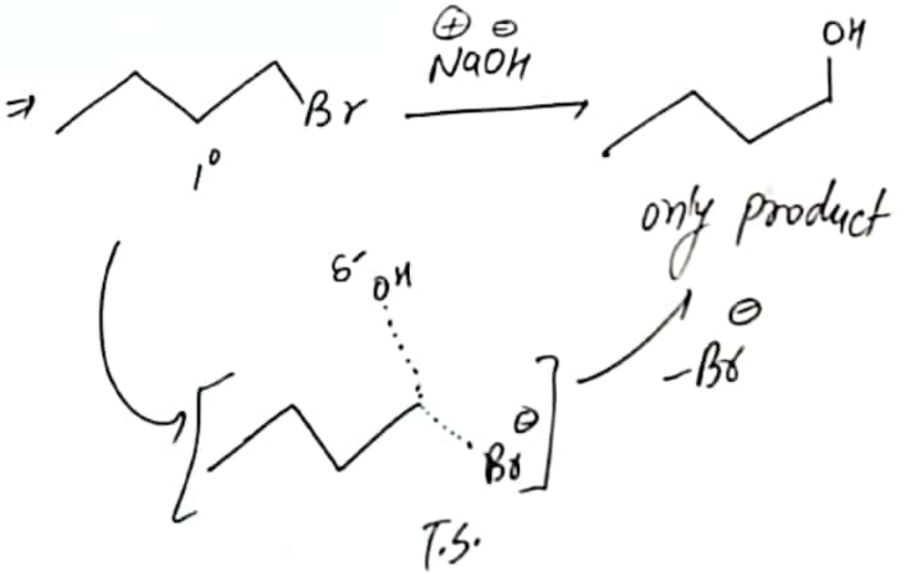


S_N2 (Bimolecular Nu. Sub. Rx^n

(7) Rearrangement

↳ No Carbocation intermediate

↳ No Rearra.



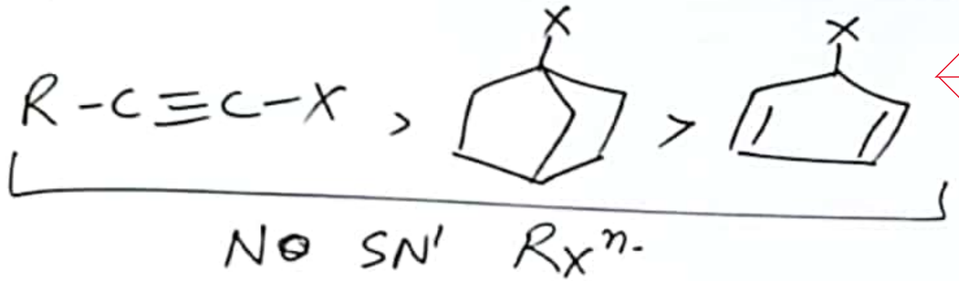
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⊗ Reactivity on the basis of alkyl group
Reactivity of SN1 ∝ Stability of carbocation intermediate of 1st step (RDS)

ORGANIC REACTION MECHANISM – III

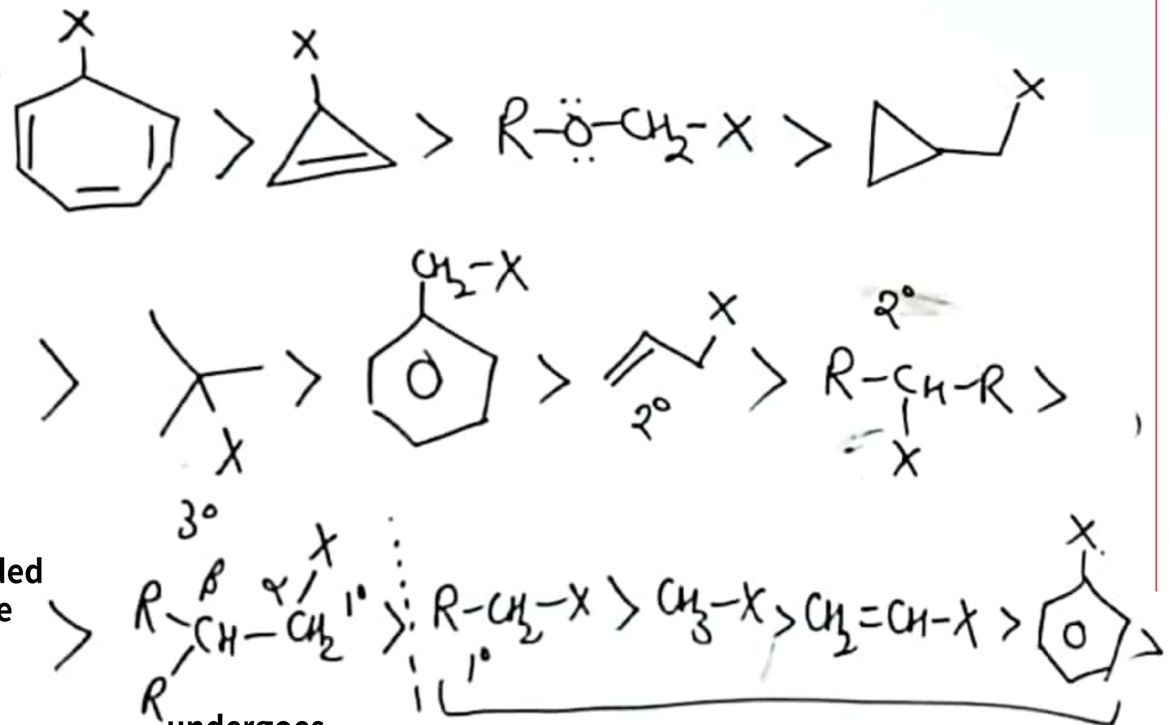
ORM-III

SN1 (Unimolecular Nu. Sub. Rxn)



No SN1 Rxn.

General order:



SN2 (Bimolecular Nu. Sub. Rxn

⊗ Reactivity for SN2 ∝ 1 / crowding on carbon of leaving group.

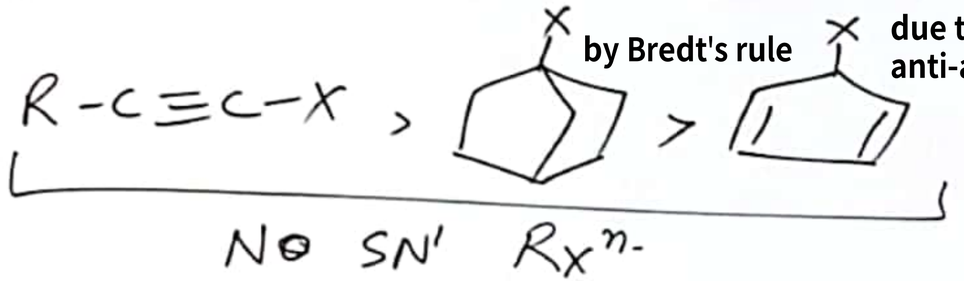
⇒ Reactivity ∝ e⁻ dificiency on carbon of l.g.

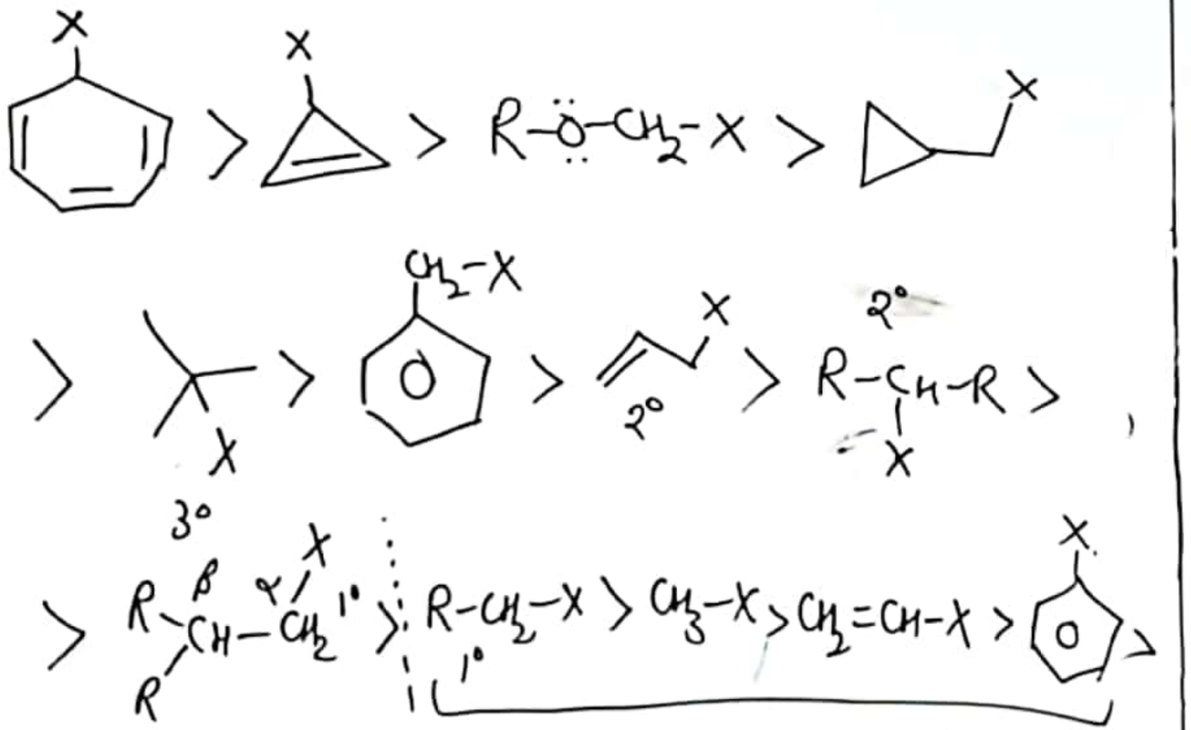
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ORGANIC REACTION MECHANISM - III

ORM-III

S_N1 (Unimolecular Nu. Sub. R_xn)

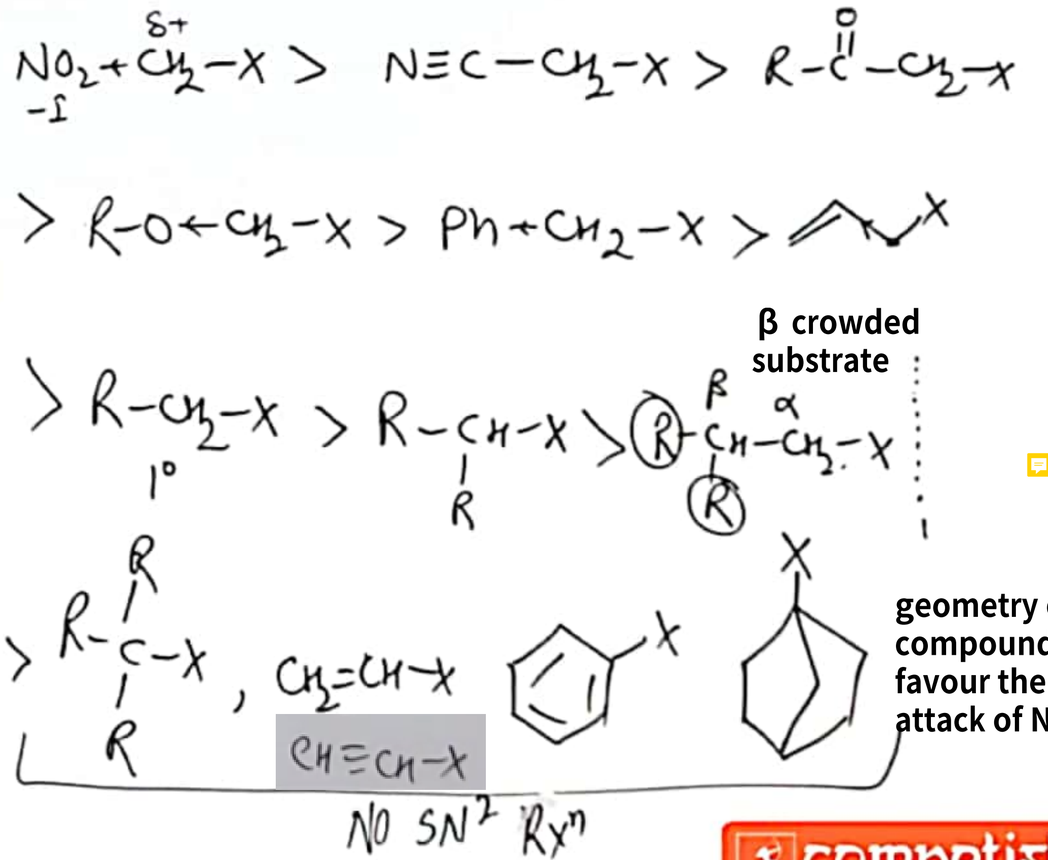




S_N2 (Bimolecular Nu. Sub. R_xn

⑧ Reactivity

General order



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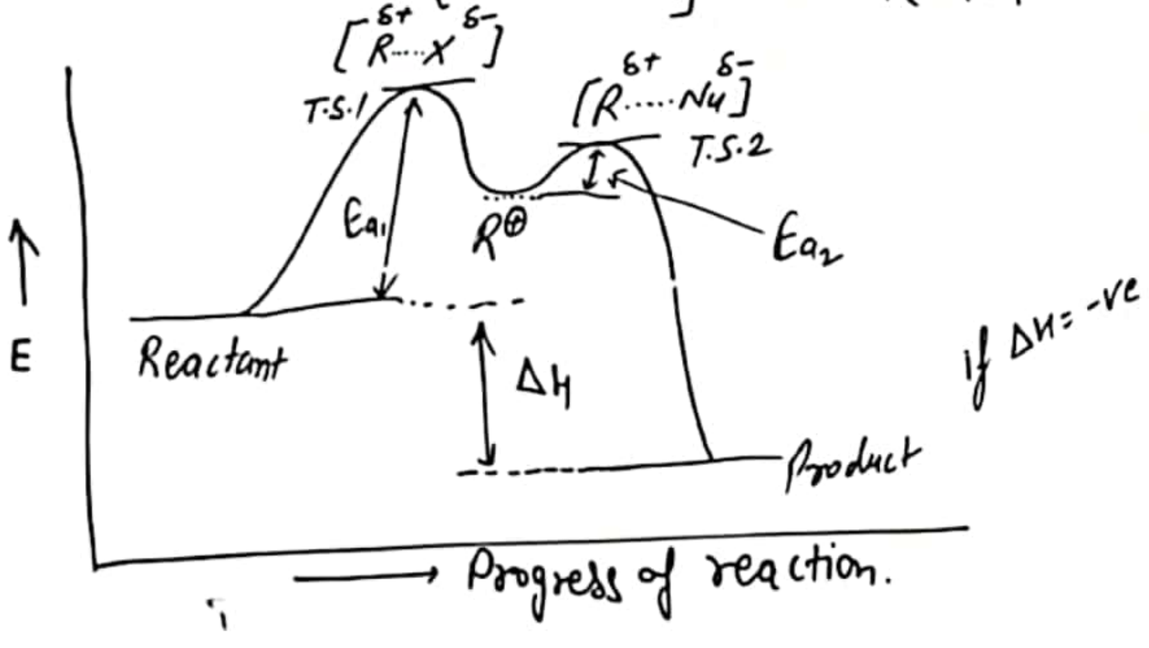
ORGANIC REACTION MECHANISM - III

S_N1 (Unimolecular Nu. Sub. Rx^n)

⑨ Energetics of S_N1

$$R-X \longrightarrow [R^{\delta+} \cdots X^{\delta-}] \longrightarrow R^{\oplus} + X^{\ominus}$$

$$R^{\oplus} + N_4^{\ominus} \longrightarrow [R^{\delta+} \cdots N_4^{\delta-}] \longrightarrow R-N_4$$

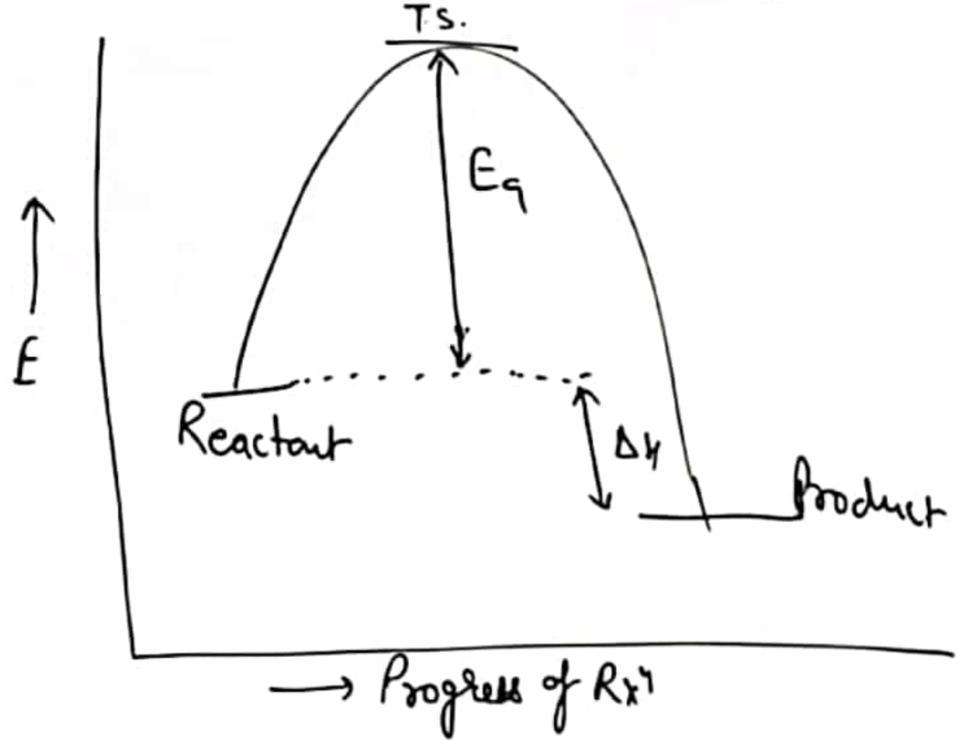


if ΔH = -ve

S_N2 (Bimolecular Nu. Sub. Rx^n)

⑨ Energetics - - - -

$$R-X + N_4^{\ominus} \longrightarrow [N_4^{\delta-} \cdots R^{\delta+} \cdots X^{\delta-}]_{TS} \rightarrow R-N_4 + X^{\ominus}$$



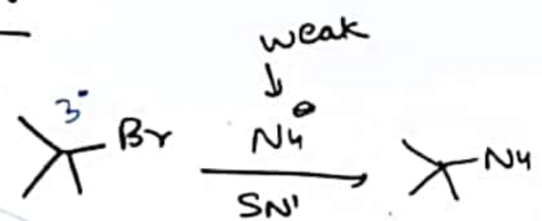
ORGANIC REACTION MECHANISM - III

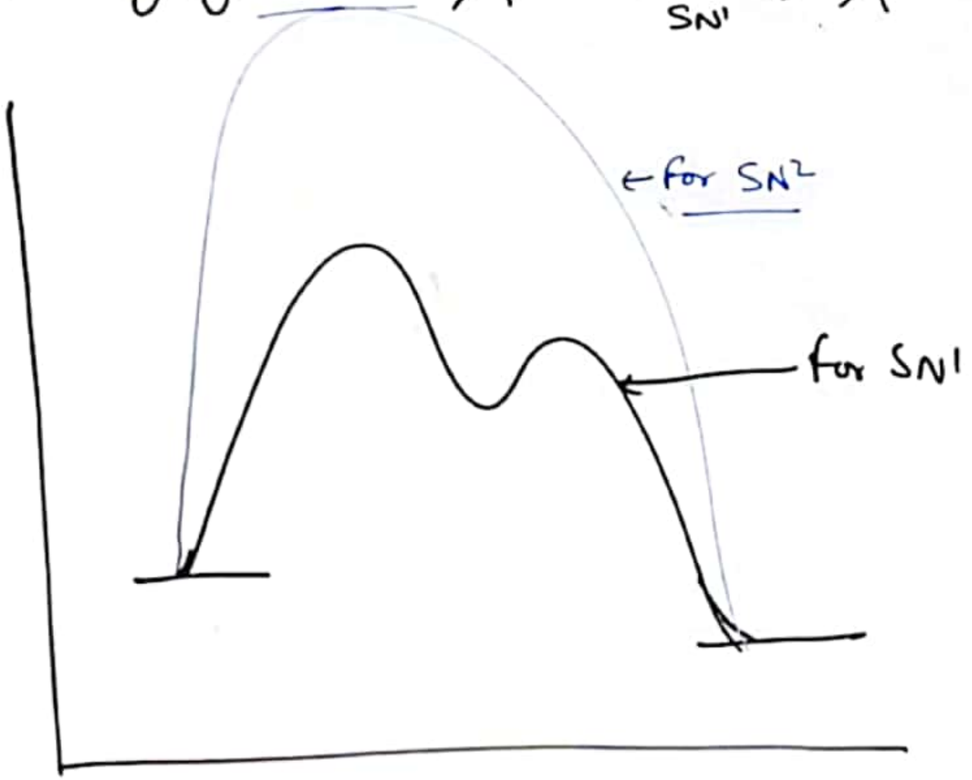
ORM-III

S_N1 (Unimolecular Nu. Sub. Rxn)

⑨ Energetics of S_N1

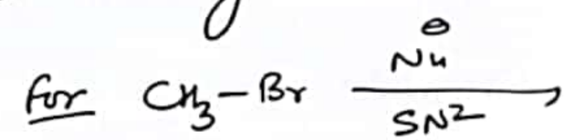
for 3° alkyl group

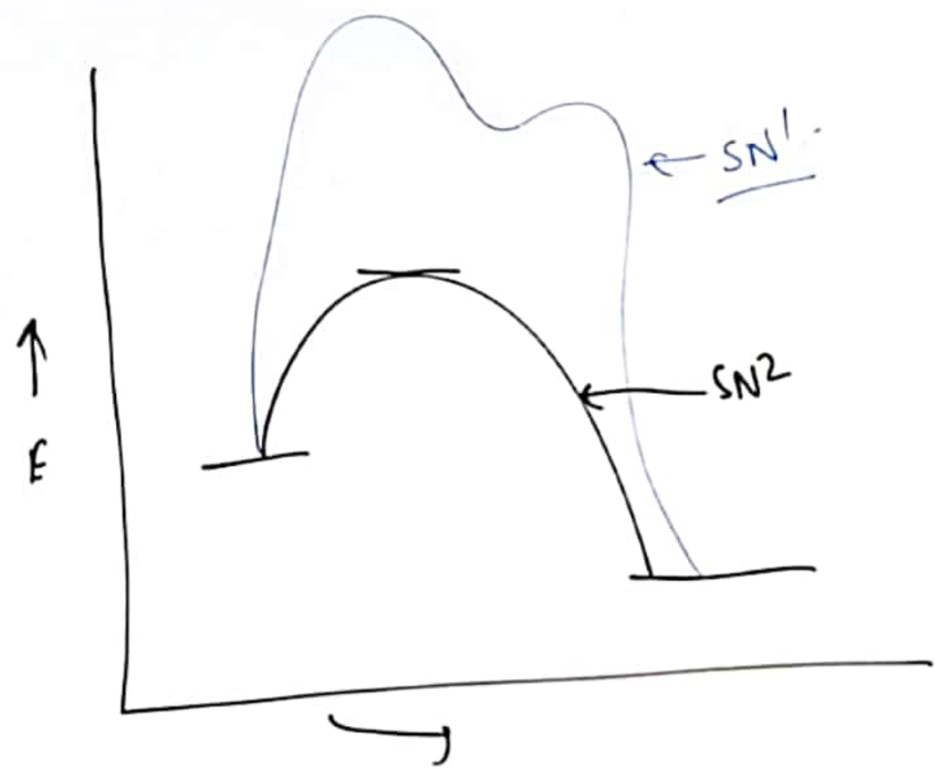




S_N2 (Bimolecular Nu. Sub. Rxn)

⑨ Energetics ----

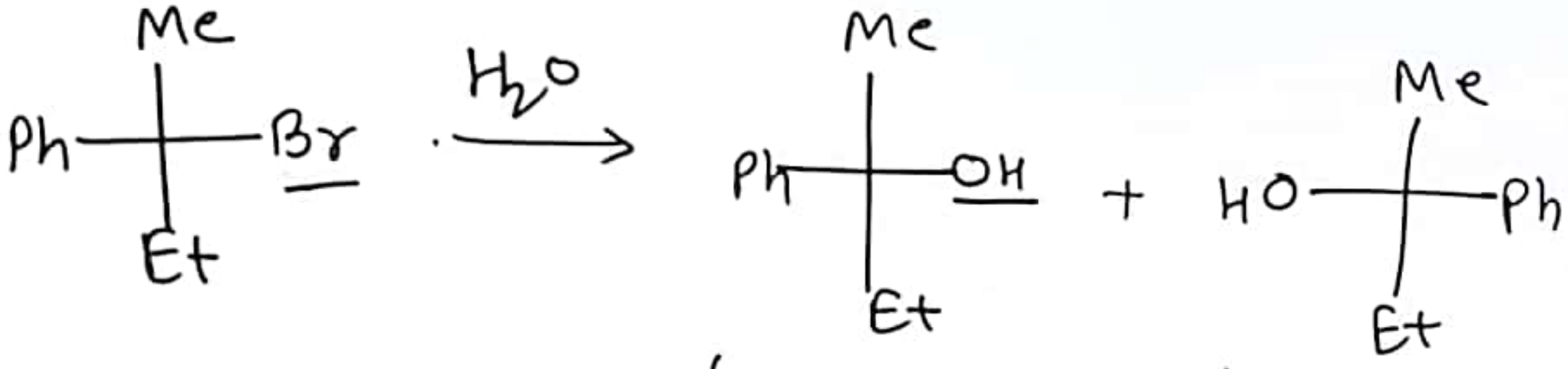




(10) Stereochemistry of SN1

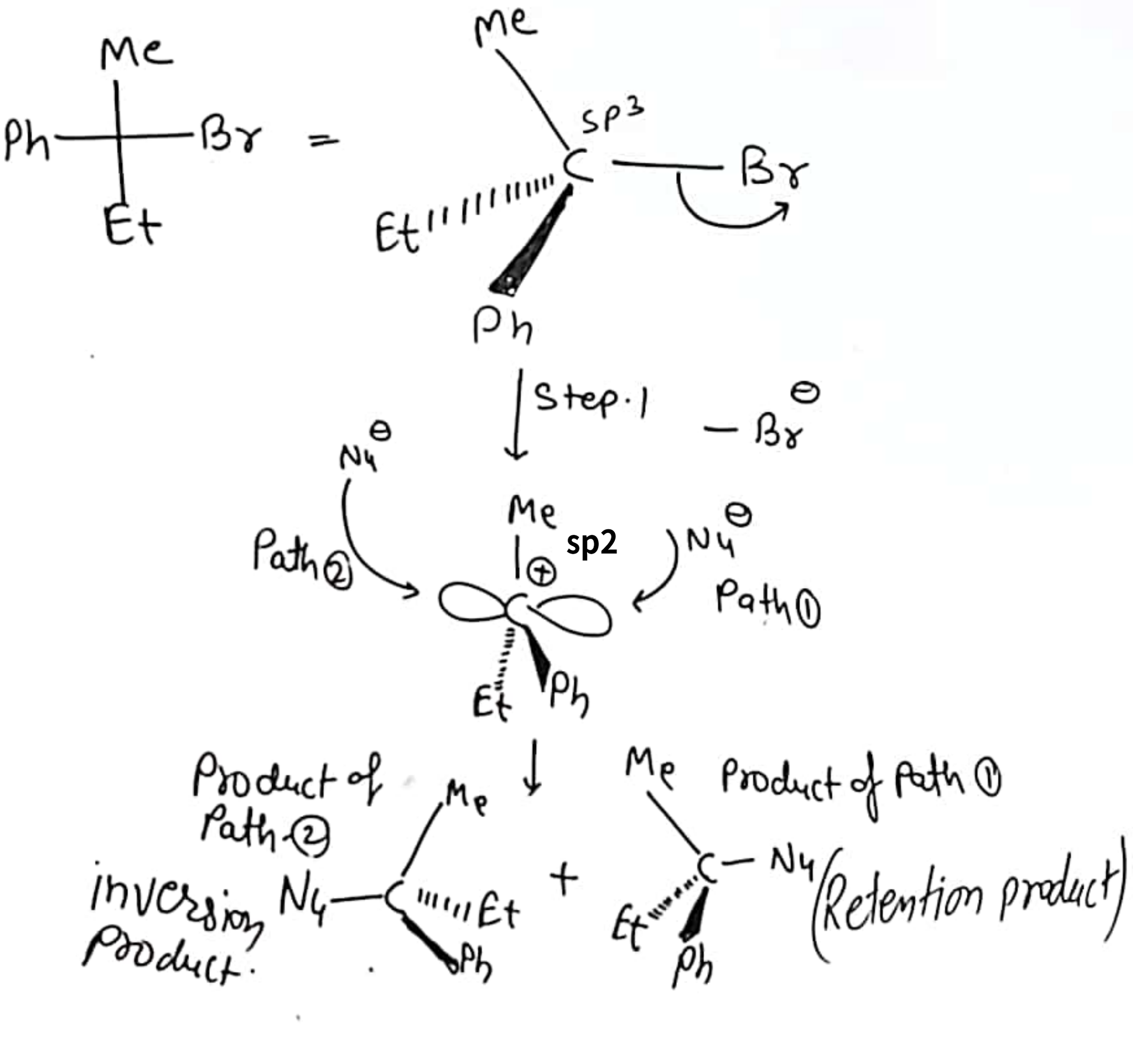
When l.g. is present on chiral C then in SN1 reaction reaction generally more than one stereoisomeric products are form (generally mix. of enantiomers) due formation of planar carbocation

(10) Stereochemistry of SN1



S_N1 (Unimolecular Nu. Sub. Rx^n)

(10) Stereochemistry of SN1



(10) Stereochemistry of S_N1

Experimently 100% racemic mix. does not form due to repulsion b/w l.g$^{\ominus}$ & Nu$^{\ominus}$ in initial moments of formation of carbocation. (% inversion product > % Retention product)

⇒ % Racemisation $\propto$ rate of formation of $C^{\oplus}$

⇒ " " " $\propto$ stability of intermediate

⇒ " " " $\propto$ stability of l.g.

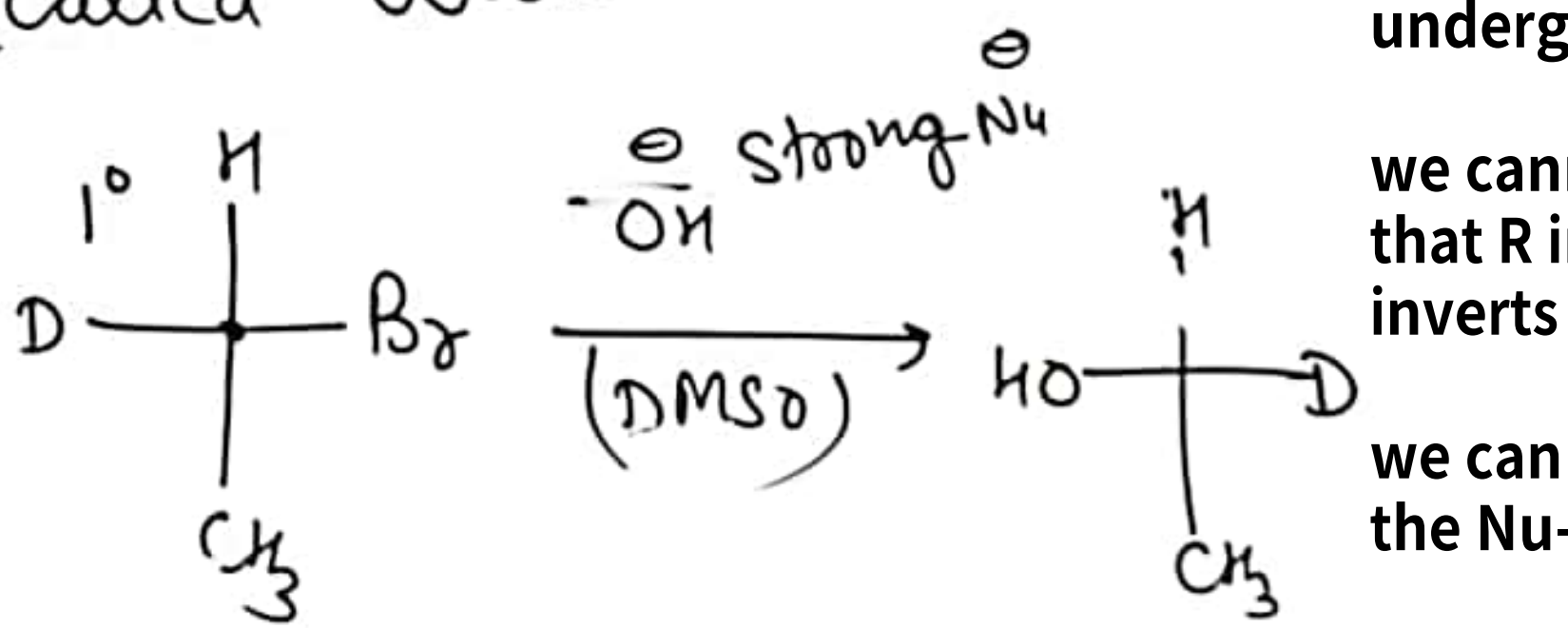
⇒ " " $\propto$ Polarity of solvent

SN^2 (Bimolecular Nu. sub. Rxn

(10) Stereochemistry for SN^2

When l.g. is present on chiral carbon then only inversion product form.

(called "walden inversion")



only inversion Product

only the side of the Nu- undergoes inversion

we cannot say with surety that R inverts to S or + inverts to -

we can only say the side of the Nu- gets inverted

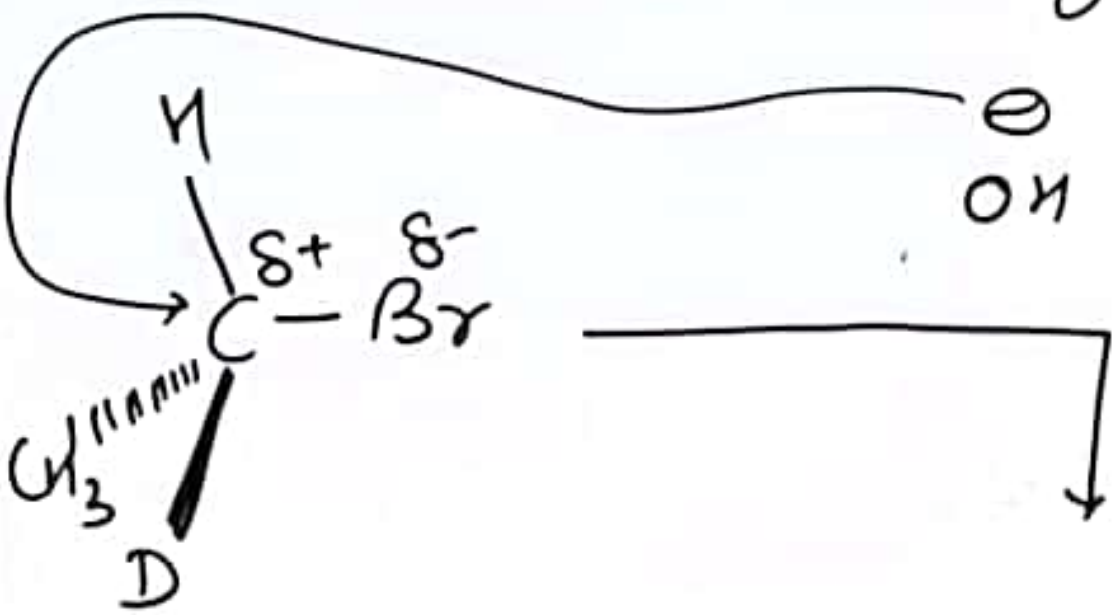
if $[\alpha] = +x^\circ$

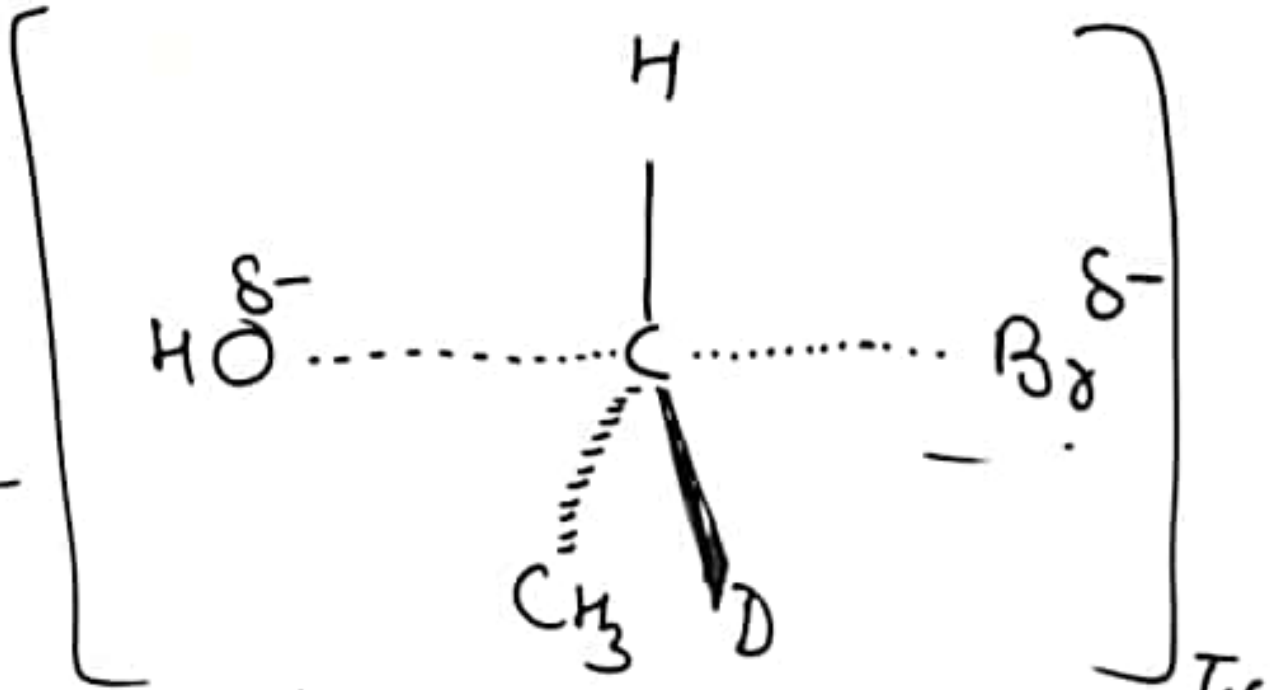
$[\alpha] \neq -x^\circ$

R S

SN^2 (Bimolecular Nu. sub. Rxn

(10) Stereochemistry for SN^2

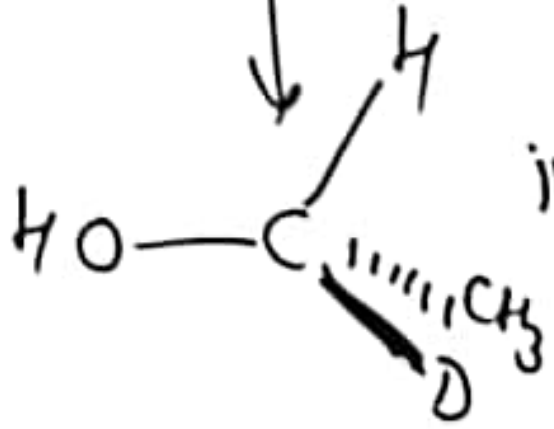




TBP

Trigonal bipyramidal

T.S

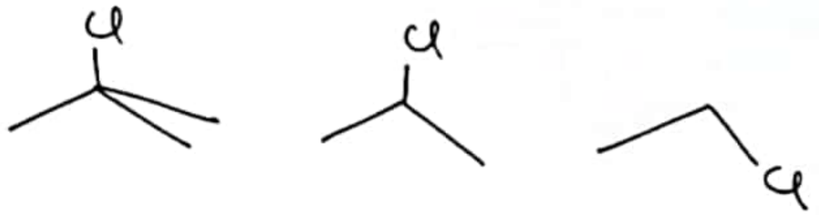


inversion product

ORGANIC REACTION MECHANISM – III

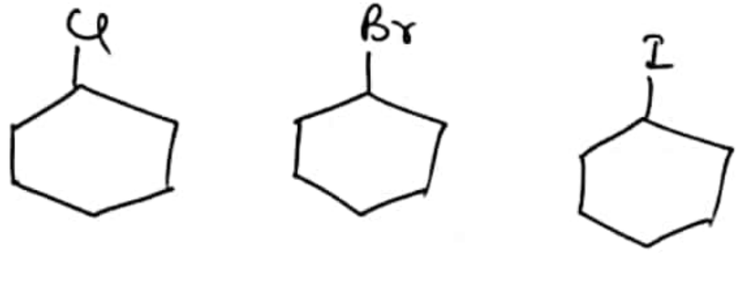
ORM-III

Q Give order of reactivity for S_N1 and S_N2 R_x^n.

① 

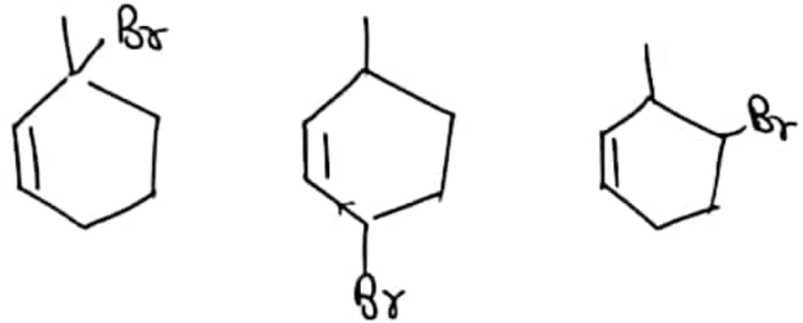
→ Reactivity of S_N1 = a > b > c (No S_N1 R_x^n)

→ " " " S_N2 = c > b > a (No S_N2 R_x^n)

② 

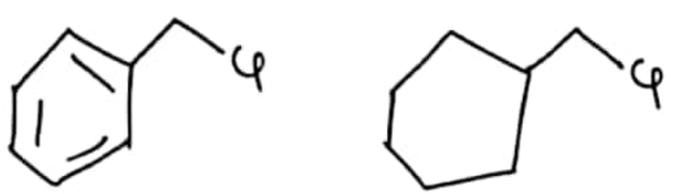
→ S_N1 → c > b > a

→ S_N2 = c > b > a

③ 

→ S_N1 → a > b > c

→ S_N2 → b > c > a a least due to steric crowding of tertiary group, despite it being an allylic position

(4) 

→ S_N1 → a > b

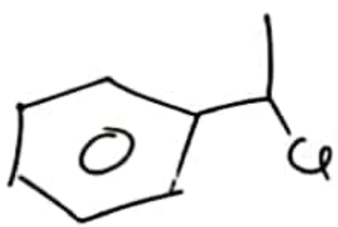
→ S_N2 → a > b

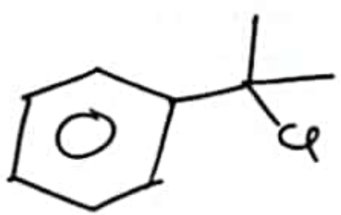
ORGANIC REACTION MECHANISM – III

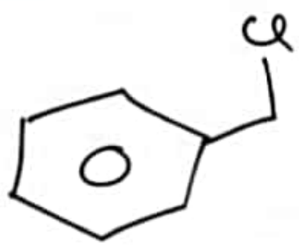
ORM-III

Give order of reactivity for S_N1 and S_N2 Rxn.

(5)





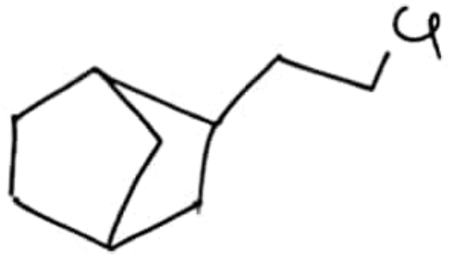


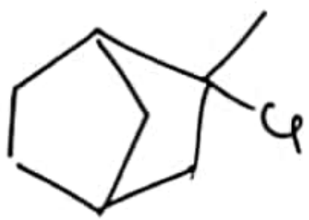
will give SN1 due to benzylic pos.

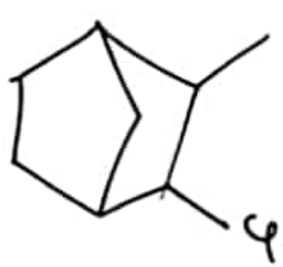
$\Rightarrow S_N1 \Rightarrow b > a > c$

$\Rightarrow S_N2 \Rightarrow c > a > b$

(6)





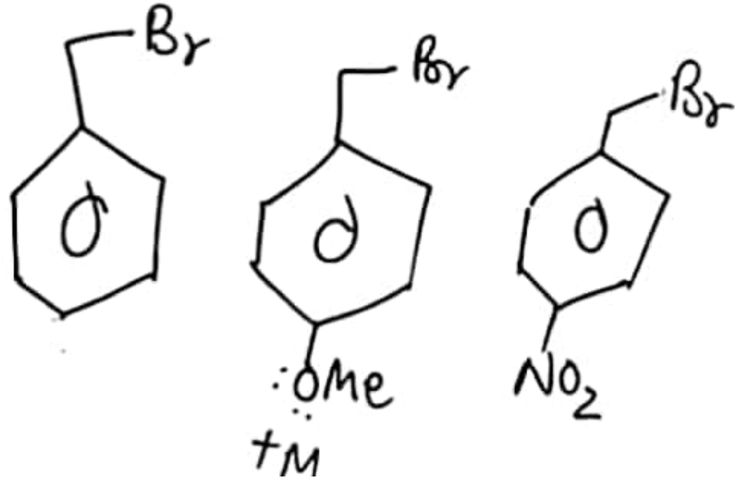




$= S_N1 \Rightarrow b > c > \underline{a > d}$ ← NO S_N1

$= S_N2 = a > c > \underline{b > d}$ ← NO S_N2

(7)



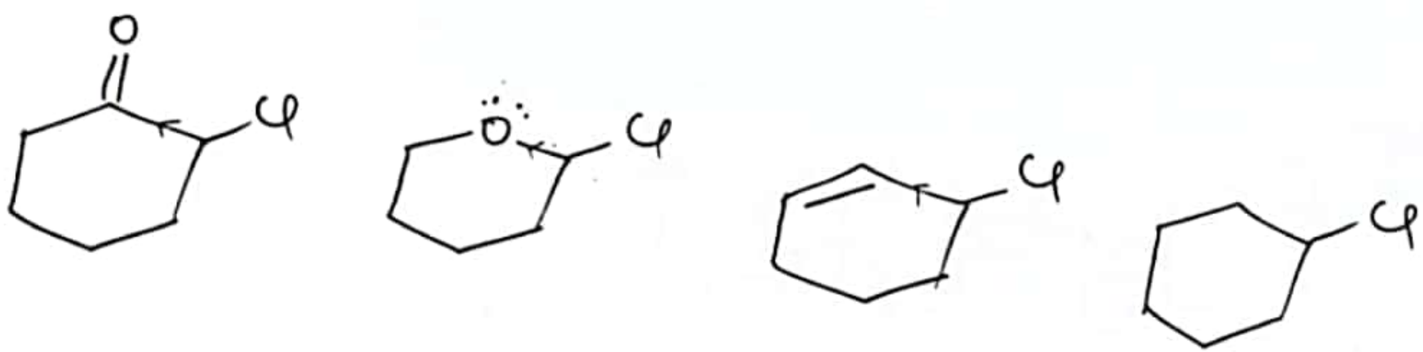
$\Rightarrow SN1 = b > a > c$

$\Rightarrow SN2 = c > a > b$

ORGANIC REACTION MECHANISM – III

Q Give order of reactivity for S_N1 and S_N2 Rxn.

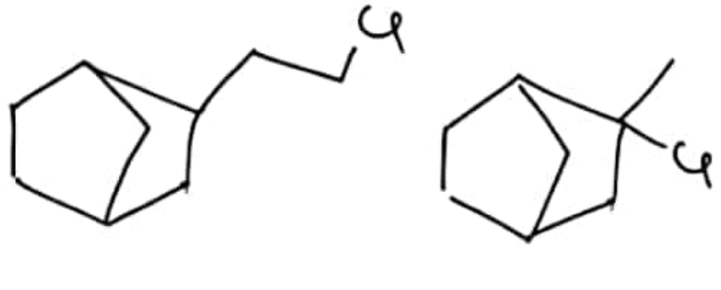
(8)

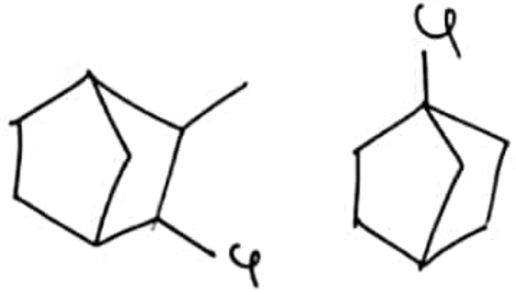


S_N1 = b > c > d > a ← No S_N1

S_N2 = a > b > c > d

(6)

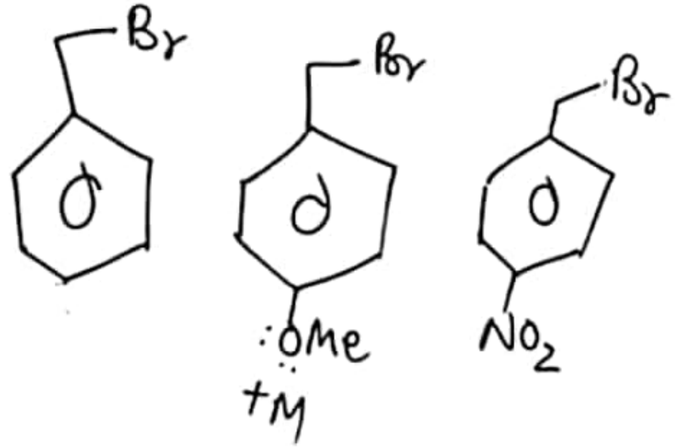




$\Rightarrow S_N1 = b > c > \underline{a > d}$ ← No S_N1

$\Rightarrow S_N2 = a > c > \underline{b > d}$ ← No S_N2

(7)



$\Rightarrow SN1 = b > a > c$

$\Rightarrow SN2 = c > a > b$

ORGANIC REACTION MECHANISM – III

ORM-III

Q How many of the following can give white ppt with $AgNO_3$ aq. solution ?

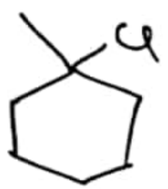
if the cation formed by the removal of Cl- is stable, then ppt. with AgNO3 will be favourable; else the rex. won't go in forward direction

AgI = yellow ppt

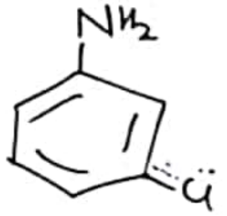
AgBr = Pale yellow ppt

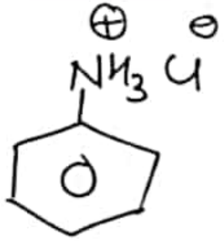
AgCl = White ppt

AgF = Soluble in water

✓ ① 

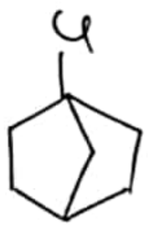
✓ ② HCl

✗ ③ 

✓ ④ 

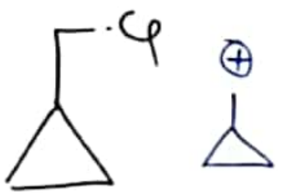
the acyl cation will be resonance stabilised

✓ ⑤ $CH_3-\overset{\overset{:O:}{\|}}{C}-Cl$

✗ ⑥ 

✓ ⑦ 

✓ ⑧ 

✓ ⑨ 

✗ ⑩ 

✗ ⑪ CH_3-Cl

✗ ⑫ 

✗ ⑬ $CH_2=CH-Cl$

Ans = 7

ORGANIC REACTION MECHANISM – III

(I) General SN reaction of alkyl halide

(II) Ammonolysis of alkyl halide.

(III) Reactions of halogen exchange in R–X

(IV) SN Rx^n of ambident nucleophile in R–X

(V) Wurtz Rx^n

(VI) S_N Rx^n of organometallic comp.

(VII) Williamson's ether synthesis (WES)

(VIII) SN Reaction of allylic position [SN' ⟨ SN'' / SN²']

(IX) Intramolecular SN^2 Rx^n

(X) SN Rx^n of R–OH

(XI) " " " ethers

(XII) " " " epoxides

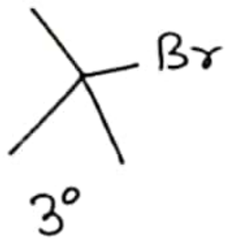
(XIII) " " Ar–X
(ArSN)

ORGANIC REACTION MECHANISM – III

ORM-III

(I) General SN Reaction of Alkyl halides:

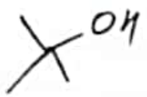
eg 1



$H_2\ddot{O}$, Δ SN'



H_2O + Acetone SN'



$H_2O + AgNO_3$ SN'

or $AgNO_{3aq.}$

" +

Ag^+, Hg^{+2}

in presence of these 2 metal ions, alkyl halide gives SN1 rex. easily (due to precipitation)

AgOH SN'

"

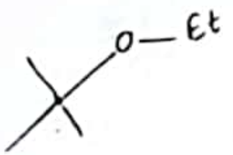
Ag_2O(moist) SN'

"

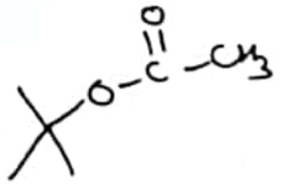
despite being a strong Nu- , it cannot make the tertiary undergo SN2

☆ KOH + H_2O SN'

EtOH SN'

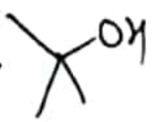


CH_3–COOH SN'

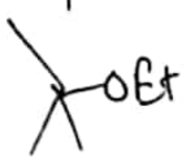


EtOH + H_2O

S_N1

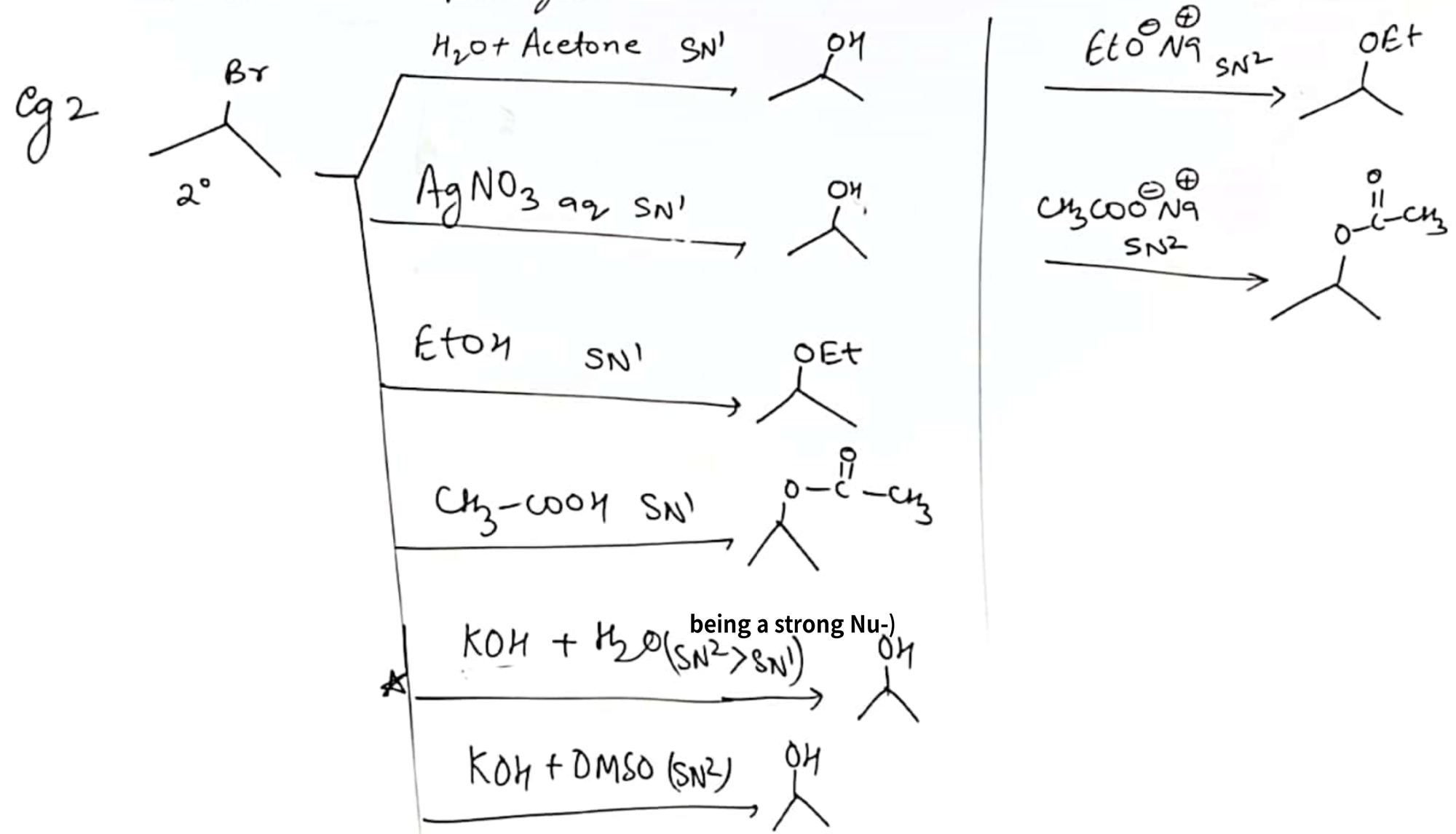


+



ORGANIC REACTION MECHANISM – III

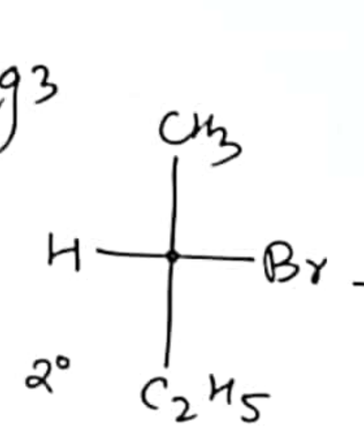
(I) General SN Reaction of Alkyl halides:



ORGANIC REACTION MECHANISM – III

(I) General SN Reaction of Alkyl halides:

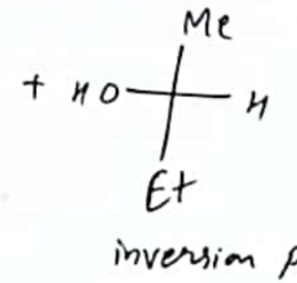
Eg 3



H_2O + Acetone
(SN1)



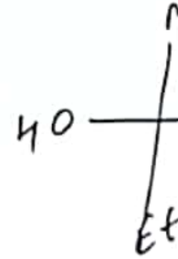
Retention



} Enantiomers.

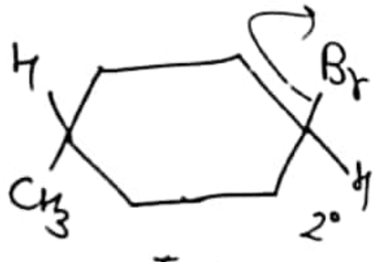
inversion product

KOH + DMSO
SN^2



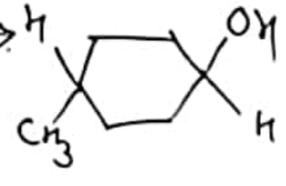
only inversion product

Eg 4

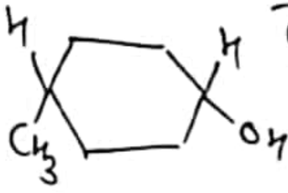


Trans

$AgNO_3 + H_2O$
(SN1)

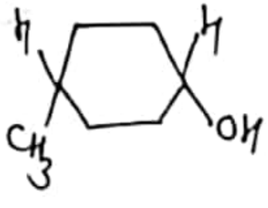


+



} mix. of geo. iso. (diastereomers)

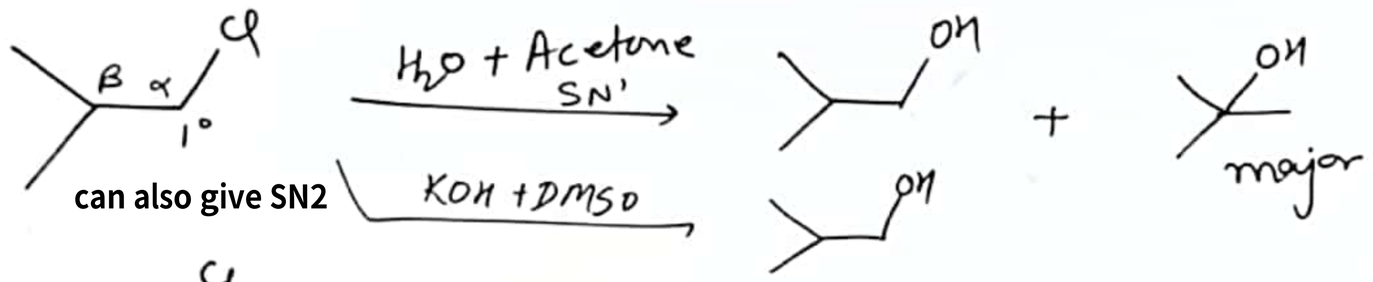
KOH + DMSO
SN2



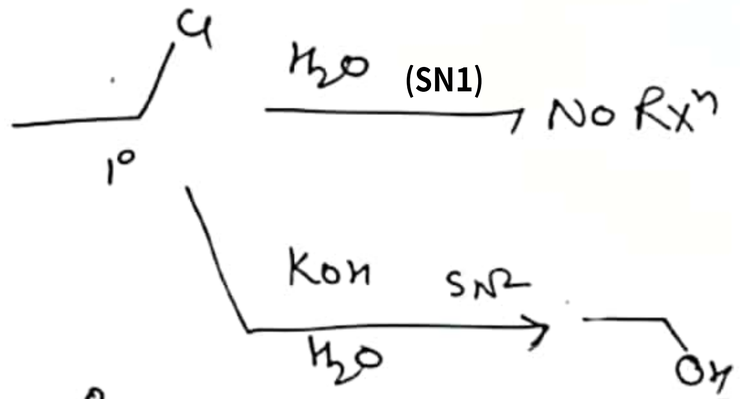
ORGANIC REACTION MECHANISM – III

(I) General SN Reaction of Alkyl halides:

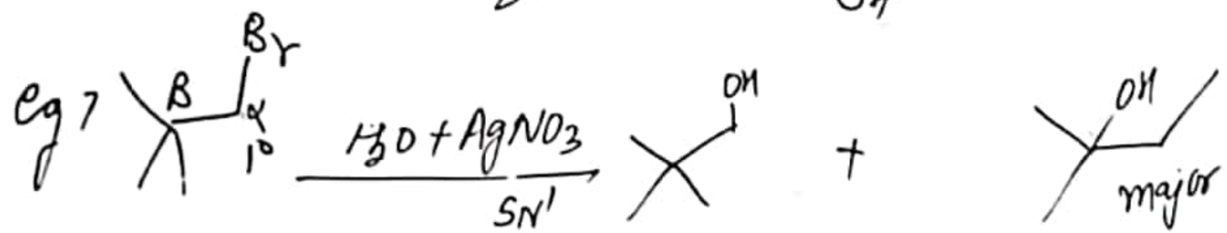
eg 5



eg 6

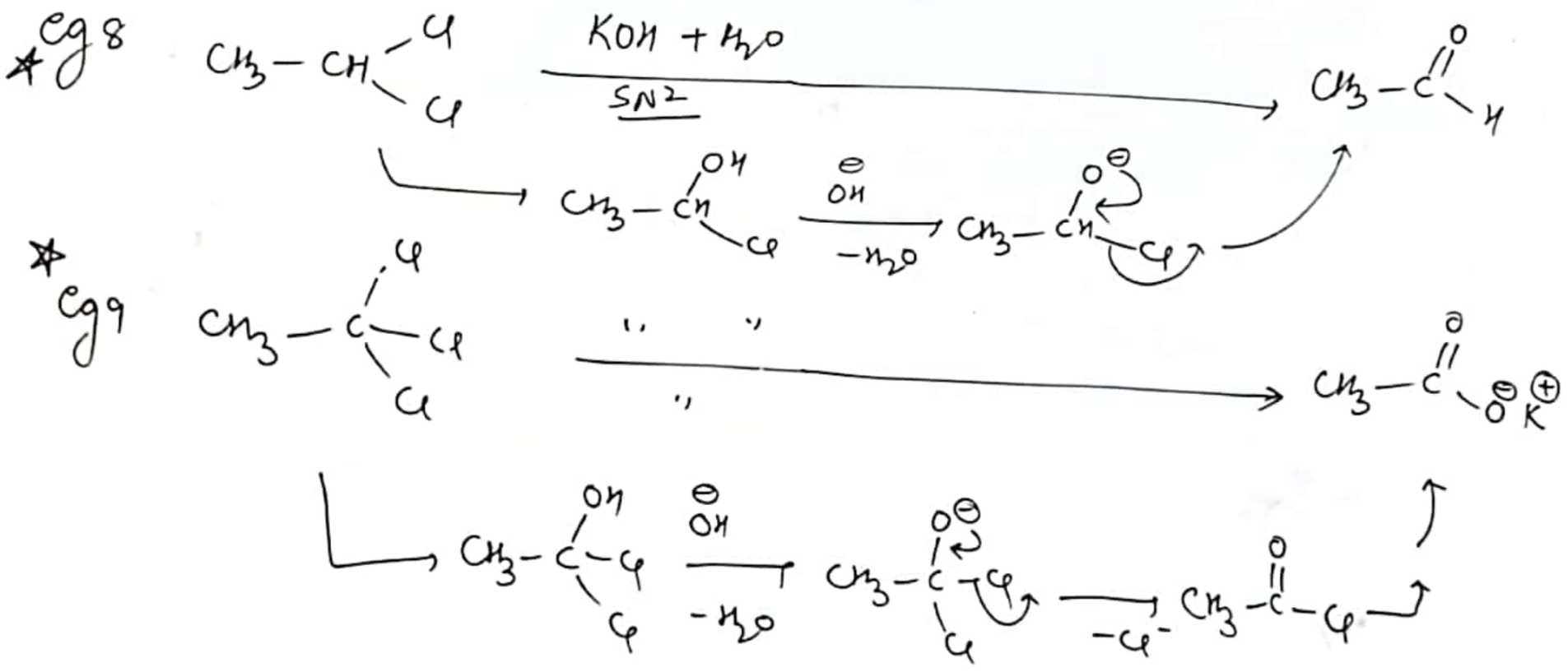


eg 7



ORGANIC REACTION MECHANISM – III

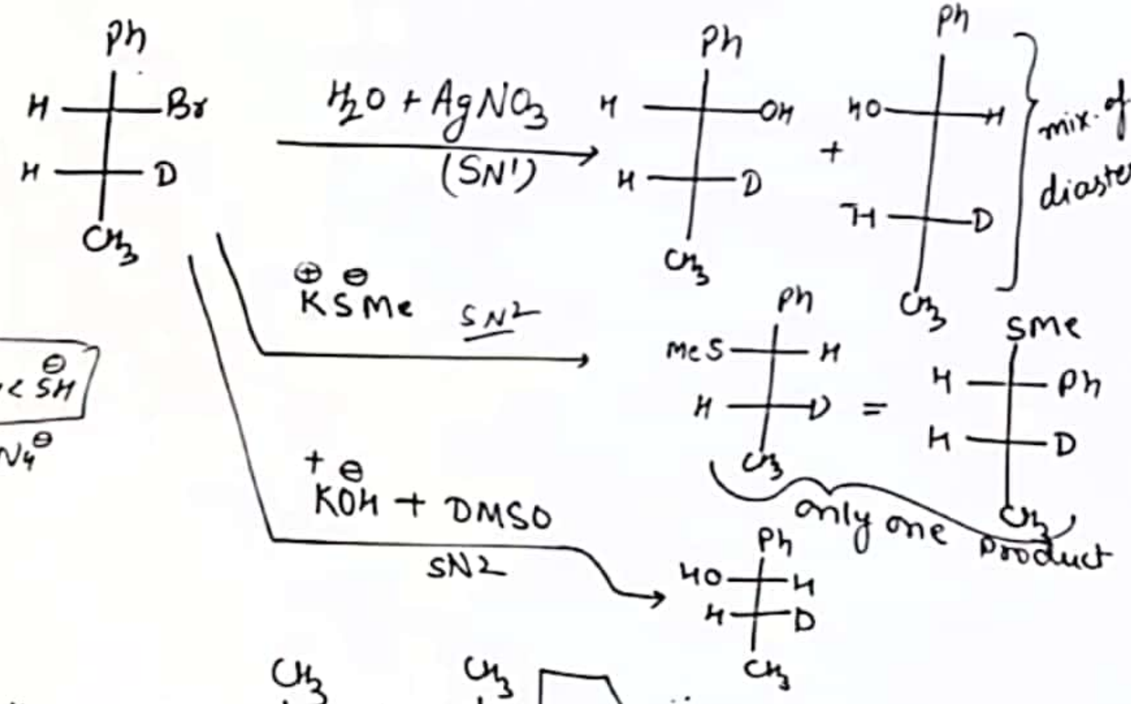
(I) General SN Reaction of Alkyl halides:



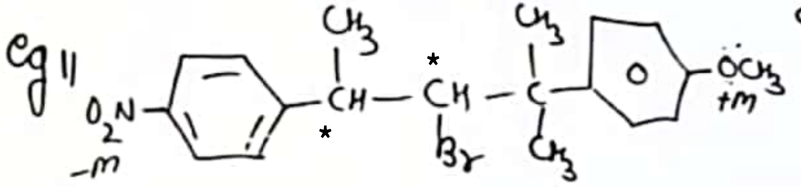
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

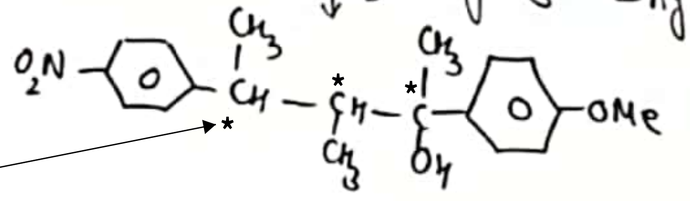
Eg 10



Eg 11



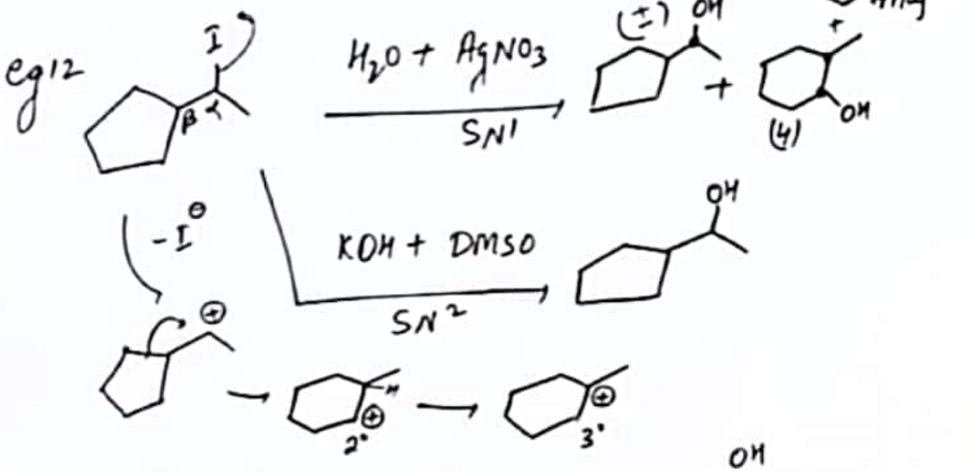
SN1 | $H_2O + AgNO_3$ $-AgBr\downarrow$



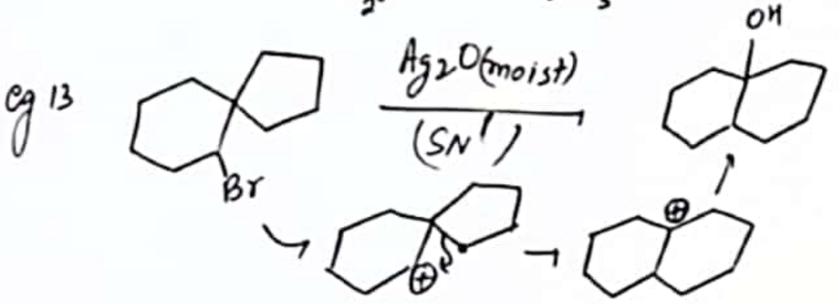
configuration is fixed here

total 4 products

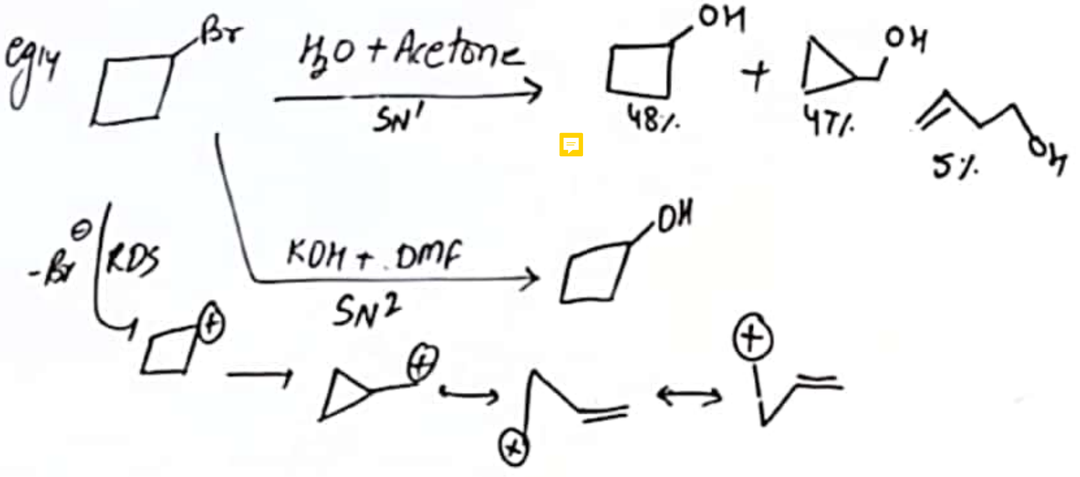
Eg 12



Eg 13

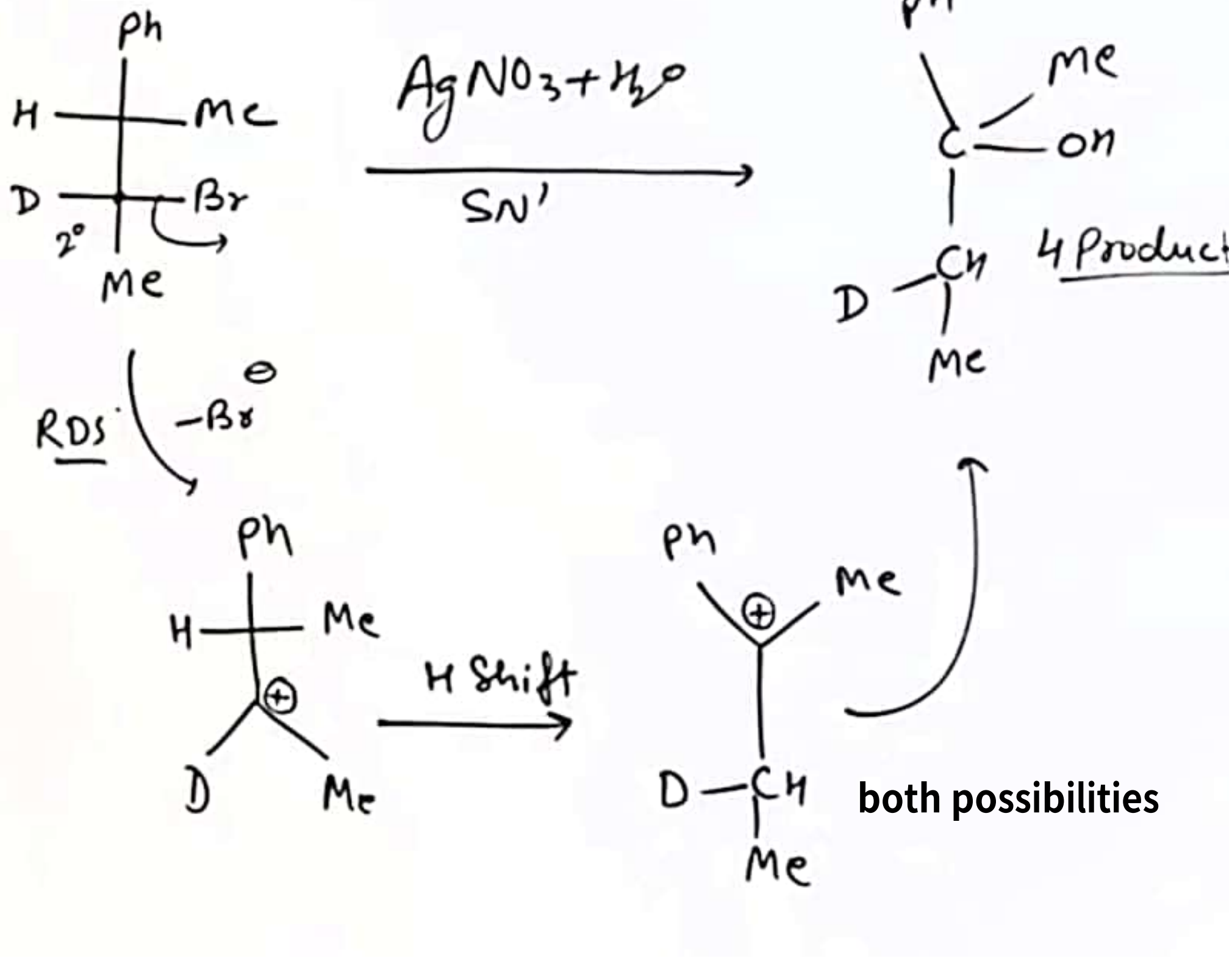


Eg 14



(I) General S_N Reactions :—

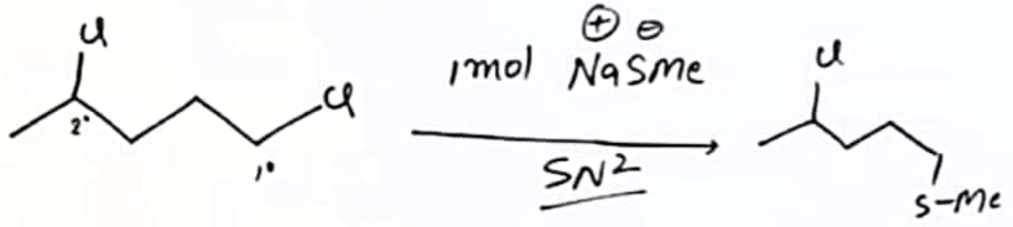
eg 15



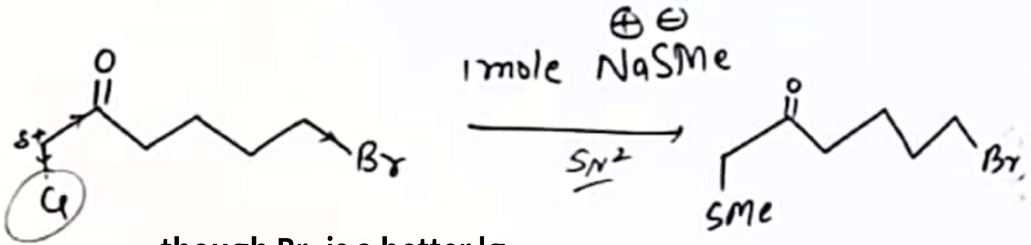
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 18

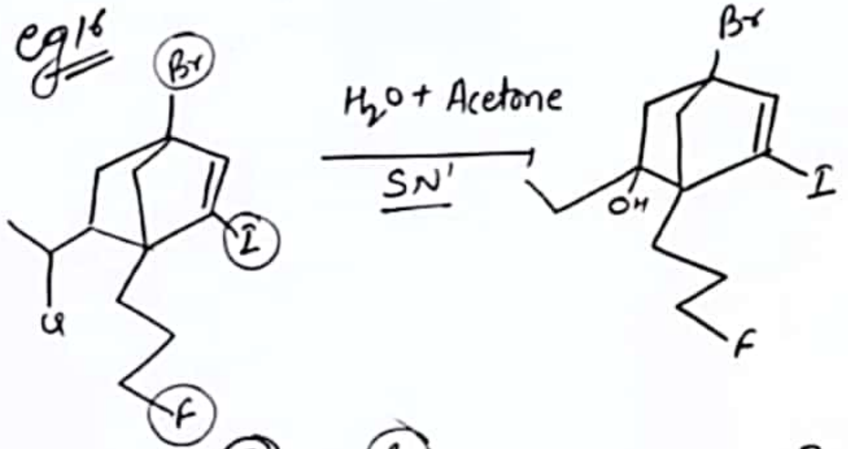


eg 19

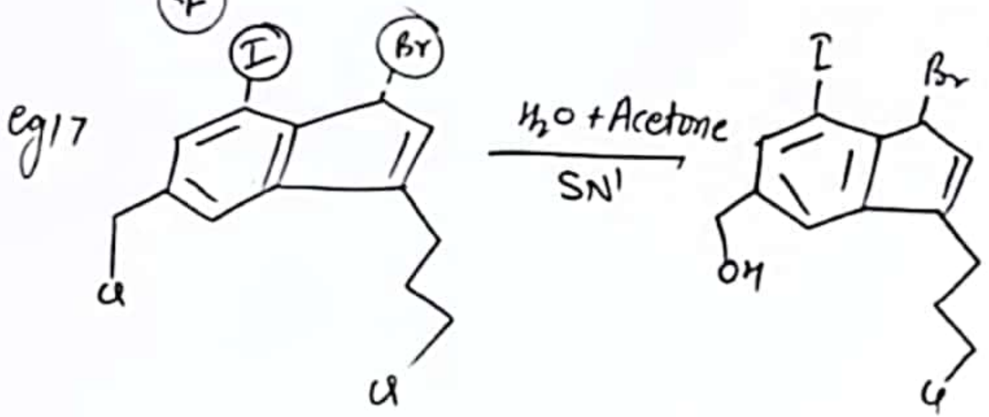


though Br- is a better lg
yet, Cl- is not that a bad leaving group and we have an EWG in adjacent position; hence it is more favoured

eg 16



eg 17



simple primary; no SN1

ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 20

CH_3-Br

1°

① H_2O + Acetone → No Rx^n

② EtOH → No Rx^n

③ KOH + H_2O S_N2 → CH_3-OH

④ KOH + DMSO (S_N2) → CH_3-OH

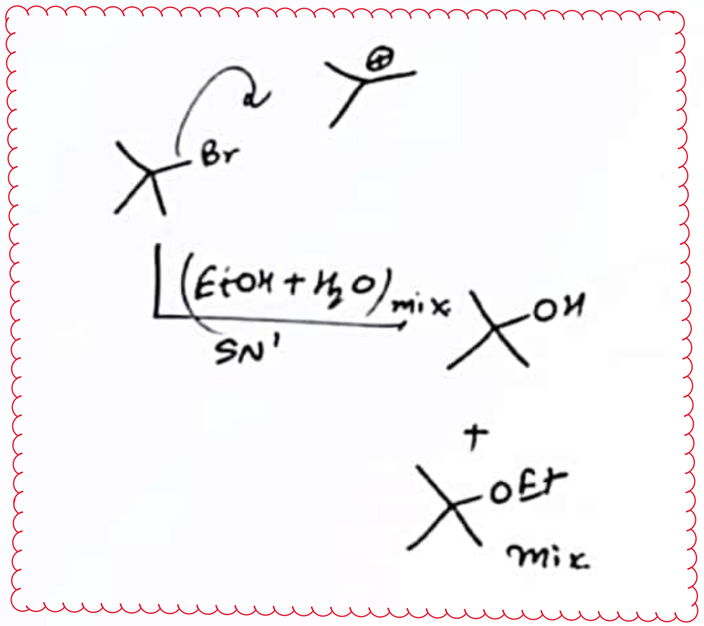
⑤ $EtO^{\ominus}Na^{\oplus}$ S_N2 → $CH_3-O-C_2H_5$

$EtO^{\ominus} > Ph-O^{\ominus}$

Nucleoph.

⑥ $Ph-O^{\ominus}Na^{\oplus}$ S_N2 → CH_3-O-Ph

⑦ $(EtO^{\ominus} + Ph-O^{\ominus})_{mix}$ → CH_3-O-Et (S_N2)



ORGANIC REACTION MECHANISM - III (ORM - III)

generally, acid-base rxn. > intramolecular > intermolecular

(I) General S_N Reactions :-

eg 20

CH_3-Br (1°)

① H_2O + Acetone → No Rx^n

② EtOH → No Rx^n

③ KOH + H_2O S_N2 → CH_3-OH

④ KOH + DMSO (S_N2) → CH_3-OH

⑤ $EtO^{\ominus}Na^{\oplus}$ S_N2 → $CH_3-O-C_2H_5$

$EtO^{\ominus} > Ph\text{-}O^{\ominus}$

Nucleoph.

⑥ $Ph\text{-}O^{\ominus}Na^{\oplus}$ S_N2 → CH_3-O-Ph

7 $(EtO^{\ominus} + Ph\text{-}O^{\ominus})_{mix.}$ S_N2 → CH_3-O-Et

WES

⑧ EtOH + Base → CH_3-OEt (S_N2; $EtO^{\ominus}$)

⑨ $(EtOH + Na)_{mix}$ → $EtO^{\ominus}Na^{\oplus} + H_2$,,

⑩ $Ph-\ddot{O}H$ → No Rx^n

⑪ $Ph-OH + Na$ → CH_3-OPh ($Ph\text{-}O^{\ominus}$)

⑫ $Ph-OH$ (1 mol) + $EtO^{\ominus}Na$ (1 mol) → CH_3-O-Ph ($Ph\text{-}O^{\ominus} + EtOH$)

⑬ $Ph-OH + EtO^{\ominus}_{excess}$ → CH_3-O-Et ($Ph\text{-}O^{\ominus} + EtOH + EtO^{\ominus}$)

Williamson's ether Synthesis.

ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-

eg 20 CH_3-Br (1°)

(14) $Na^{+}\,{}^{\ominus}SH$ S_N2 → CH_3-SH thioalcohol

(15) Na_2S^{-2} S_N2 → CH_3-S-CH_3 thioether

(16) $MeSH + Na$ S_N2 → CH_3-S-CH_3 ; $MeSH + Na \rightarrow Me-S^{\ominus}$

(17) NaN_3 S_N2 → CH_3-N_3

(18) $NaCN$ S_N2 → CH_3-CN

(19) $NaNH_2$ → CH_3-NH_2

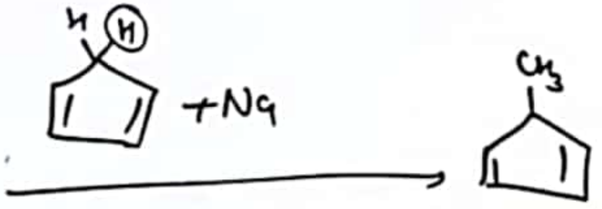
(20) CH_3Na or CH_3Li → CH_3-CH_3

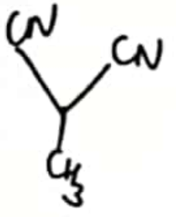
(21) $R\,MgX$ S_N2 → $R-CH_3 + Mg(X)Br$

22 R_2CuLi S_N2 → CH_3-R Corey-House Synthesis of alkane

(23) $R-C\equiv C^{\ominus}\,Na^{\oplus}$ S_N2 → $R-C\equiv C-CH_3$

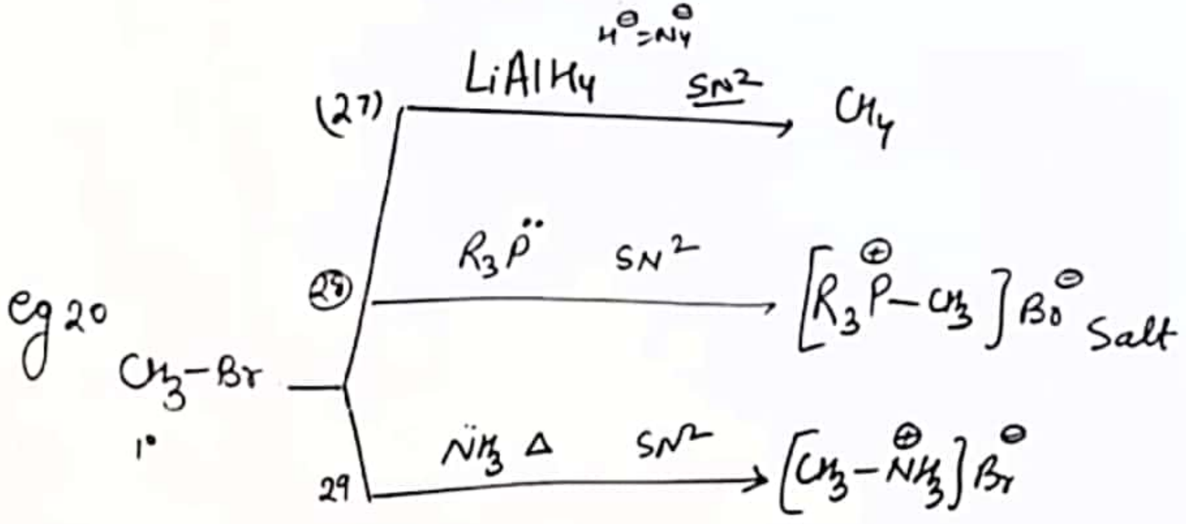
(24) $R-C\equiv C-H$ (C is sp; H is acid) $+ Na$ (S_N2) → $R-C\equiv C-CH_3$; $R-C\equiv C^{\ominus}\,Na^{\oplus}$

(25)  $+ Na$ →

(26)  $+ NaOH$ → 

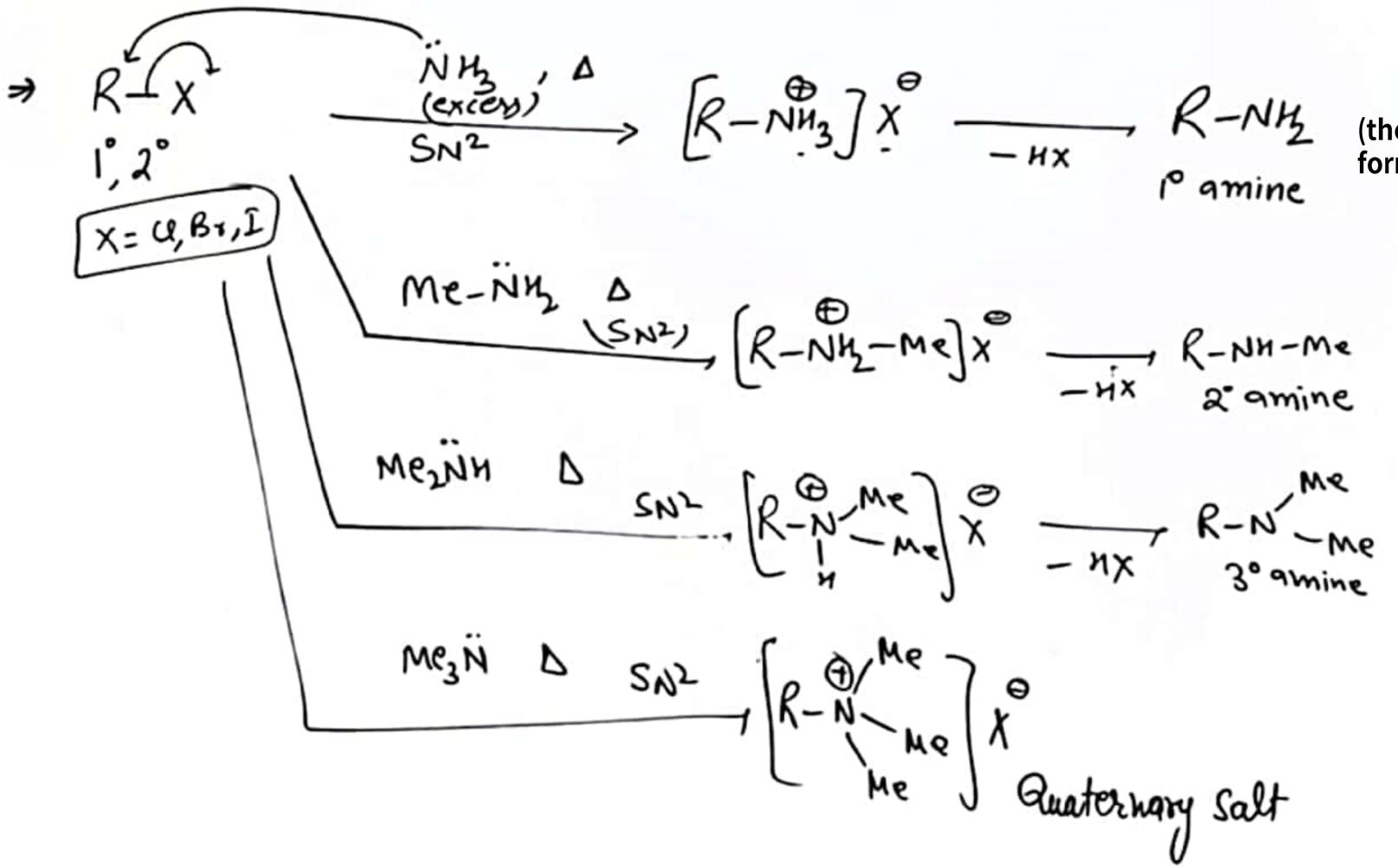
ORGANIC REACTION MECHANISM - III (ORM - III)

(I) General S_N Reactions :-



ORGANIC REACTION MECHANISM - III (ORM - III)

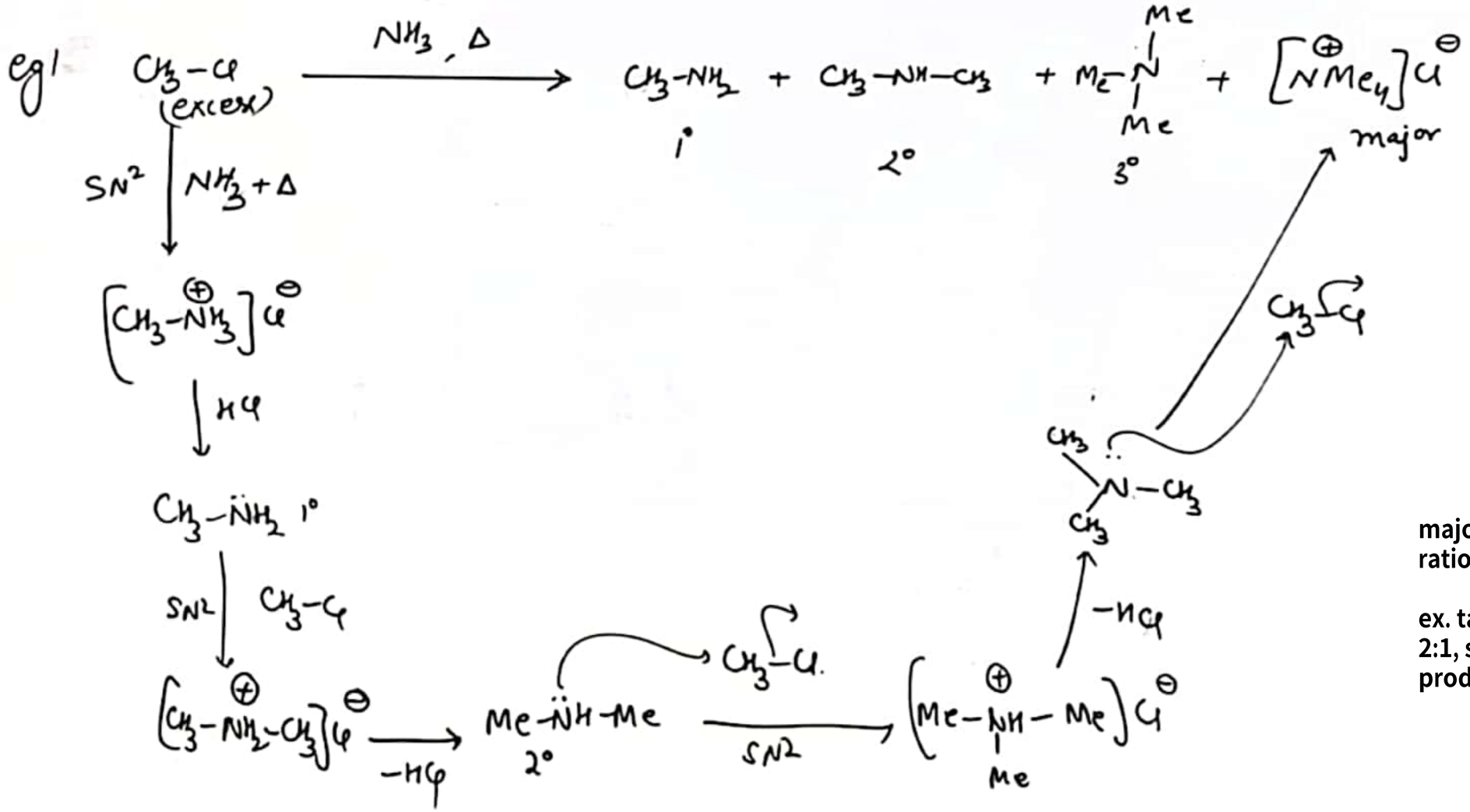
(II) Ammonolysis of alkyl halide :- (S_N2)



(the formed HX reacts with NH3 forming a salt)

ORGANIC REACTION MECHANISM - III (ORM - III)

(II) Ammonolysis of alkyl halide :- (SN^2)

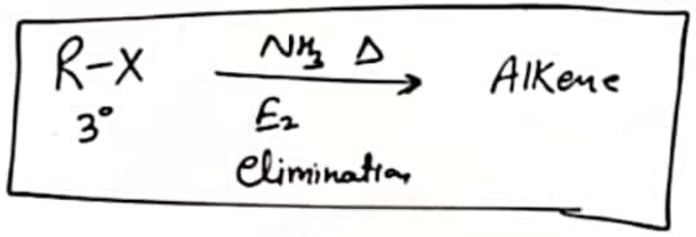


major product depends on the ratio of RX and NH3 taken

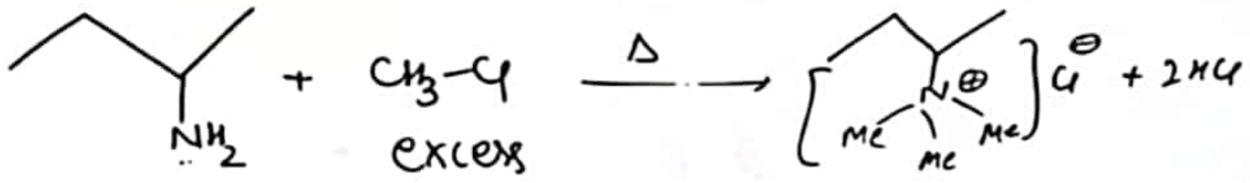
ex. taking RX and NH3 in the ratio 2:1, secondary amine will be major product

ORGANIC REACTION MECHANISM - III (ORM - III)

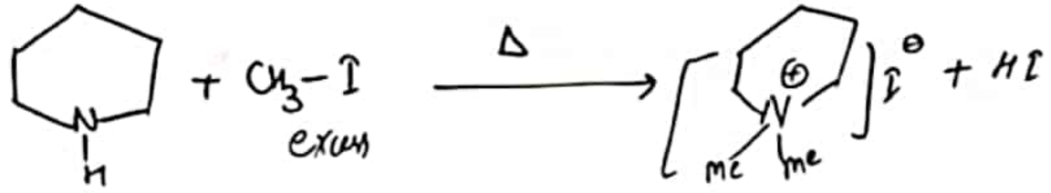
(II) Ammonolysis of alkyl halide :- (SN^2)



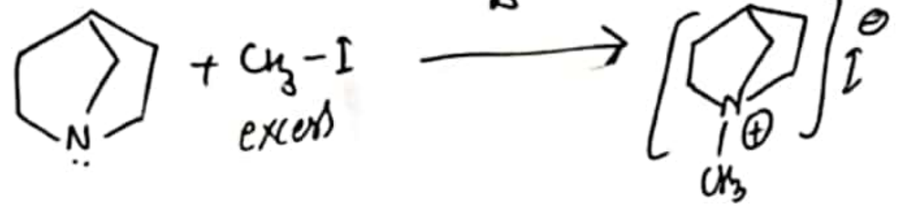
eg 2



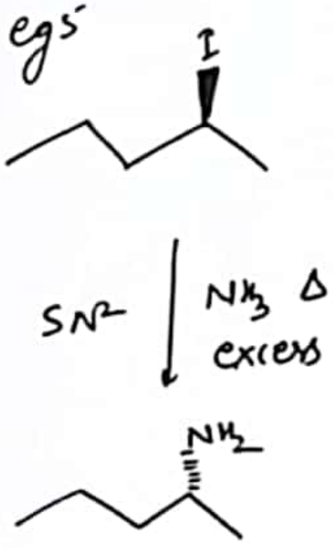
eg 3



eg 4



eg 5



ORGANIC REACTION MECHANISM - III (ORM - III)

(III) S_N Reactions of halogen exchange:—

the driving force behind these reactions is mainly the precipitation of the salts

$\Rightarrow$ $R\overset{\delta+}{-}X$ ($X = Cl, Br$) $\xrightarrow[\text{(Finkelstein Reaction)}]{NaI + \text{Acetone}\ \ S_N2}$ $R-I + NaX\downarrow$ (Ppt)

- NaI soluble in acetone
- NaCl insoluble in acetone
- NaBr insoluble " "

$F^{\ominus}\cdots\overset{\delta+}{R}-\overset{\delta-}{X}$ ($X = Cl, Br, I$) $\xrightarrow[\text{(Swartz Reaction)}]{AgF_{aq.} \,/\, Hg_2F_2 \,/\, CoF_2}$ $R-F + AgX\downarrow$ (Ppt)

- AgF soluble in H_2O
- AgCl insoluble in H_2O
- AgBr " "
- AgI " "

Nucleophilicity $F^{\ominus} < Cl^{\ominus} < Br^{\ominus} < I^{\ominus}$ in polar protic solvent

$R-X \xrightarrow{NaF_{aq.}}$ ND $R_X{}^n$ (weak $N^{\ominus}$ in H_2O)

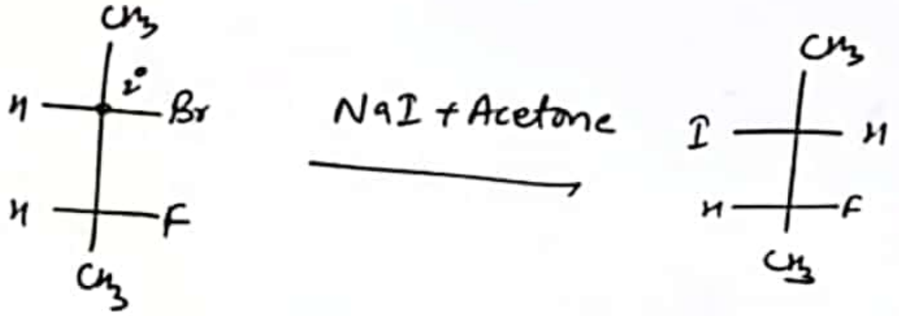
$R-X \xrightarrow[S_N2]{NaF + DMF}$ $R-F$ (strong $N^{\ominus}$)

Nucleophilicity in polar aprotic solvent $\Rightarrow F^{\ominus} > Cl^{\ominus} > Br^{\ominus} > I^{\ominus}$

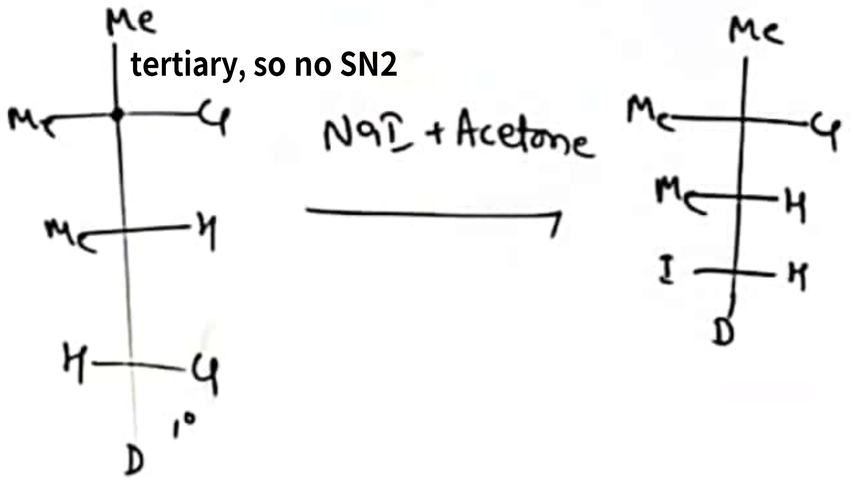
ORGANIC REACTION MECHANISM - III (ORM - III)

(III) S_N Reactions of halogen exchange :—

eg 1

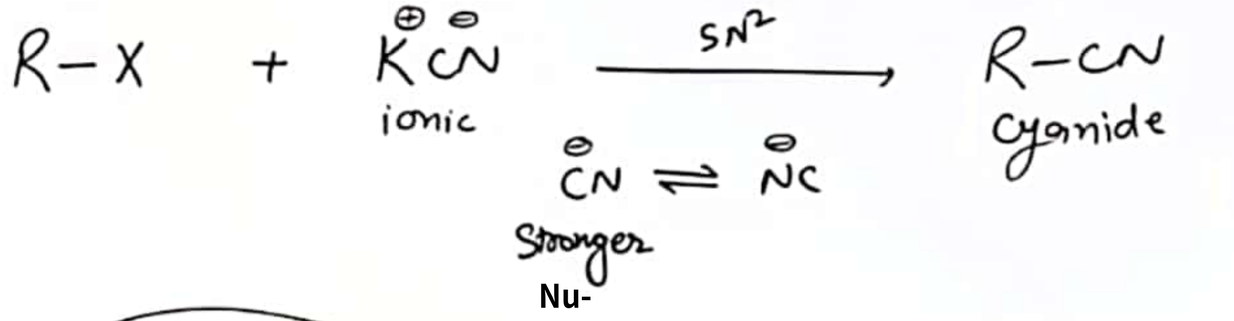


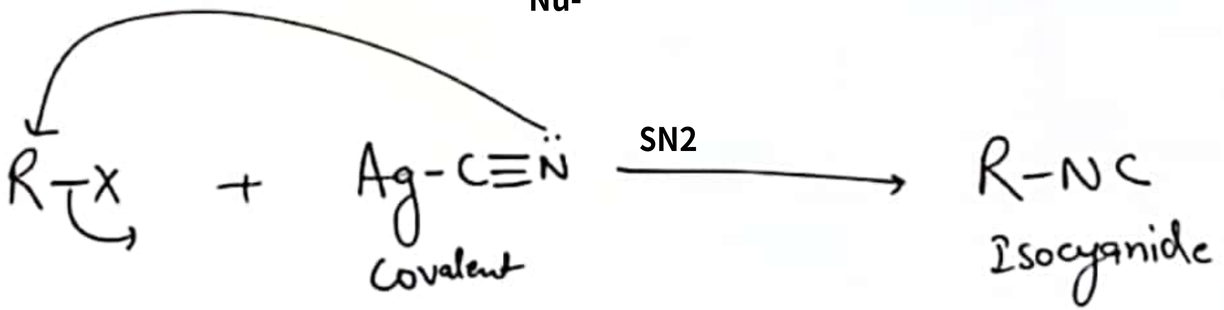
eg 2



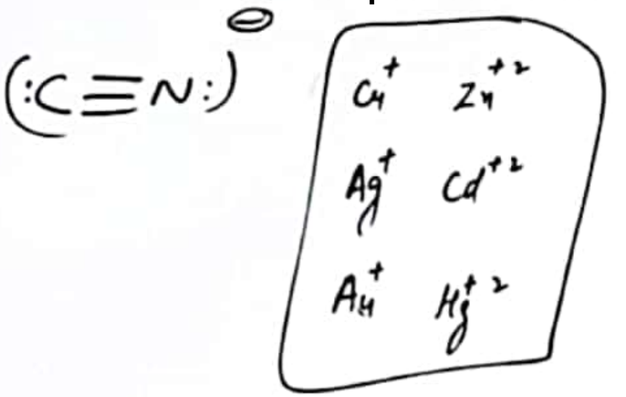
ORGANIC REACTION MECHANISM - III (ORM - III)

(IV) SN Reaction of ambident nucleophiles



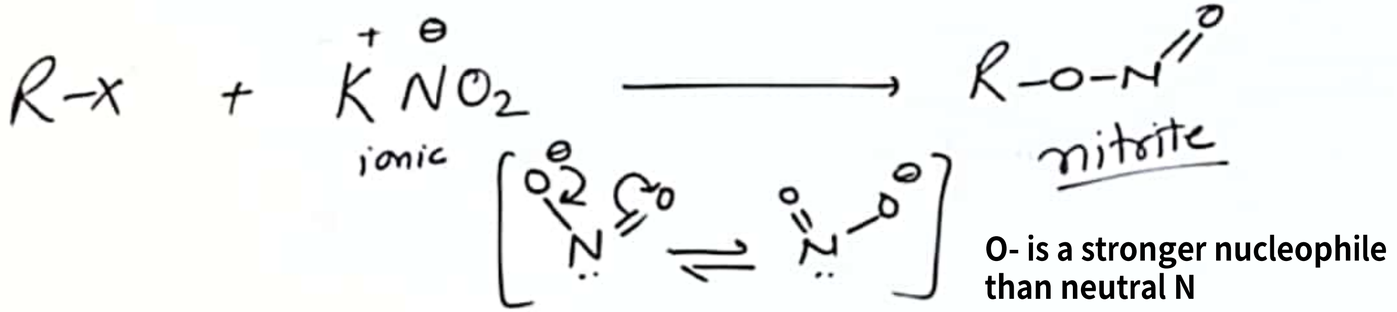


pseudo-inert gas configuration (high covalent character in compounds of these ions)

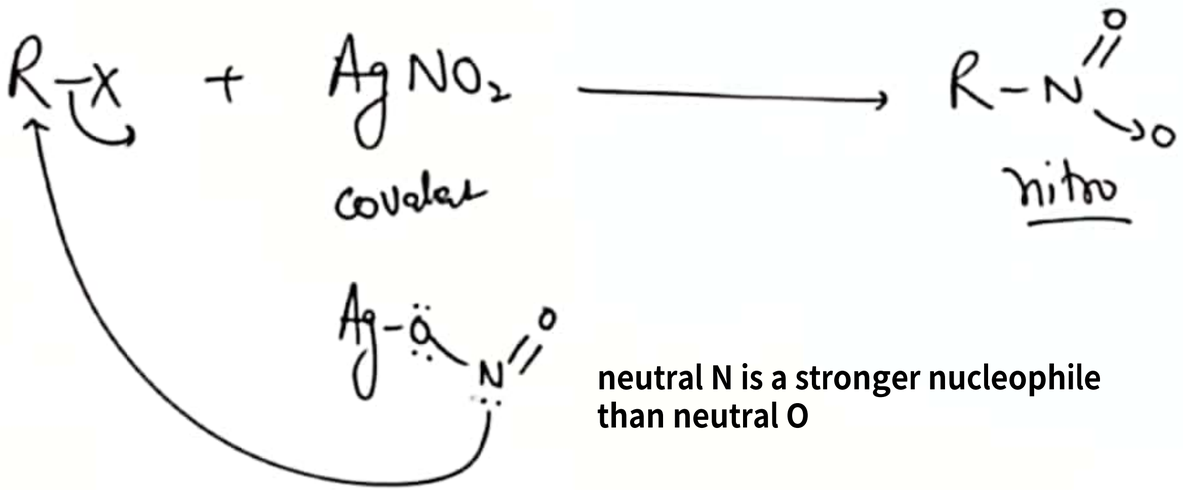


ORGANIC REACTION MECHANISM - III (ORM - III)

(IV) SN Reaction of ambident nucleophiles



O- is a stronger nucleophile than neutral N



neutral N is a stronger nucleophile than neutral O

ORGANIC REACTION MECHANISM - III (ORM-III)

(V) **Wurtz Reaction:-**

$$R-X + 2Na + X-R \xrightarrow[\text{solvent}]{\text{dry ether}} R-R + 2NaX$$

R–R : alkane

mechanism

(a) Ionic mechanism (S_N2)

$$2Na \longrightarrow 2Na^{\oplus} + 2e^{\ominus}$$

$$R-X + 2e^- \longrightarrow R^{\ominus} + X^{\ominus}$$

$$R^{\ominus} + R-X \xrightarrow{S_N2} R-R + X^{\ominus}$$

Strong Nucleophile

$1^\circ > 2^\circ > 3^\circ$ probability of SN2

(b) Free-Radical mechanism

$$\cdot Na \longrightarrow Na^+ + e^-$$

$$R-X + e^- \longrightarrow R^\cdot + X^{\ominus}$$

$$R^\cdot + R^\cdot \longrightarrow R-R$$

$3^\circ > 2^\circ > 1^\circ$ probability of free radical mechanism

ORGANIC REACTION MECHANISM - III (ORM-III)

(V) Wurtz Reaction:-

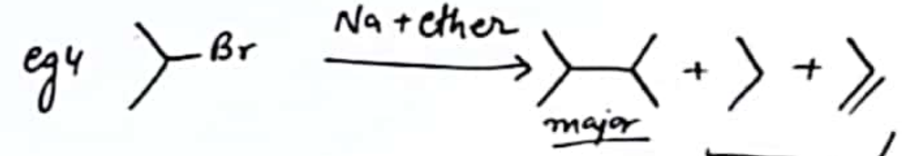
eg 1 $CH_3-Cl \xrightarrow{Na + ether} CH_3-CH_3$

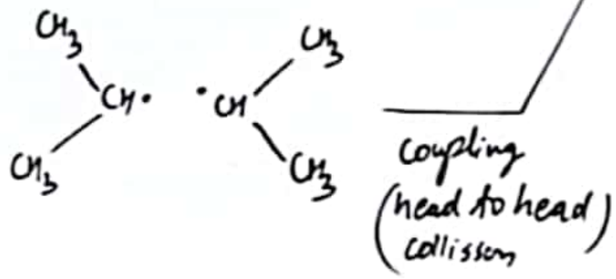
eg 2 $C_2H_5-Cl \xrightarrow{Na + ether} C_2H_5-C_2H_5$

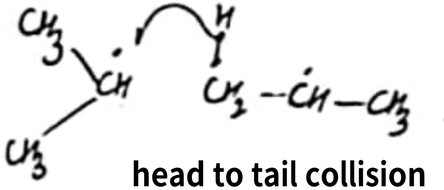
eg 3 $[CH_3-Cl + C_2H_5-Cl]_{mix} \xrightarrow{Na + Et_2O}$ CH_3-CH_3 + $C_2H_5-C_2H_5$ + $CH_3-CH_2-CH_3$

$R-X + R'-X \xrightarrow{Na+ether}$ $R-R$, $R-R'$, $R'-R'$

thus, alkanes of the type R-R' cannot be formed exclusively using Wurtz reaction

eg 4 

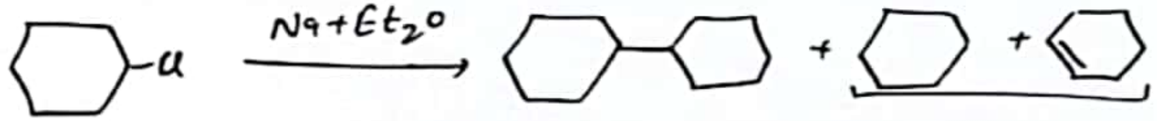


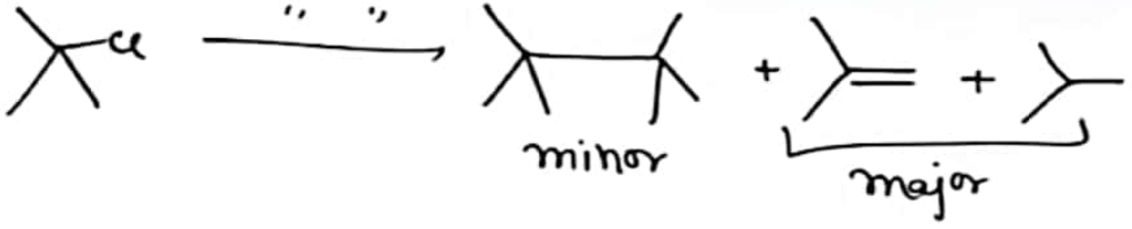


Note for 3° alkyl halide in wurtz Rx^n ⇓ Major product (disproportionation)

ORGANIC REACTION MECHANISM - III (ORM-III)

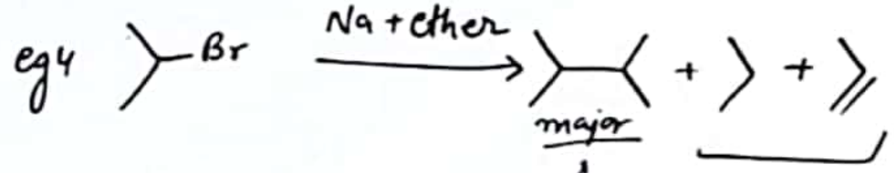
(V) Wurtz Reaction:-

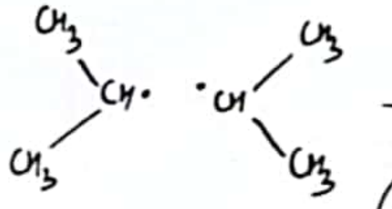
eg 5 

eg 6 

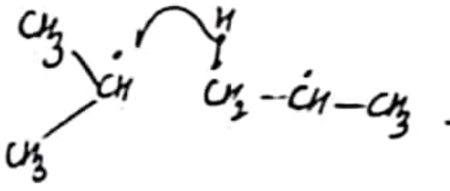
eg 7 $Ph-CH_2-Cl \xrightarrow{Na,\ Et_2O} Ph-CH_2-CH_2-Ph$

1,2 - Diphenyl ethane

eg 4 



Coupling (head to head collision)



Disproportionation.

Note for 3° alkyl halide in wurtz Rxn

⇓

Major product (disproportionation)

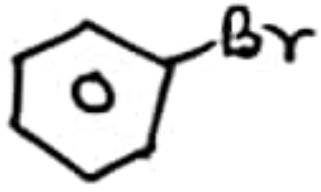
(V) Wurtz Reaction:-

Wurtz- Fittig Rxn

$$Ar-X + R-X \xrightarrow{Na + ether} Ar-R$$

Fittig Rxn

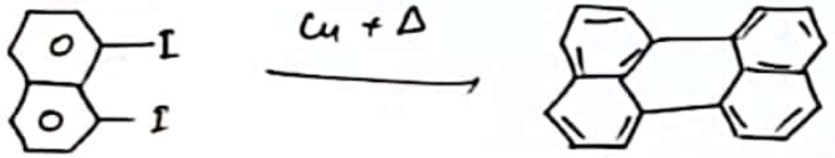
$$Ar-X \xrightarrow{Na + ether} Ar-Ar$$

Eg  $\xrightarrow{Na + ether}$ 

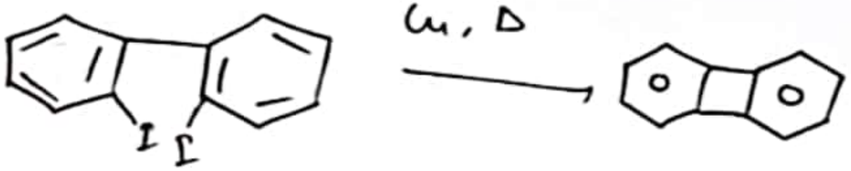
Biphenyl

ORGANIC REACTION MECHANISM - III (ORM-III)

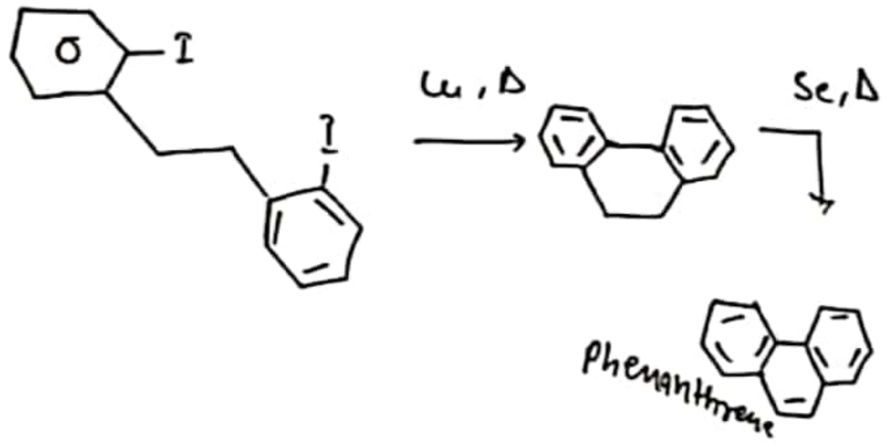
eg 4



eg 5



eg 6



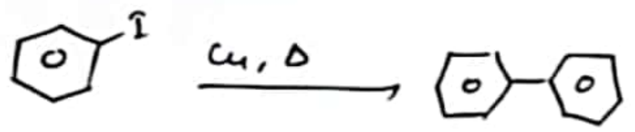
Ullamann's Reaction:-

$$2Ar\text{-}I \xrightarrow[-CuI]{Cu\ \Delta} Ar\text{-}Ar$$

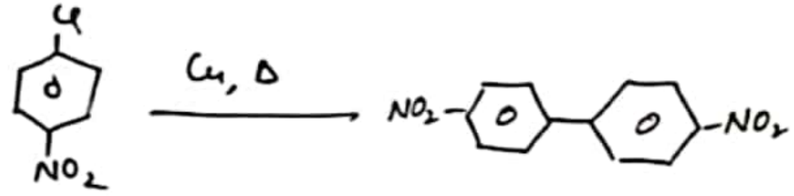
Radical mech.

rex. with Iodides takes place easily
for Chlorides and Bromides, EWG must be present at o,p on the ring

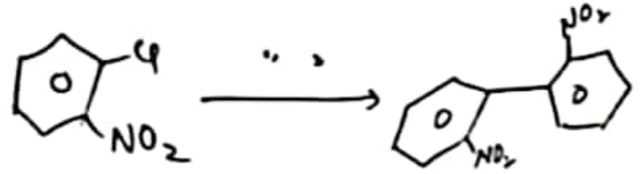
eg 1



eg 2

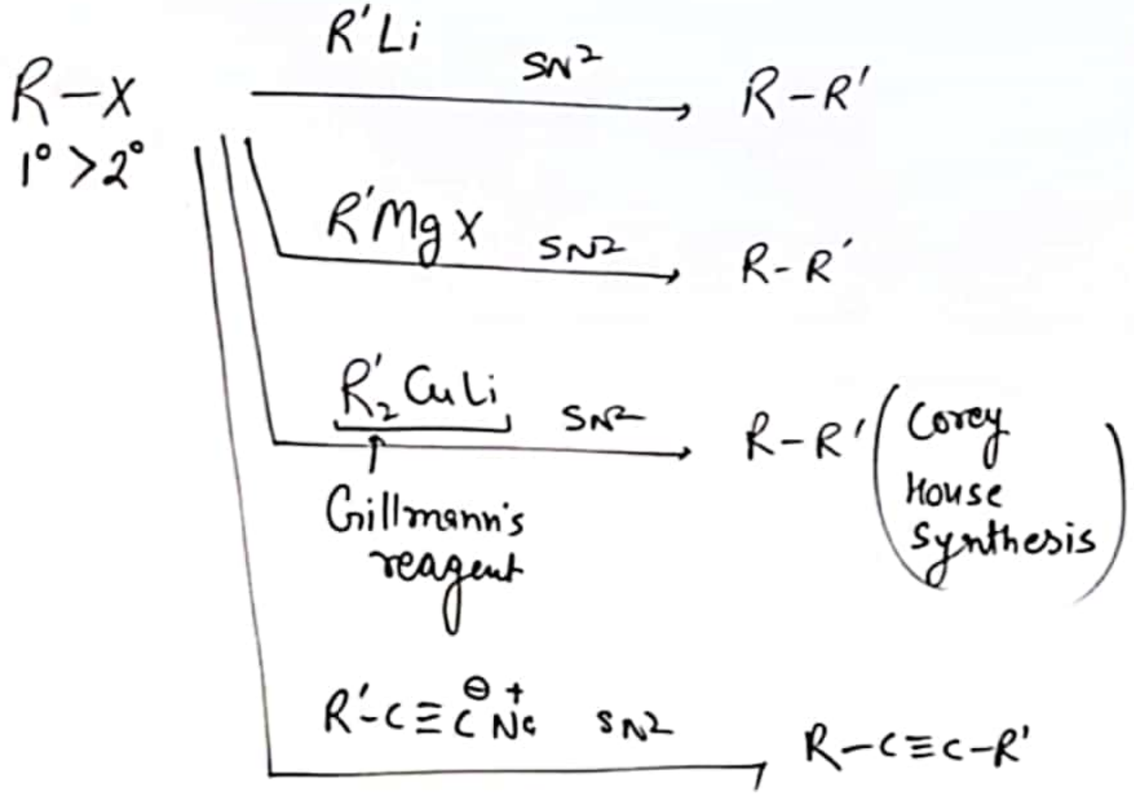


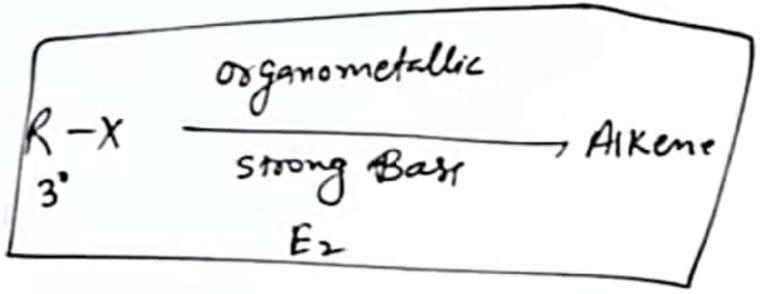
eg 3

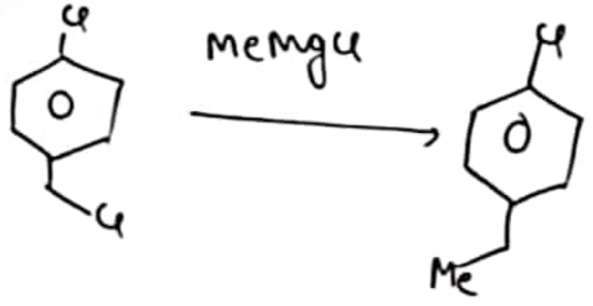


ORGANIC REACTION MECHANISM - III (ORM-III)

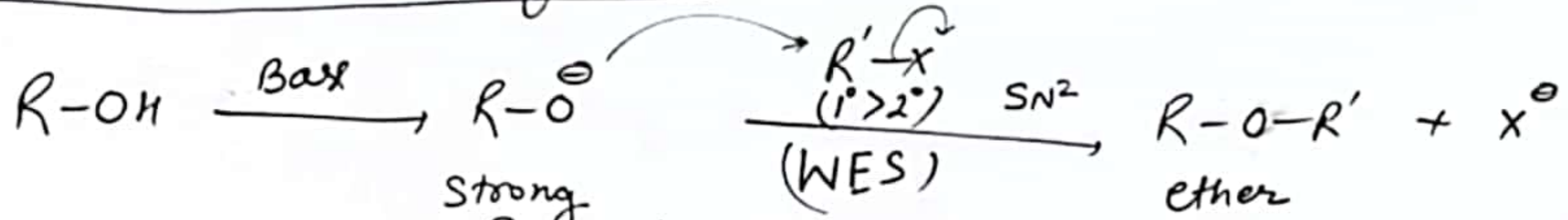
(VI) S_N2 Reaction of organometallic compound.

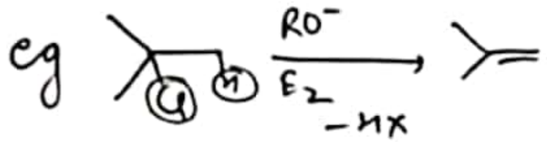




eg 

(VII) Williumson's Ether Synthesis:- (WES)



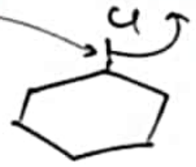
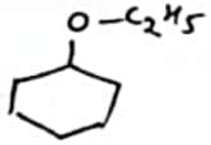
eg 

R-X 1°, 2°	$R'O^{\ominus}$ / NH_3, Δ / $R'MgX$ / R'_2CuLi / $LiAlH_4$ (SN2 Reaction)	R-O-R' / R-NH₂ / R-R' / R-R' / R-H
R-X 3°	" / " / " / ; / ,, E2 = Elimination (-HX)	Alkene

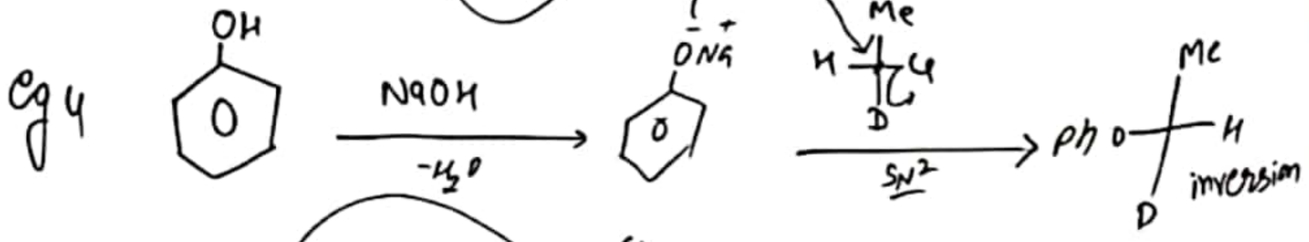
ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

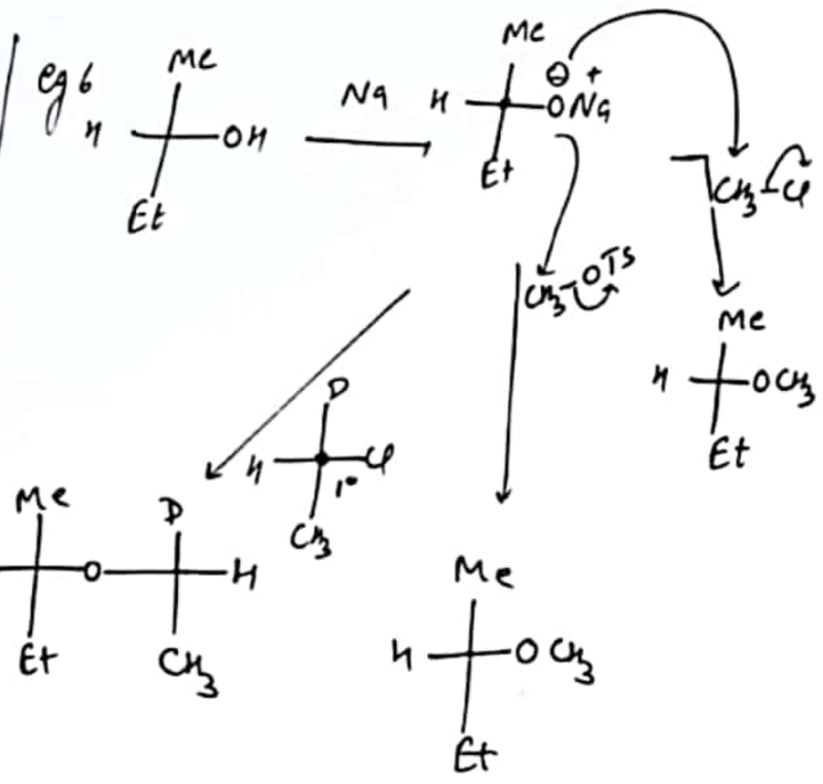
eg 1 $CH_3-\overset{\ominus}{O}Na + \text{C}_6\text{H}_5(CH_2-Cl) \longrightarrow CH_3-O-CH_2-Ph$

eg 2 $C_2H_5-\overset{\ominus}{O}Na +$  $\longrightarrow$ 

eg 3 

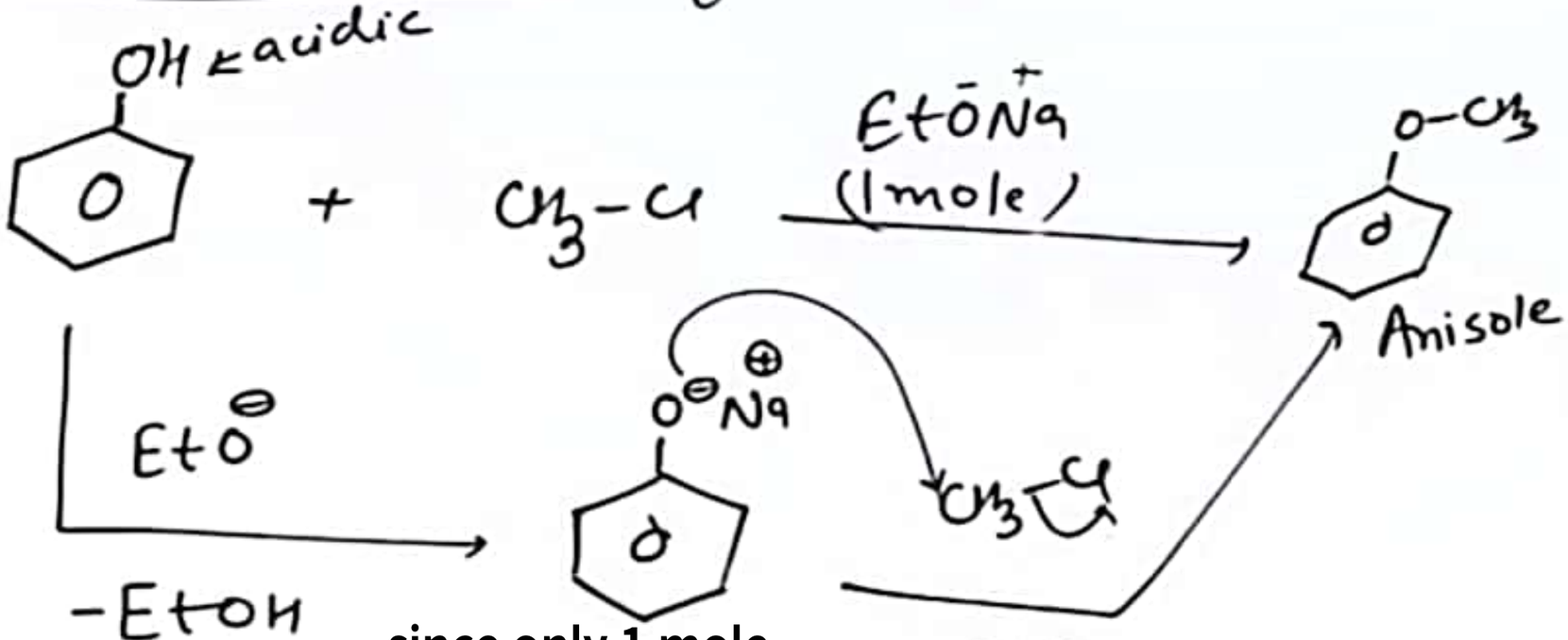
eg 4 

eg 5 $C_2H_5O^-Na^+ + Cl-CH_2-\underset{D}{\overset{CH_3}{C}}(H) \longrightarrow C_2H_5O-CH_2-\underset{D}{\overset{CH_3}{C}}(H)$

eg 6 

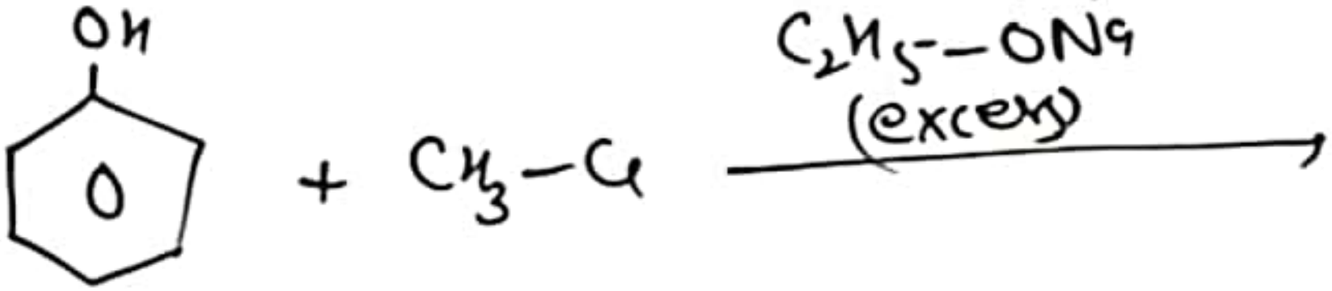
(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

eg 7



since only 1 mole EtO- was taken, hence only PhO- is the final Nu-

eg 8

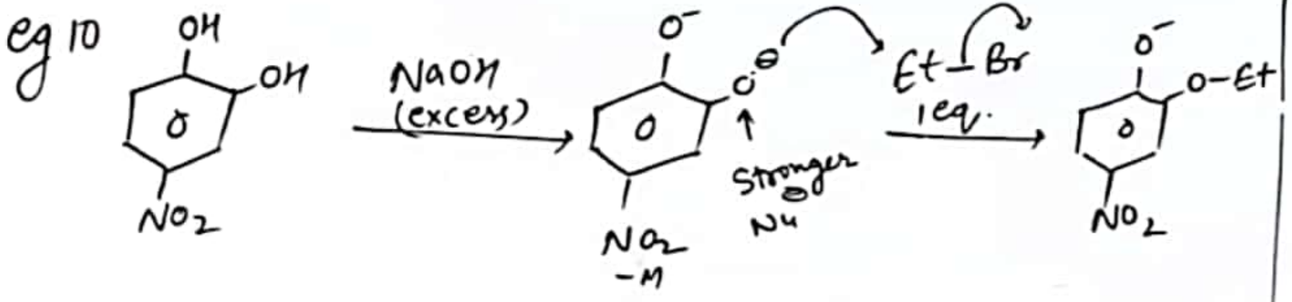


eg 9

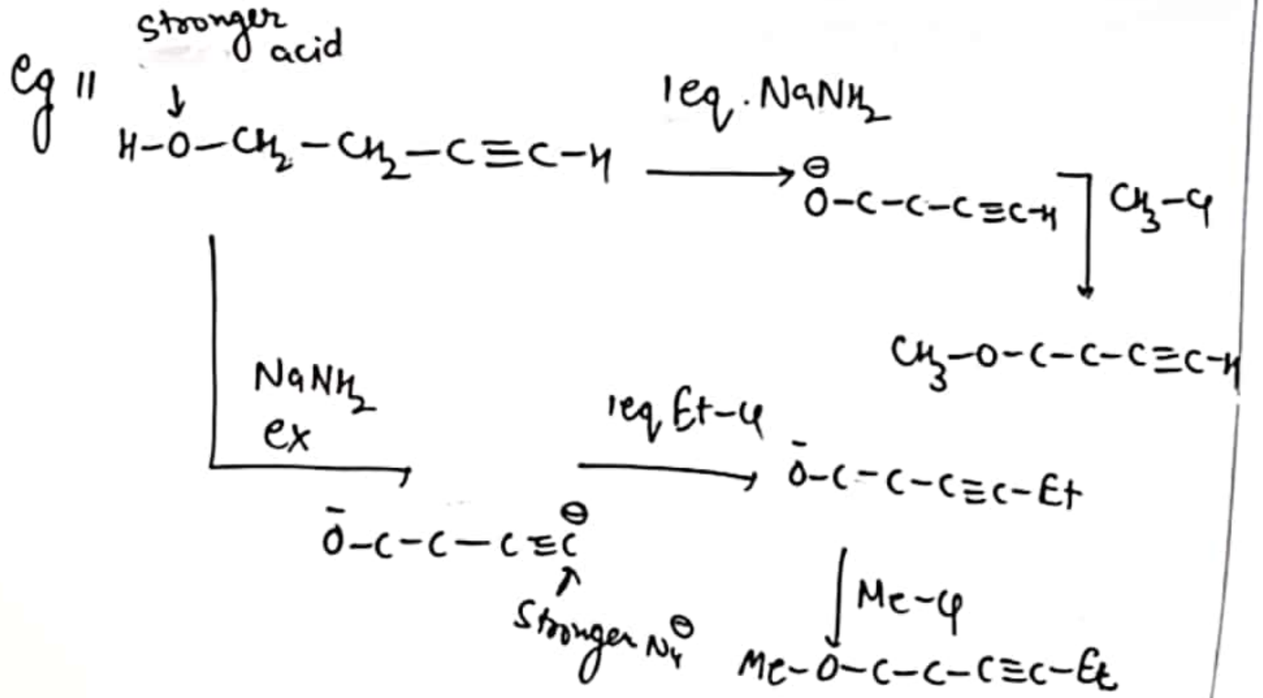
ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

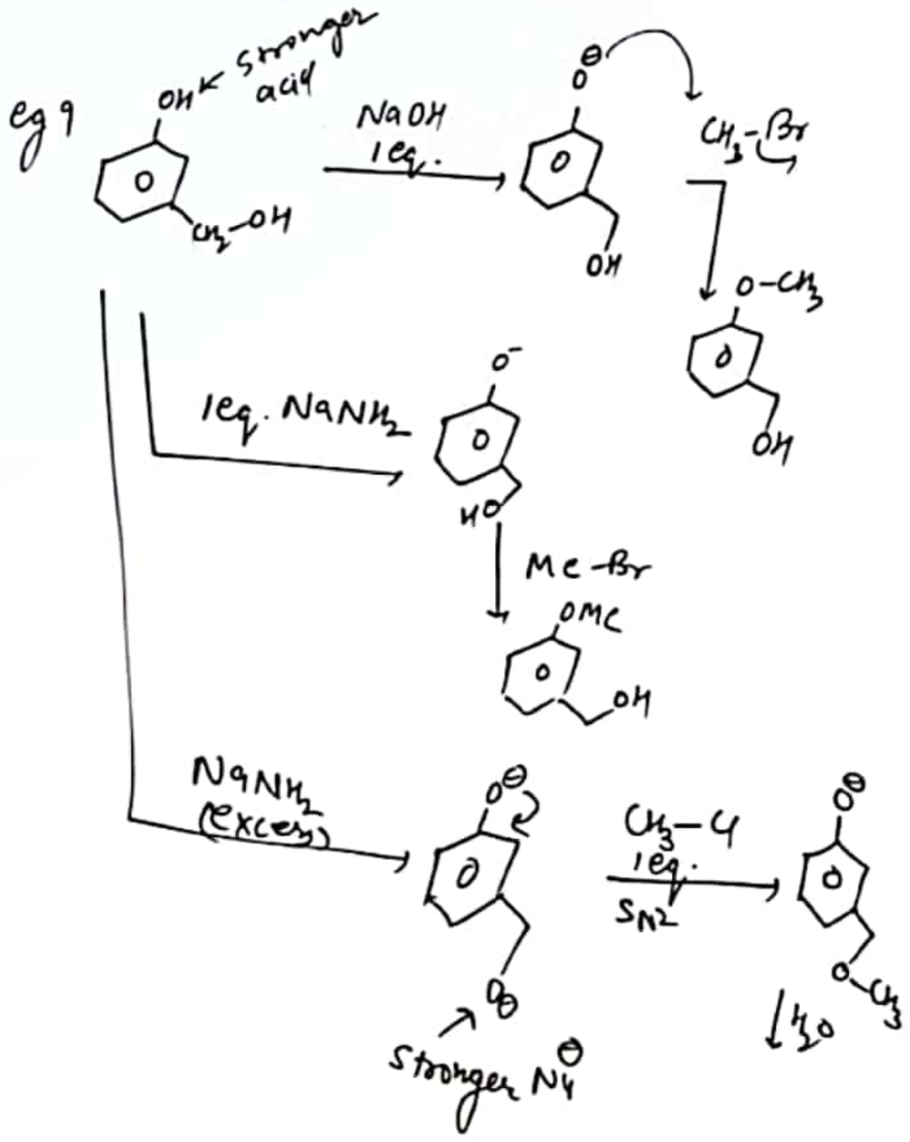
eg 10



eg 11



eg 9



ORGANIC REACTION MECHANISM - III (ORM-III)

(VII) Williumson's Ether Synthesis:- (WES) (S_N2)

Q Give most suitable reactants for formation of following ether (by EWS)

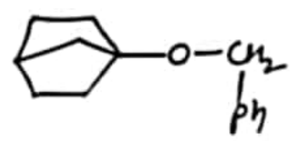
① 

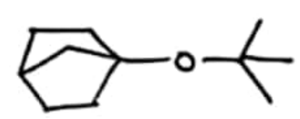
② 

③ 

④ 

⑤ 

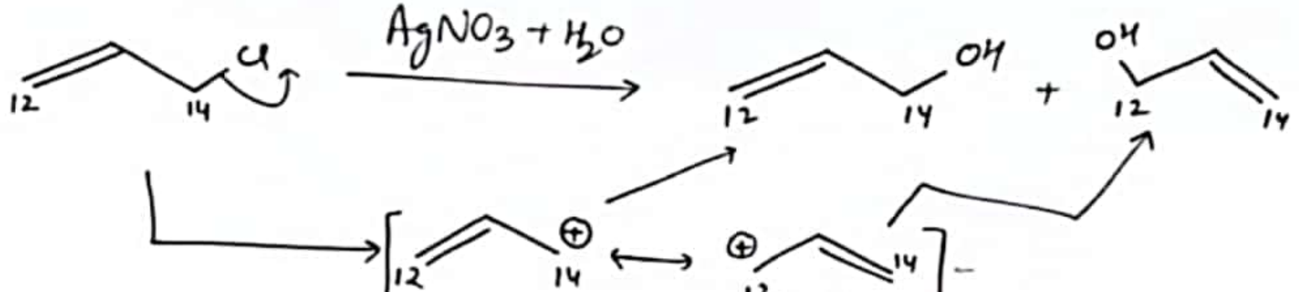
⑥ 

⑦ 

(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

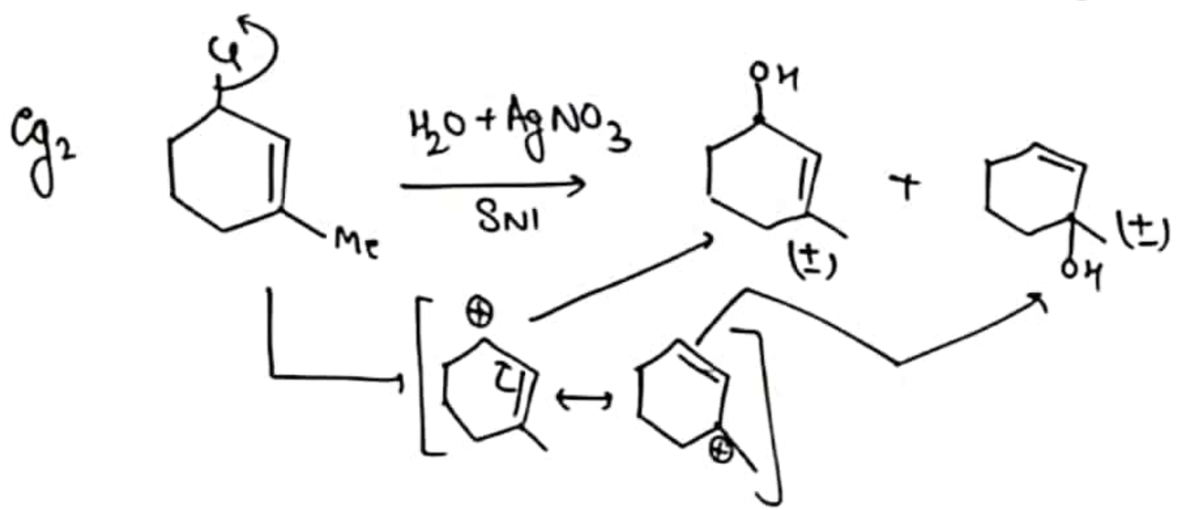
→ S_N1'

Eg 1



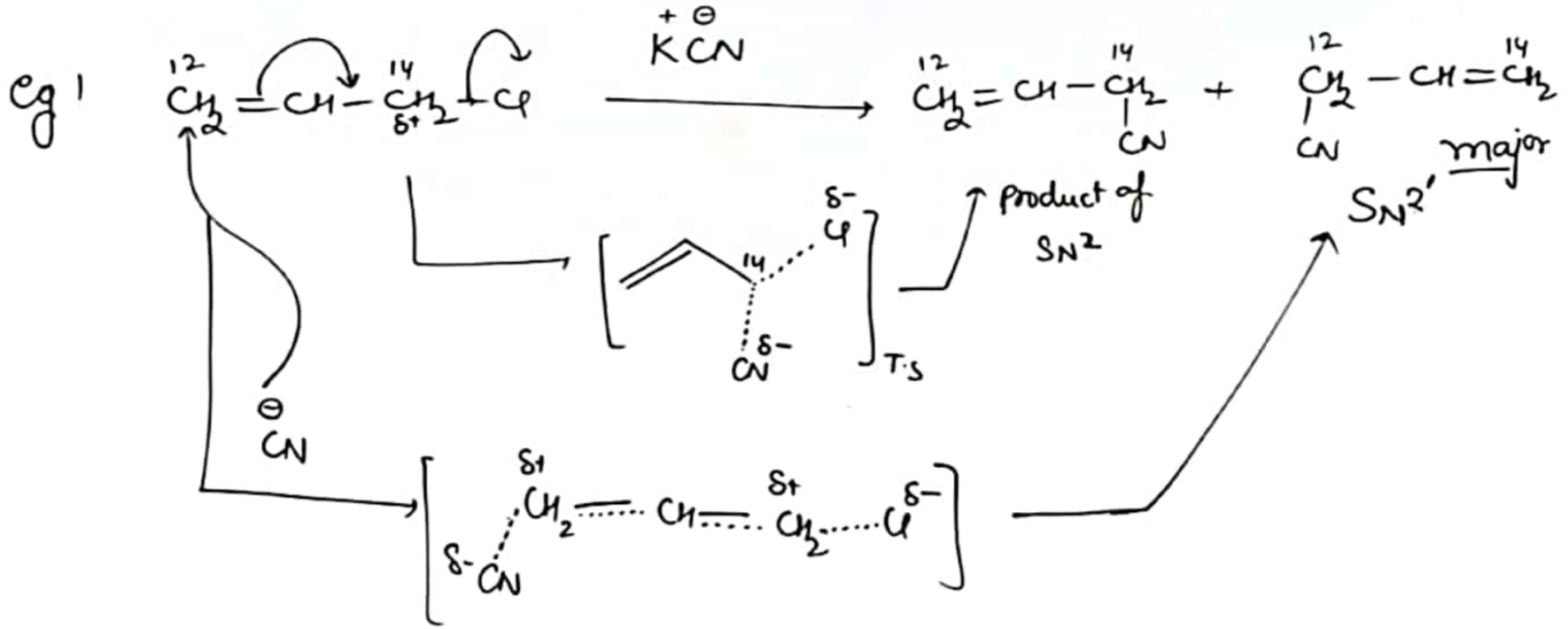
both products obtained in equal amounts

Eg 2



(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

⇒ S_N2' Reaction.

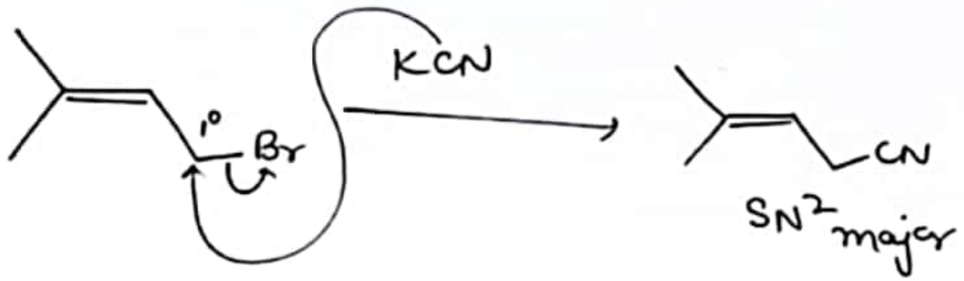


due to the comparatively less crowding on C12 than on C14

(VIII) Allylic S_N Reaction = S_N' → S_N1' / S_N2'

⇒ S_N2' Reaction.

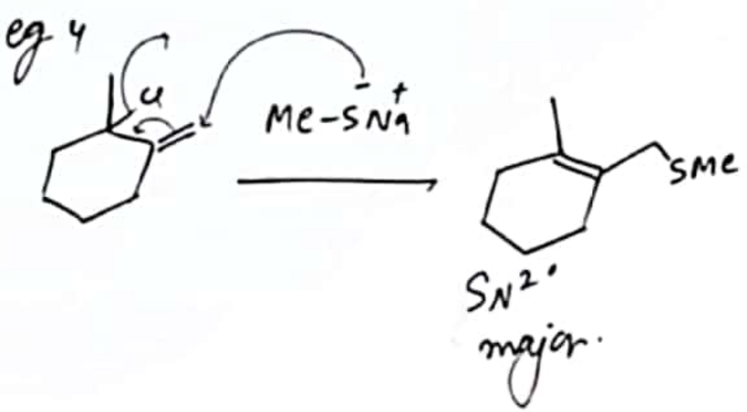
eg 2



eg 3



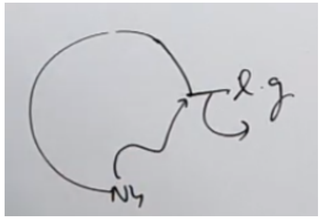
eg 4



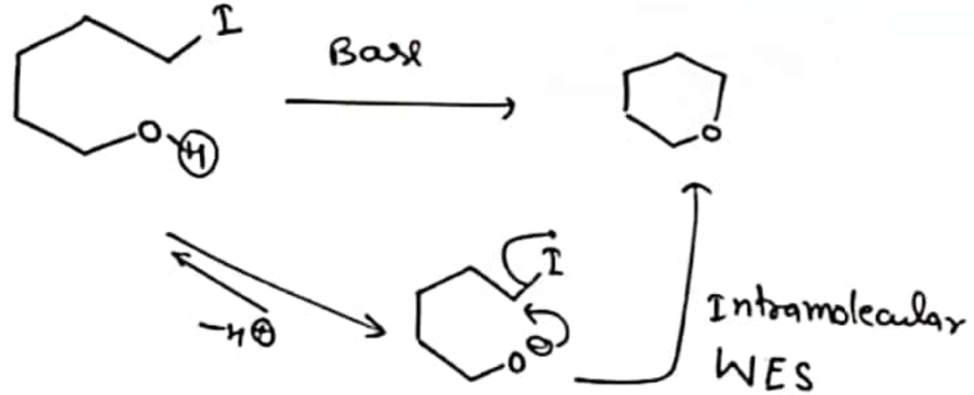
ORGANIC REACTION MECHANISM - III (ORM-III)

(IX) Intramolecular S_N2 Reaction:-

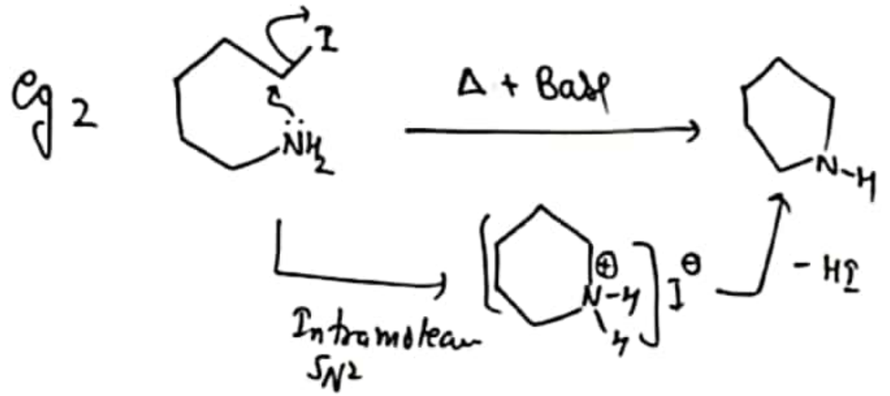
When l.g and strong nucleophile both are present in same molecule then intramolecular S_N2 reaction possible.



eg 1

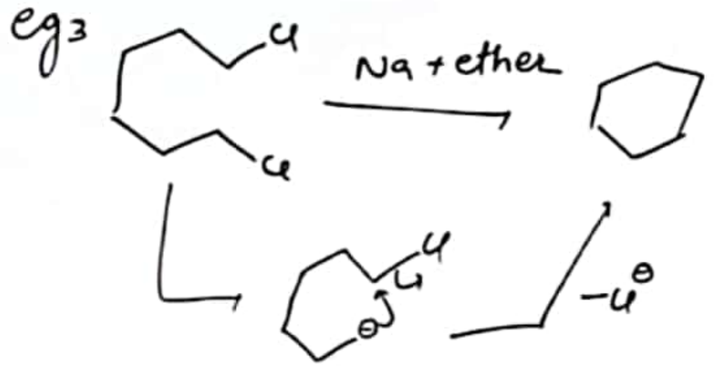


eg 2



Base given to absorb HI

eg 3

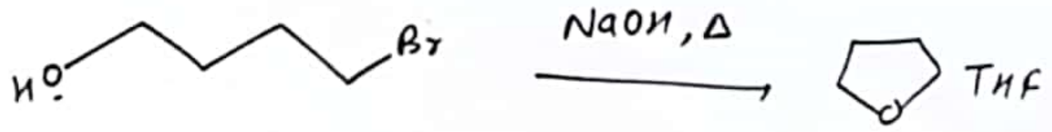


Intramolecular reactions are more favourable than intermolecular reactions

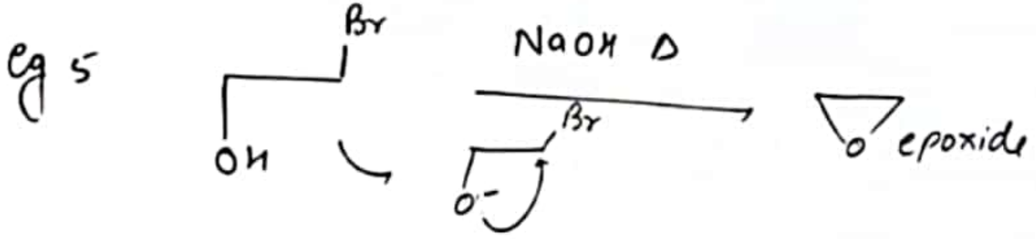
ORGANIC REACTION MECHANISM - III (ORM-III)

(IX) Intramolecular SN^2 Reaction:-

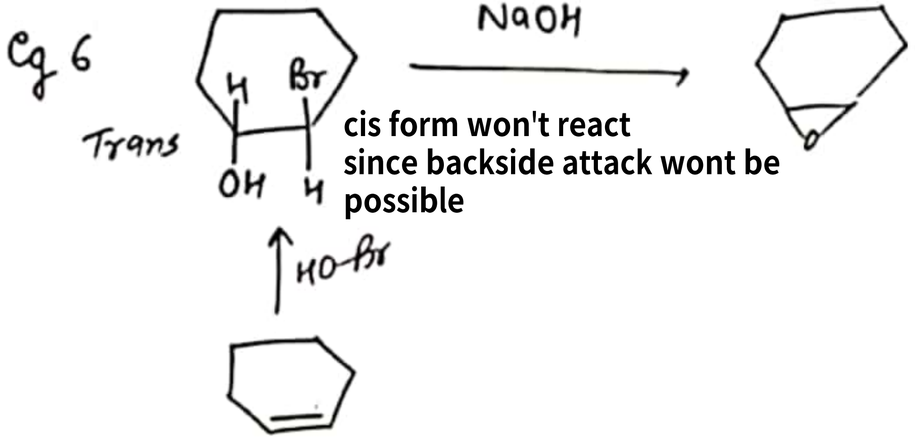
eg 4



eg 5

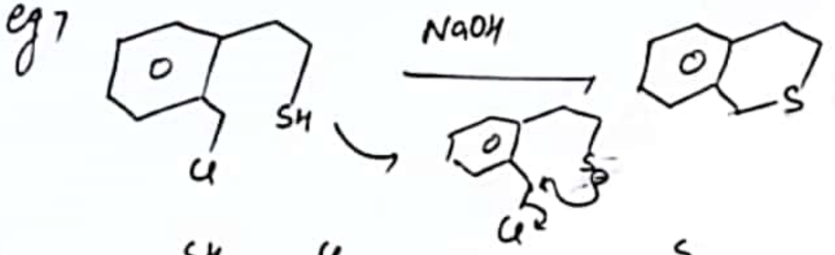


eg 6

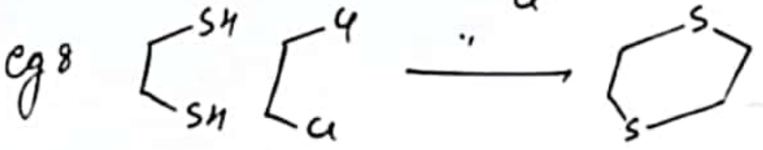


cis form won't react since backside attack wont be possible

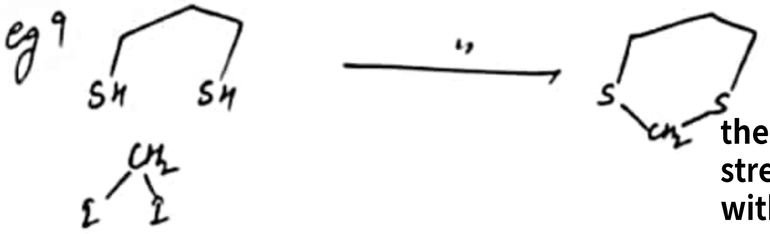
eg 7



eg 8

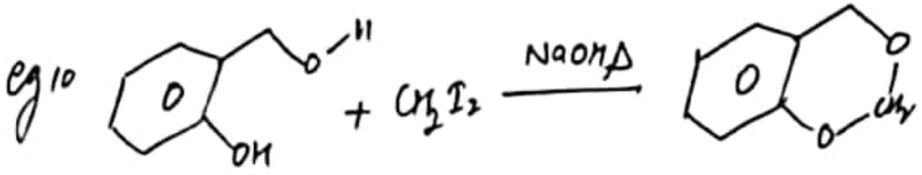


eg 9



the H also has some acidic strength (undergoes resonance with vacant d-orbitals of S)

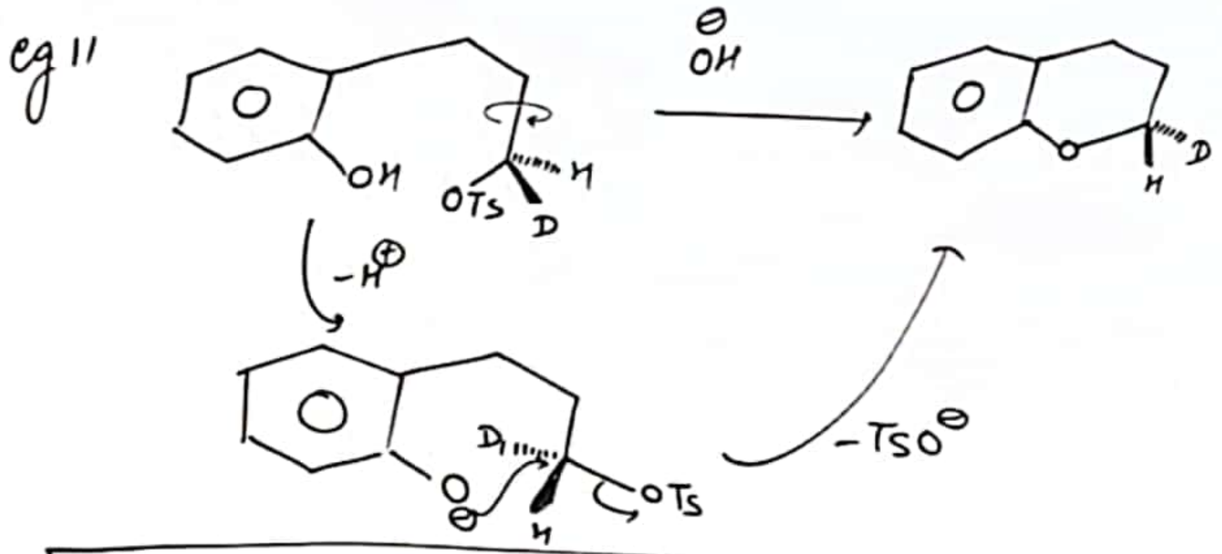
eg 10

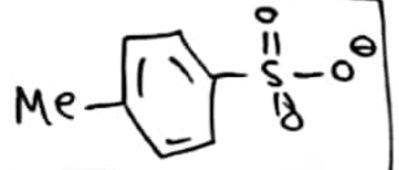


ORGANIC REACTION MECHANISM - III (ORM-III)

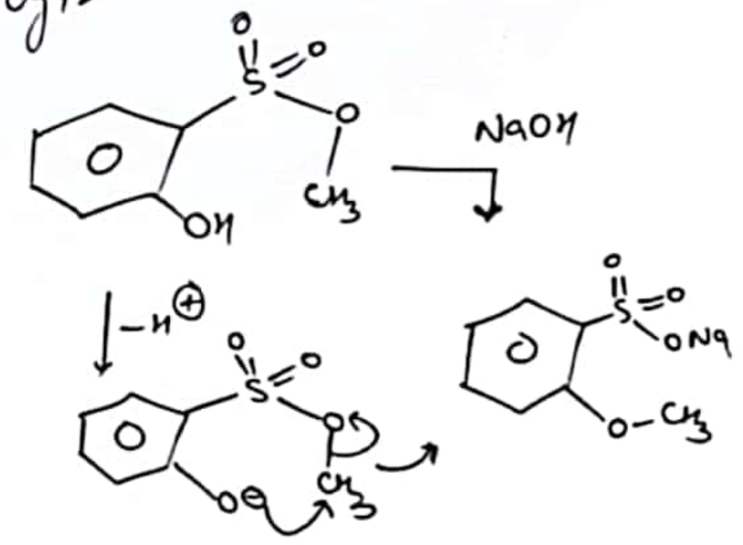
(IX) Intramolecular SN^2 Reaction:-

eg 11



$TsO^{\ominus}$ = 

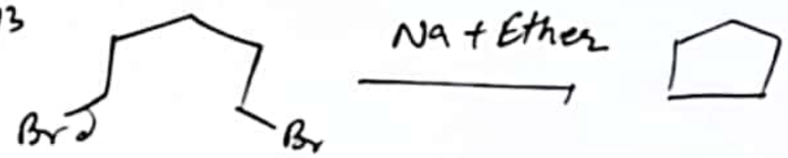
eg 12



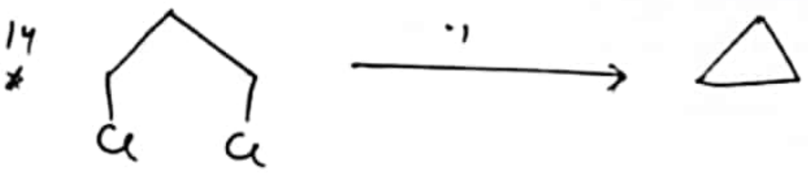
(IX) Intramolecular S_N2 Reaction:-

Examples of Intramolecular Wurtz Reaction

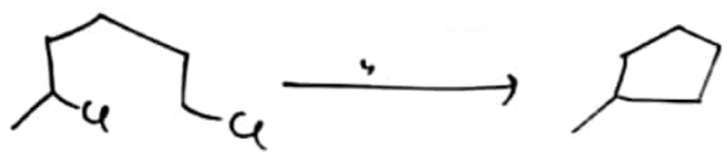
eg 13



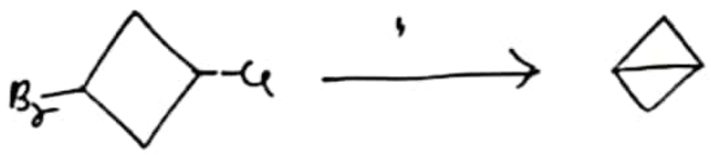
eg 14 *



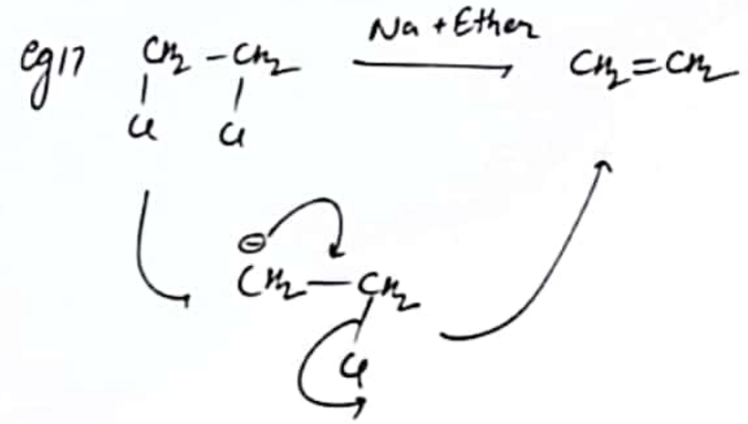
eg 15



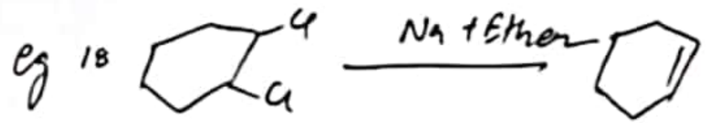
eg 16



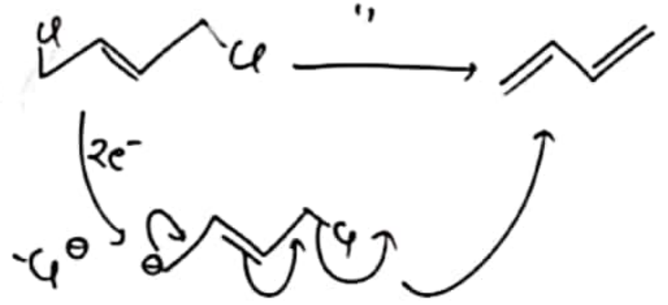
eg 17



eg 18



eg 19



ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

- [A] → Reaction with HX
- [B] → " " PX_3 / PX_5
- [C] → " " $SOCl_2 / SOBr_2$

[X.A] Reaction of alcohols with HX :-

$$R\text{-}OH \xrightarrow{\text{Conc. HX}} R\text{-}X + H_2O$$

alkyl halide

lg (good)

$$R\text{-}OH \xrightarrow{\oplus\ \ominus} \text{(X)}$$

$\oplus$ Na $X^{\ominus}$

$$R\text{-}X + OH^{\ominus}$$

Poor l.g

⇒ Reactivity of HX ⇒ HI > HBr > HCl > HF

HF → No R_x^n

⇒ Lucas Reagent = anhydrous $ZnCl_2$ + conc. HCl

⇒ 3° R-OH / 2° R-OH / B-crowded alcohol / Benzylic / Allylic

↓ Conc. HX

S_N1

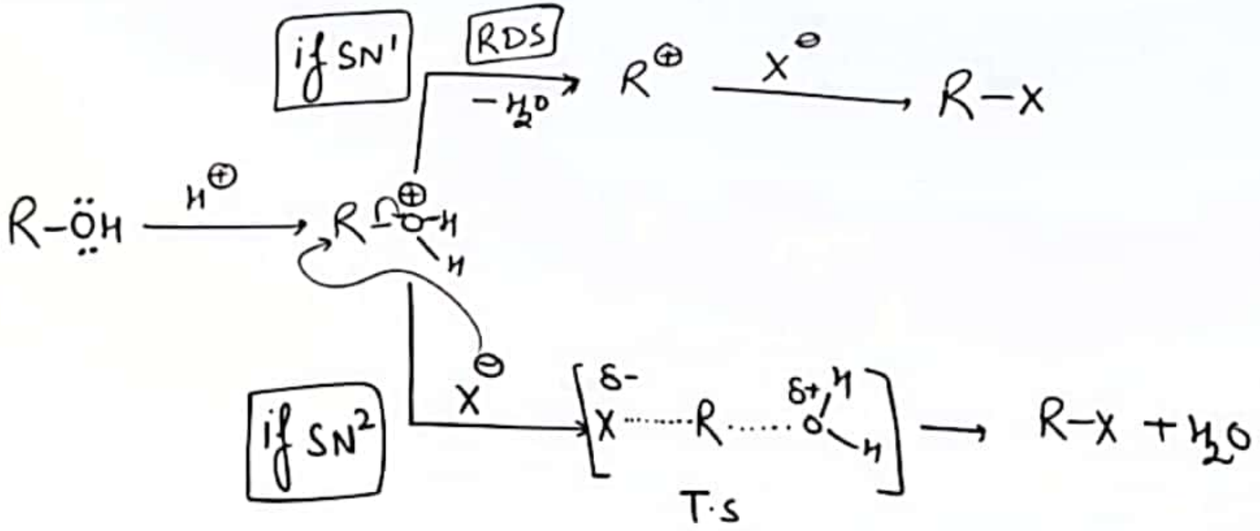
⇒ 1° R-OH $\xrightarrow{\text{Conc. HX}}$ S_N2

the rex. of primary alcohols and some secondary alcohols with HCl is v slow

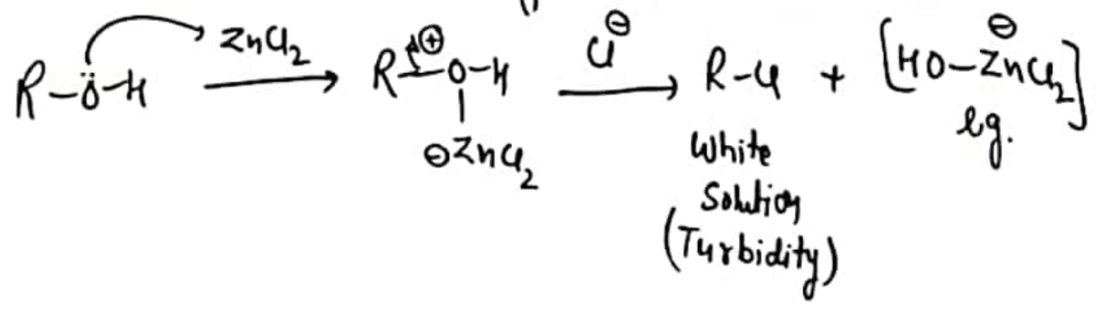
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

Mechanism



Mechanism with Lucas Reagent :-



⇒ Reactivity of HX ⇒ HI > HBr > HCl > HF

(No R_x^n)

⇒ Lucas Reagent = anhydrous $ZnCl_2$ + conc. HCl

⇒ 3° R-OH / 2° R-OH / B-Crowded alcohol / Benzylic / Allylic

↓ conc. HX

S_N1

⇒ 1° R-OH $\xrightarrow{\text{Conc. HX}}$ S_N2

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ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

⇒ Lucas Test

3° R-OH / Benzylic / Allylic alcohol $\xrightarrow{ZnCl_2 + HCl}$ R-Cl

instant White turbidity

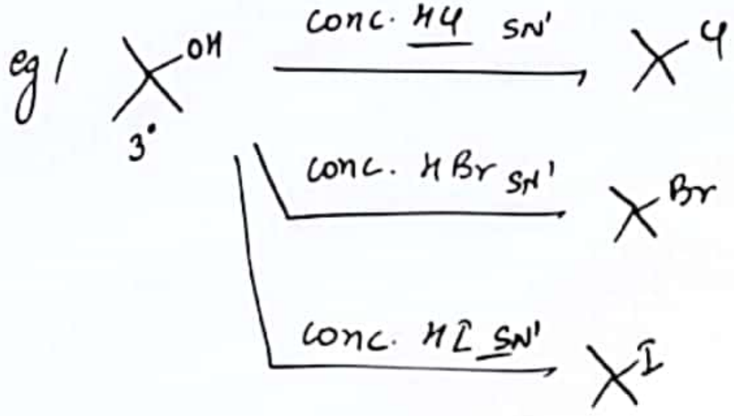
2° alcohol $\xrightarrow{(ZnCl_2 + HCl)}$ R-Cl

white turbidity after 5 to 10 min.

1° Alcohol $\xrightarrow{HCl + ZnCl}$ R-Cl

White turbidity (after 30 min / on heat)

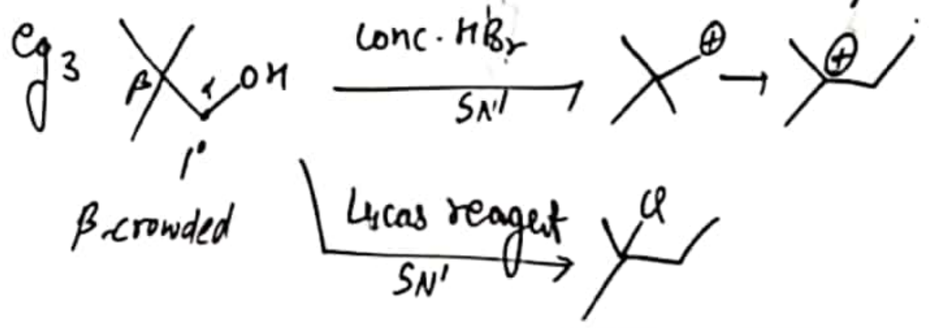
eg 1



eg 2



eg 3

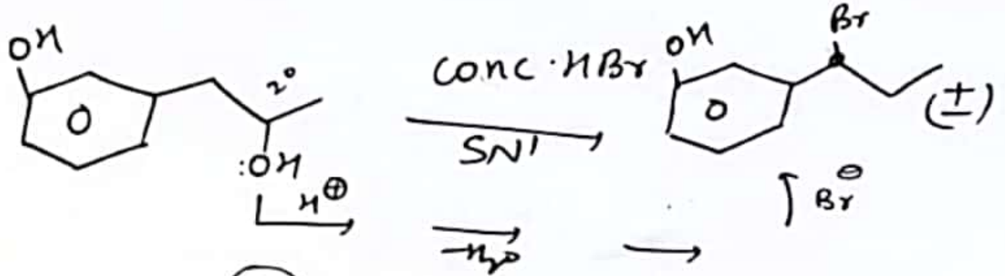


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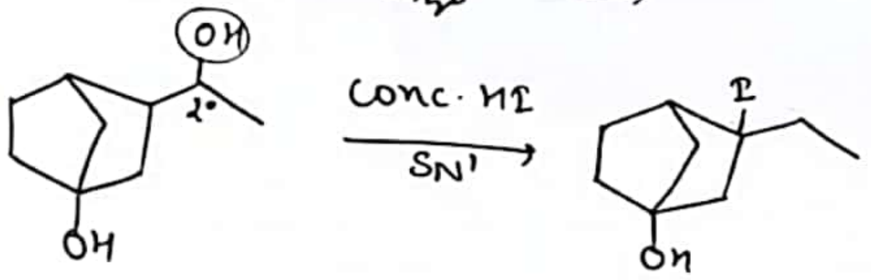
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

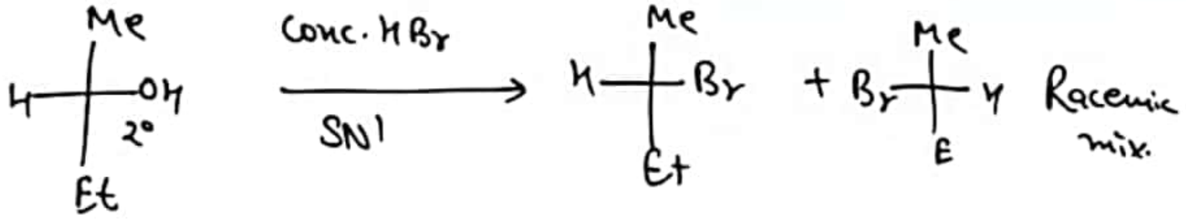
Eg 4



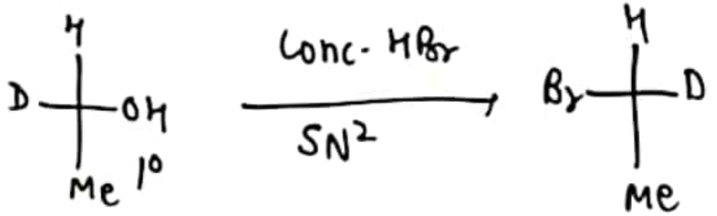
Eg 5



Eg 6



Eg 7



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(X) S_N Reactions of Alcohols :-

[X.B] S_N Reactions of alcohols with PX_3, PX_5

$$\left[\begin{array}{l} PX_3 = PCl_3,\ PBr_3,\ PI_3 \\ PX_5 = PCl_5,\ PBr_5 \\ P + I_2 \longrightarrow PI_3 \end{array}\right]$$

PI5 nahi bnta due to the very large size of I

$$3R\text{-}OH + PX_3 \xrightarrow{\text{mainly } SN^2} 3R\text{-}X + \underset{\text{Phousphorous acid (l.g)}}{H_3PO_3}$$

$$R\text{-}OH + PX_5 \xrightarrow{\text{" " } SN^2} R\text{-}X + HX + \underset{\text{l.g.}}{POX_3}$$

$1^\circ R\text{-}OH,\ 2^\circ R\text{-}OH \xrightarrow{PX_3/PX_5} SN^2$

$3^\circ R\text{-}OH \xrightarrow{\text{" "}} SN^1$

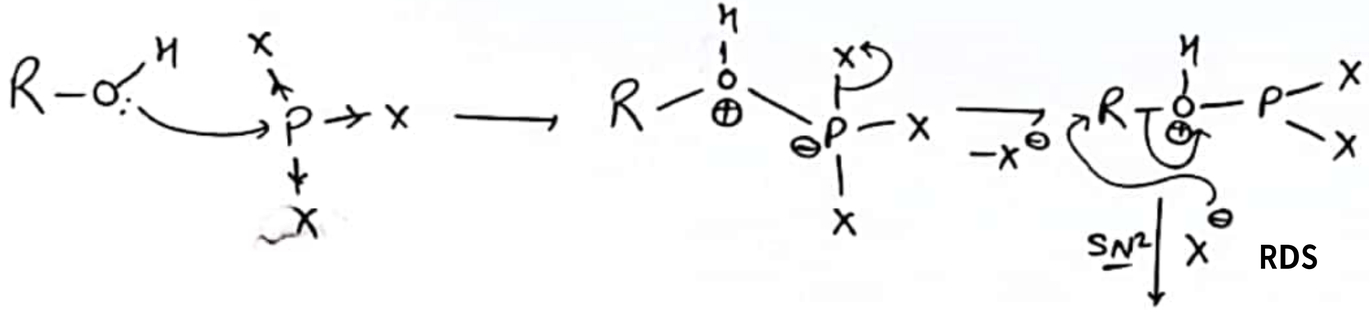
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ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

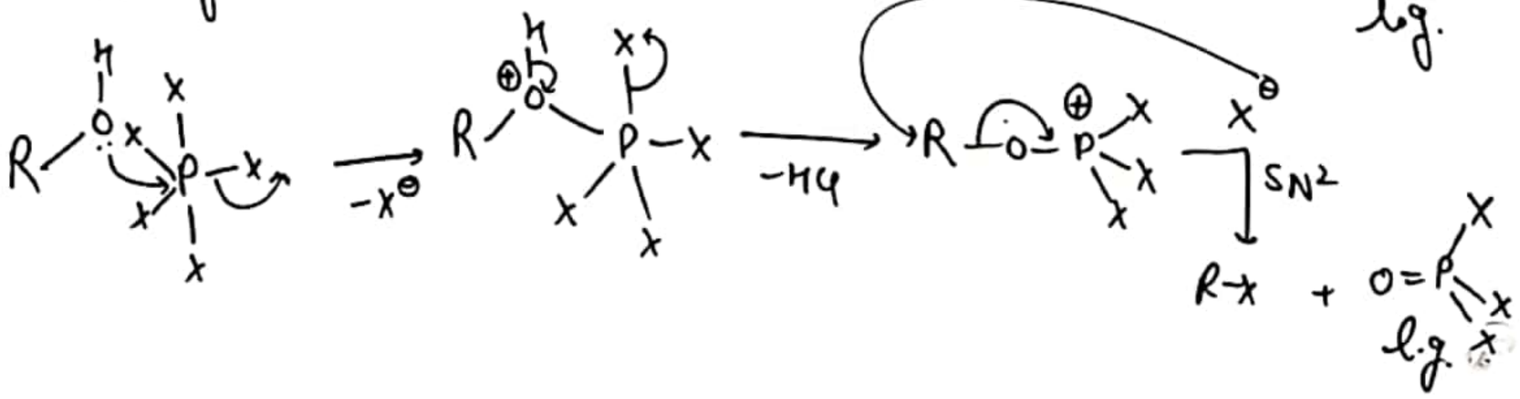
[X.B] S_N Reactions of alcohols with PX_3 , PX_5-

mechanism with PX_3



now, this will go and attack two more molecules of alcohol and finally form H3PO3 at the last step

mech. for PX_5



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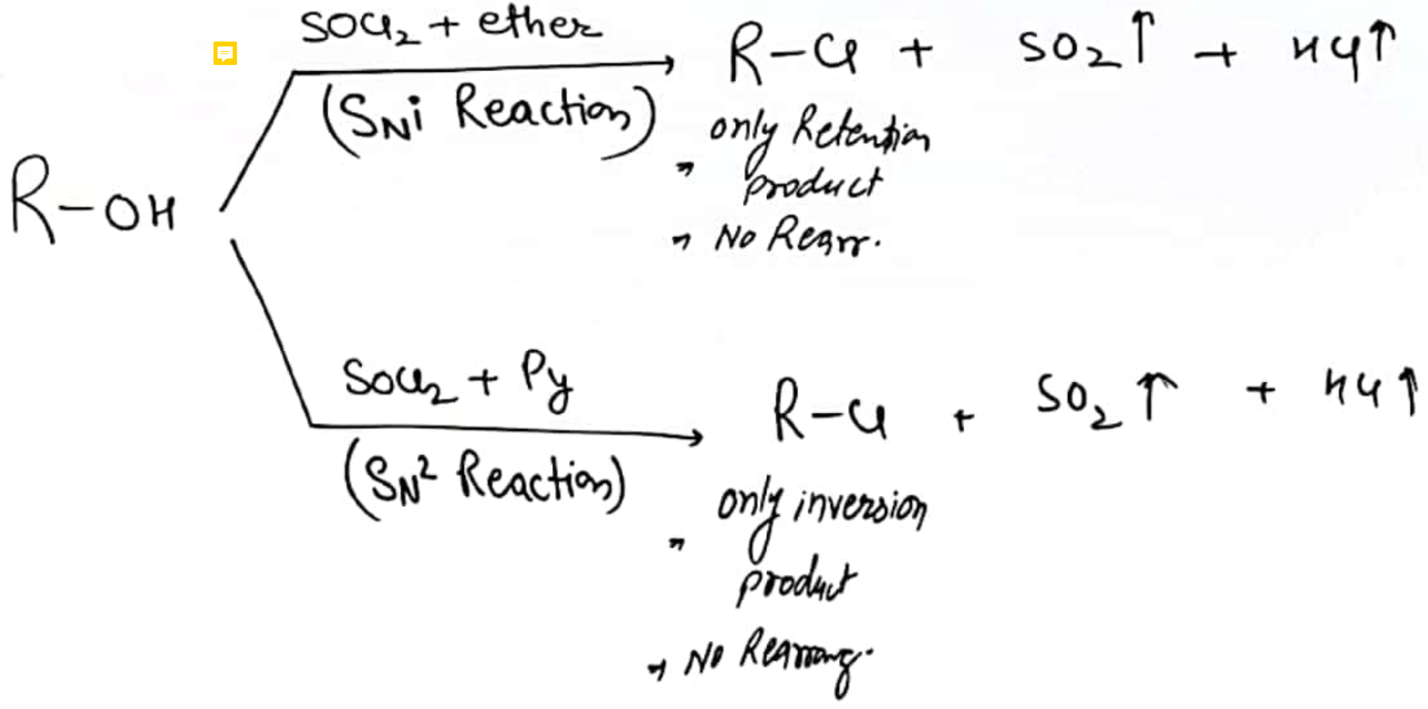
ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

$SOCl_2$ ↑ Thionyl chloride

⇒ Darzen's process. = industrial method for formation of alkyl chloride due to formation of gaseous product

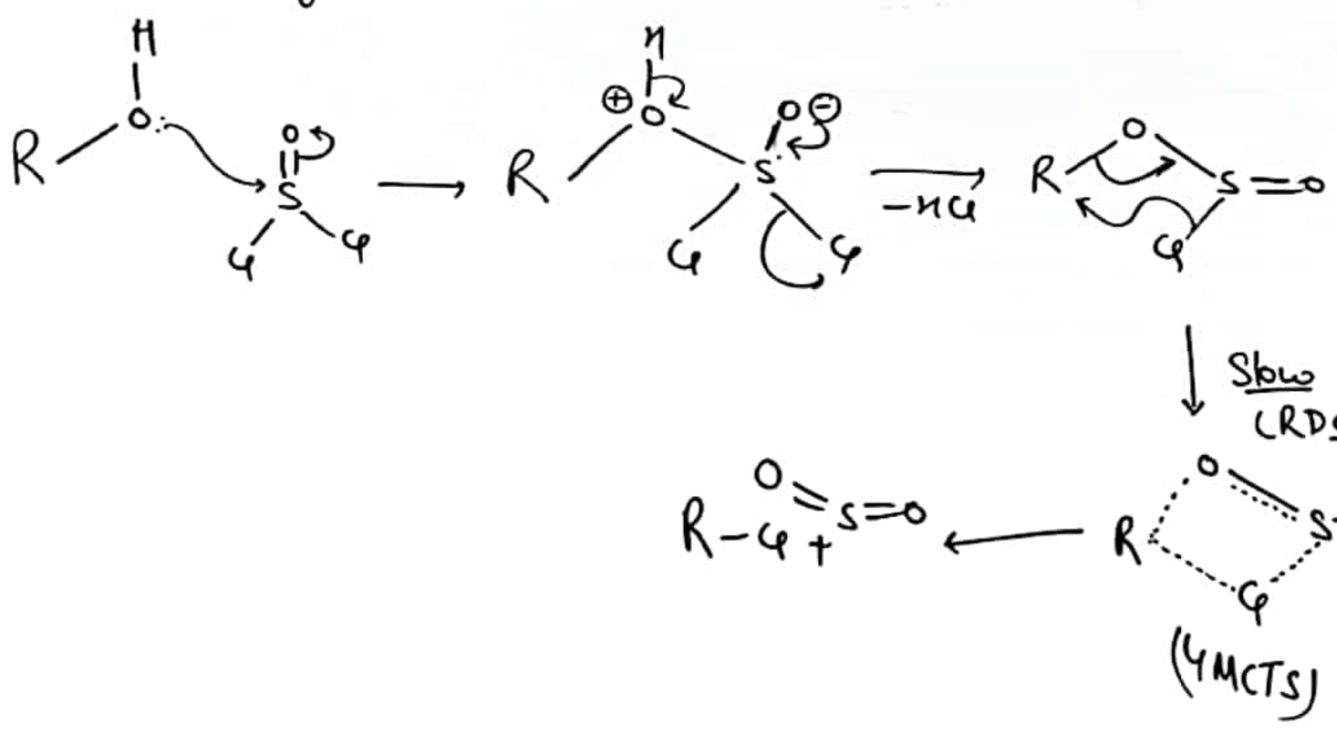


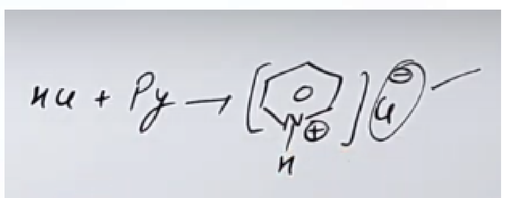
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(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

mechanism of $S_N i$ (i = internal)





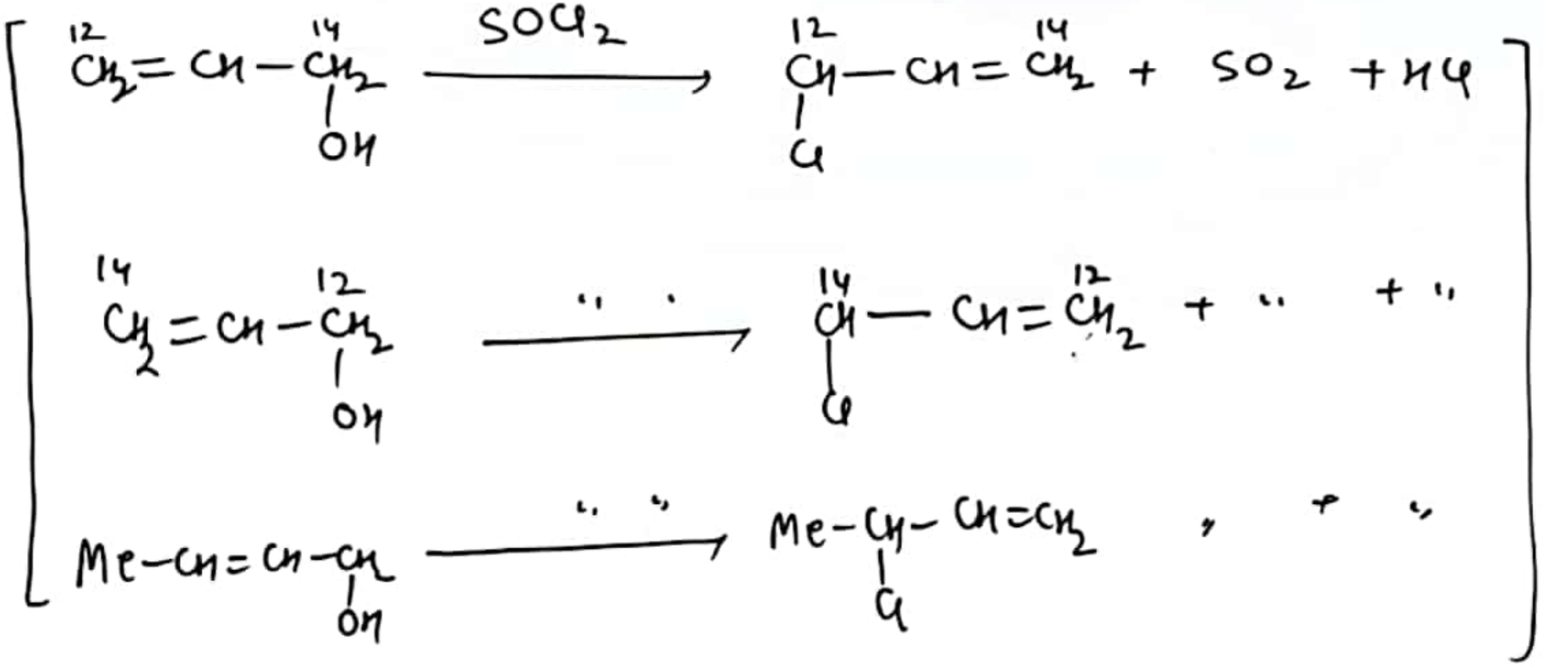
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ORGANIC REACTION MECHANISM - III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

⇒ Reaction of $SOCl_2$ on allylic alcohols.



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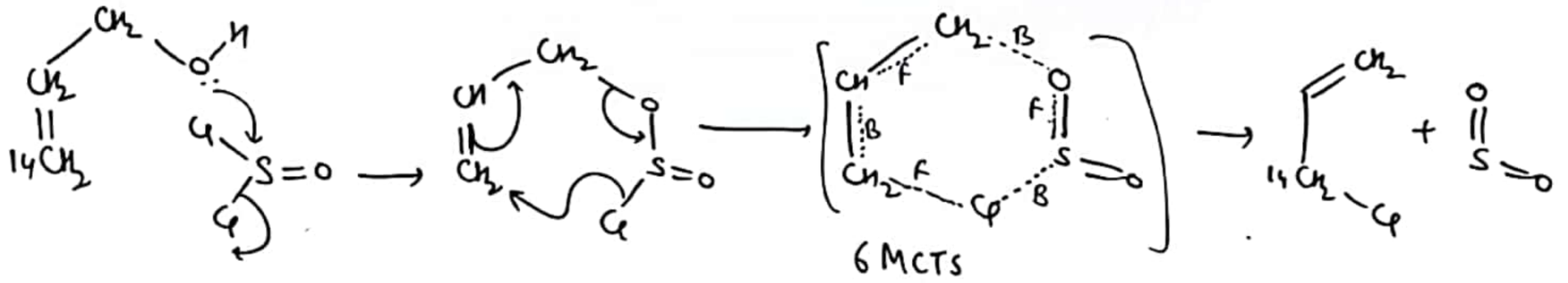
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

(X.c) Reaction with $SOCl_2$ / $SOBr_2$

⇒ Reaction of $SOCl_2$ on allylic alcohols.

mechanism



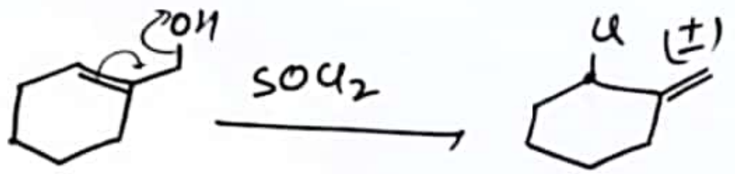
6 MCTS is more stable than 4 MCTS

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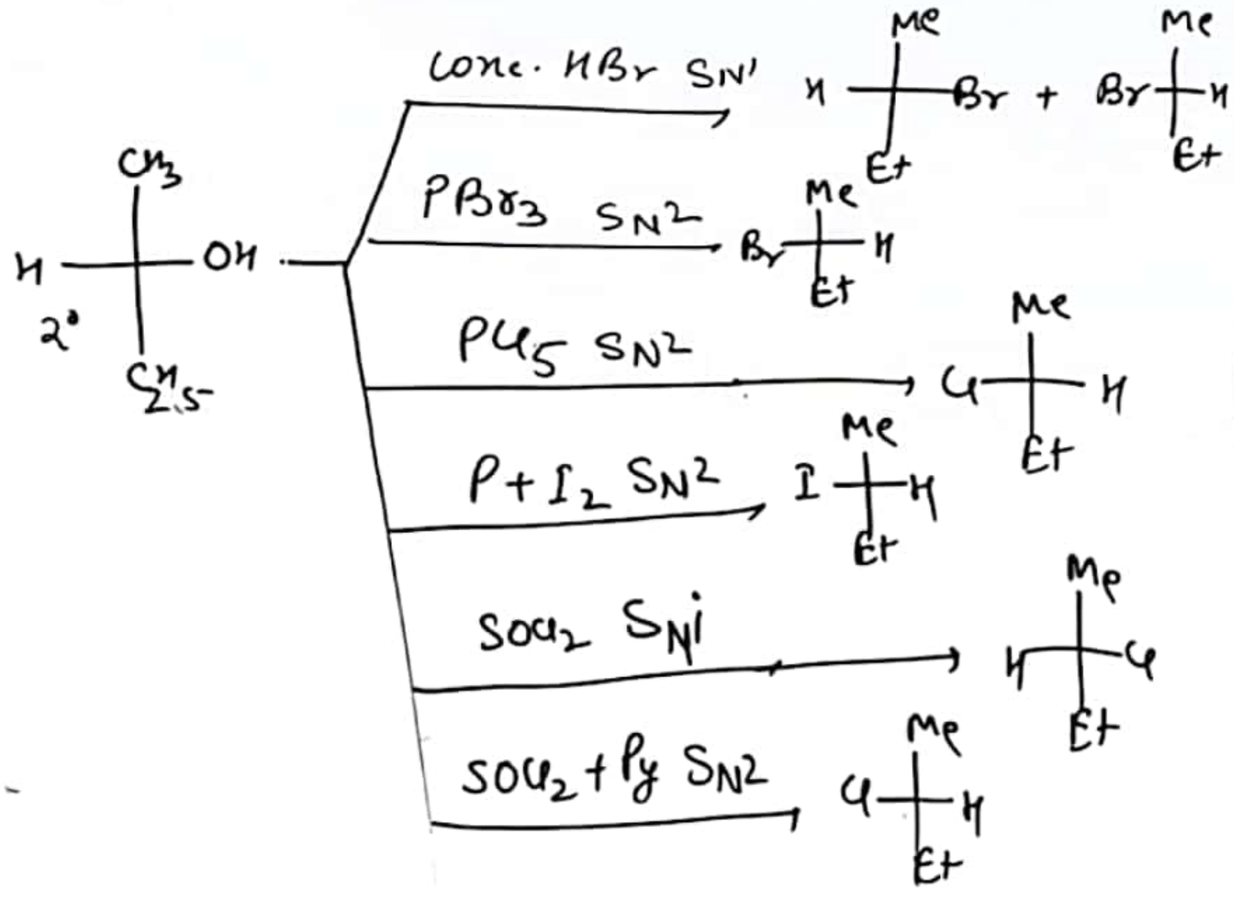
ORGANIC REACTION MECHANISM -III

X) S_N Reactions of Alcohols :-

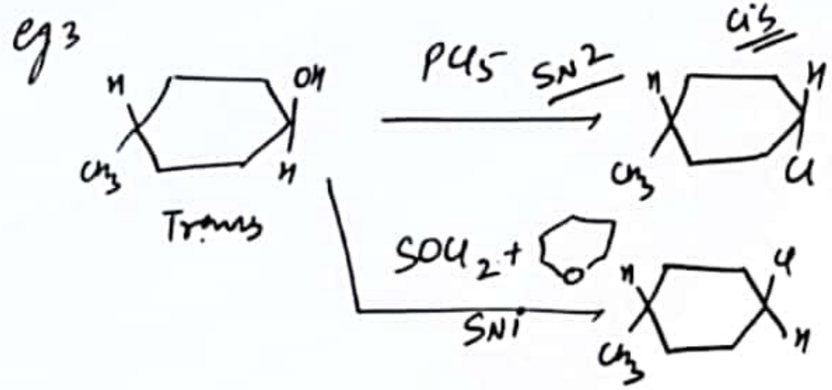
eg 1



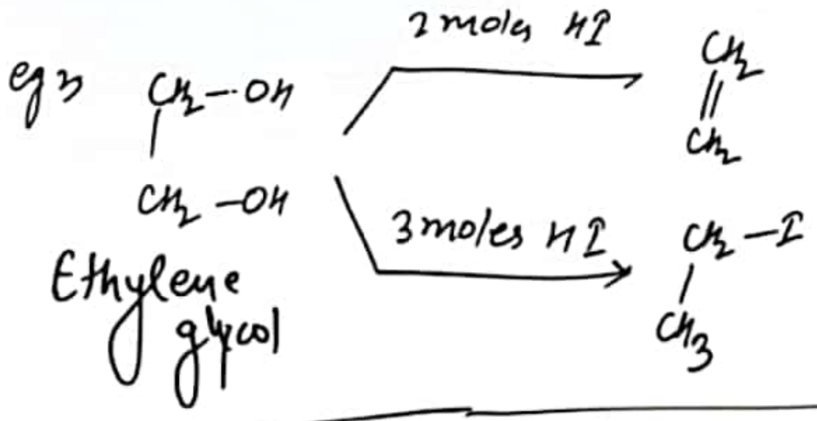
eg 2



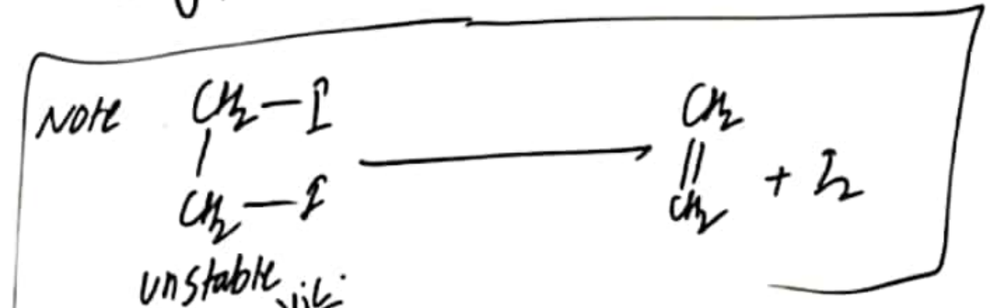
eg 3



eg 3



Note

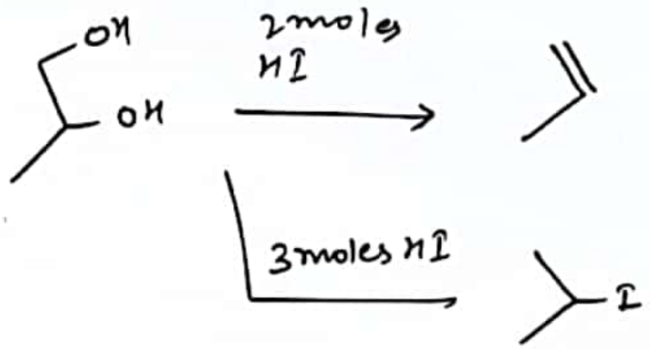


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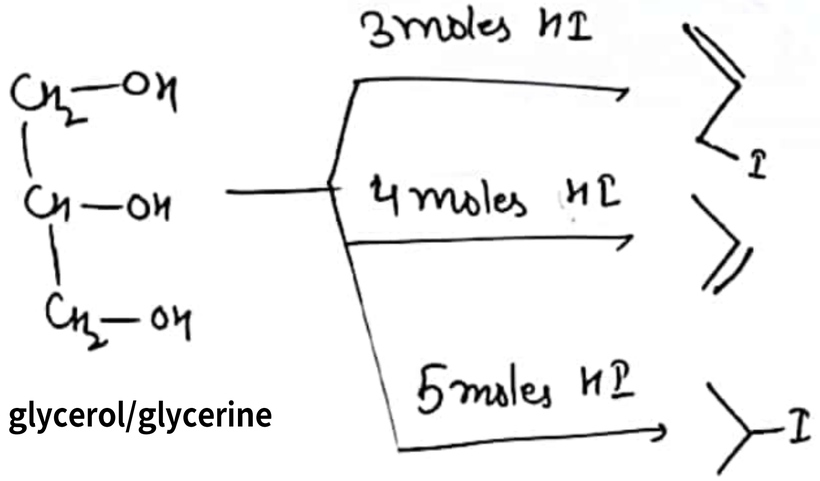
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

eg 5

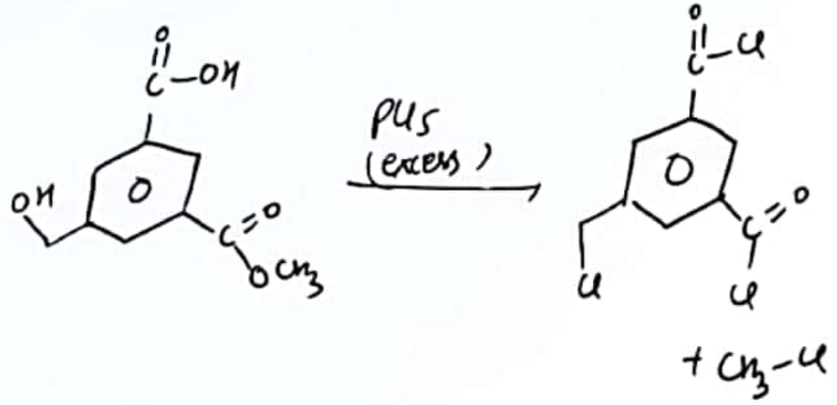


eg 6

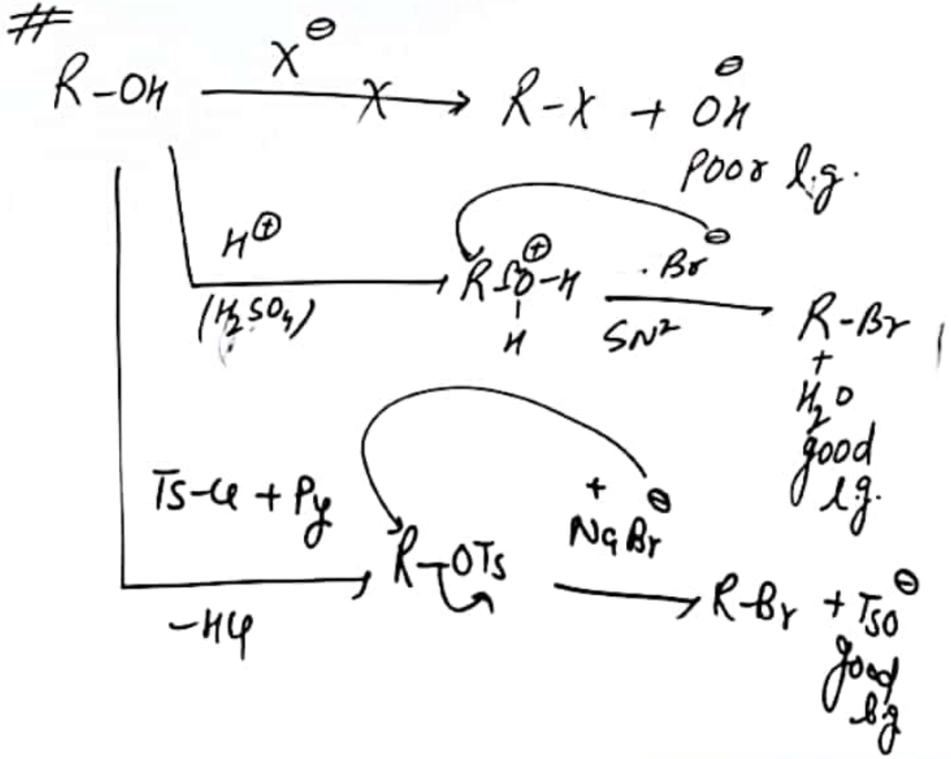


glycerol/glycerine

eg 1



#

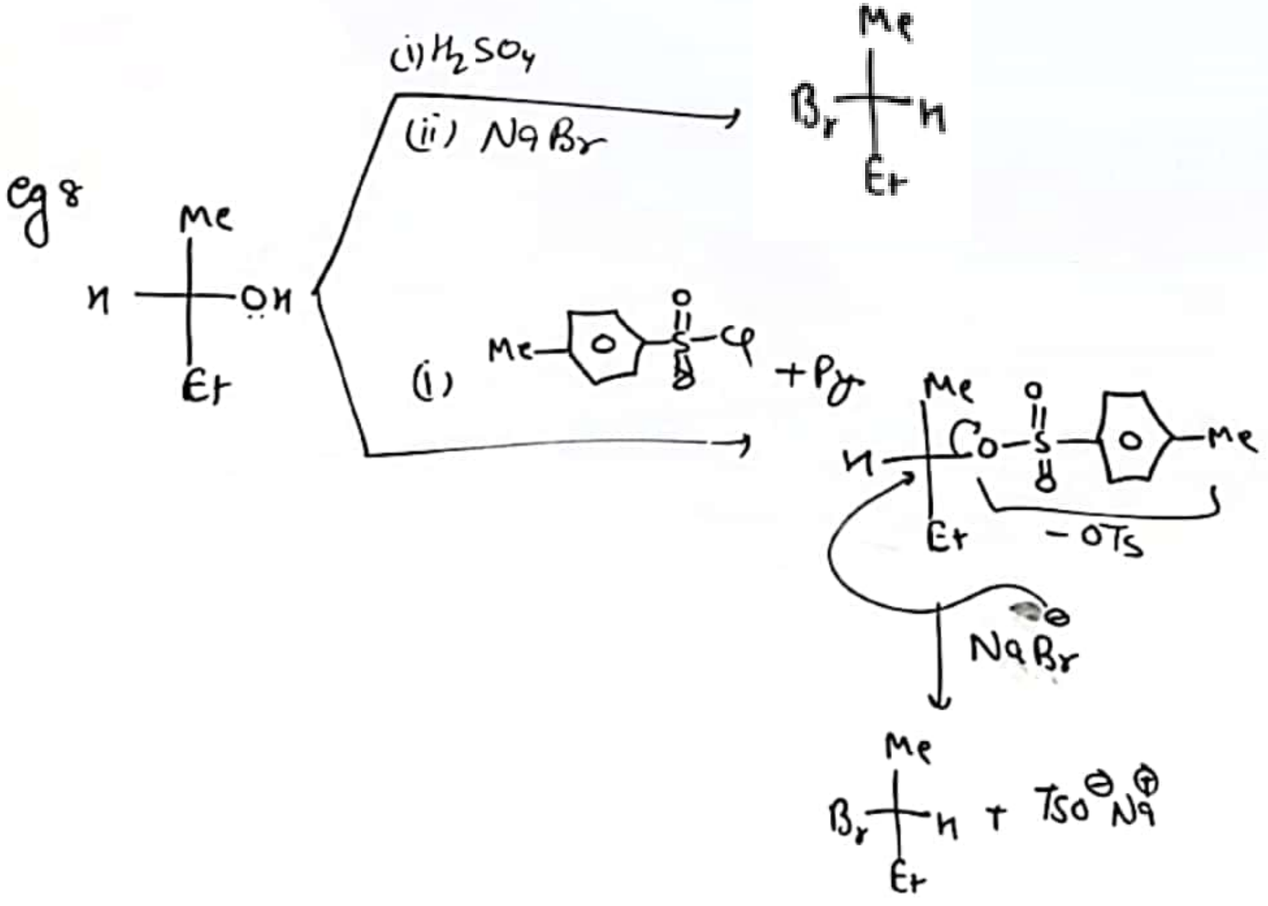


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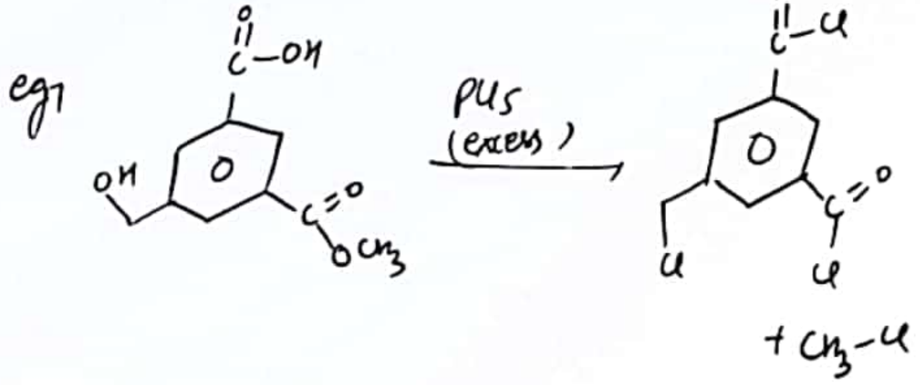
ORGANIC REACTION MECHANISM -III

(X) S_N Reactions of Alcohols :-

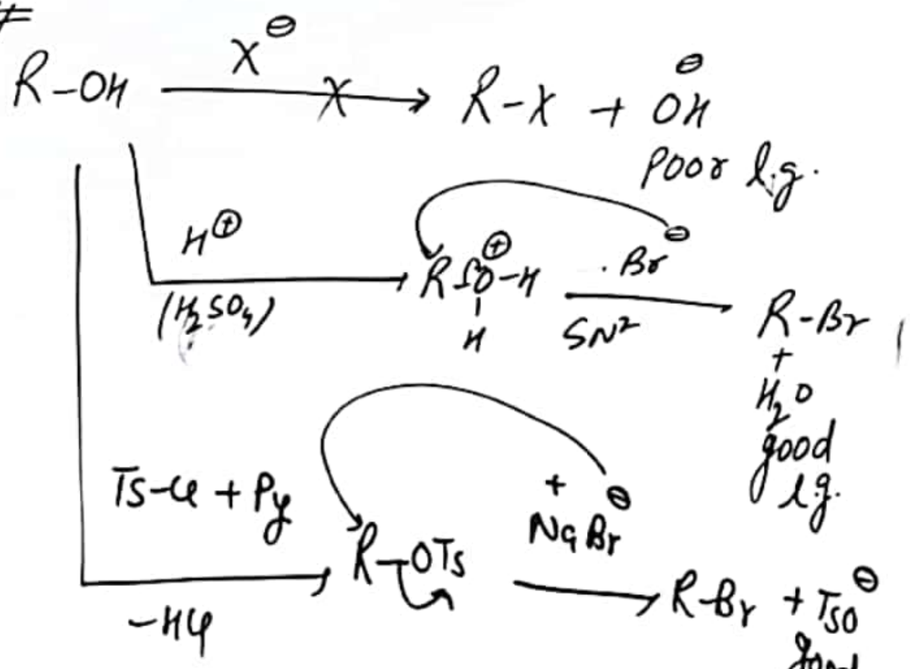
eg 8



eg 7



#



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ORGANIC REACTION MECHANISM - III

(XI) Nucleophilic Sub. Rxn of Ethers:-

- [A] → Reaction with HX
- (B) → Hydrolysis of ethers

[XI.A.] Reaction of ethers with HX

$$R-\ddot{O}-R \xrightarrow{\text{Conc. HX (1 mol)}} \underset{\text{l.g}}{R-OH} + R-X$$

$$\xrightarrow{\text{Conc. HX (excess)}} 2\,R-X + H_2O$$

$$\xrightarrow{NaX} \text{(X)} \quad R-X + \underset{\text{poor l.g.}}{R-O^{\ominus}}$$

⇒ Reactivity of HX

$$HI > HBr > HCl > \underset{\text{No Reaction}}{HF}$$

⇒ When any alkyl group of ether is 3° alkyl or benzylic (or resonance stabilised) then S_N1 mech. take place. otherwise S_N2 take place.

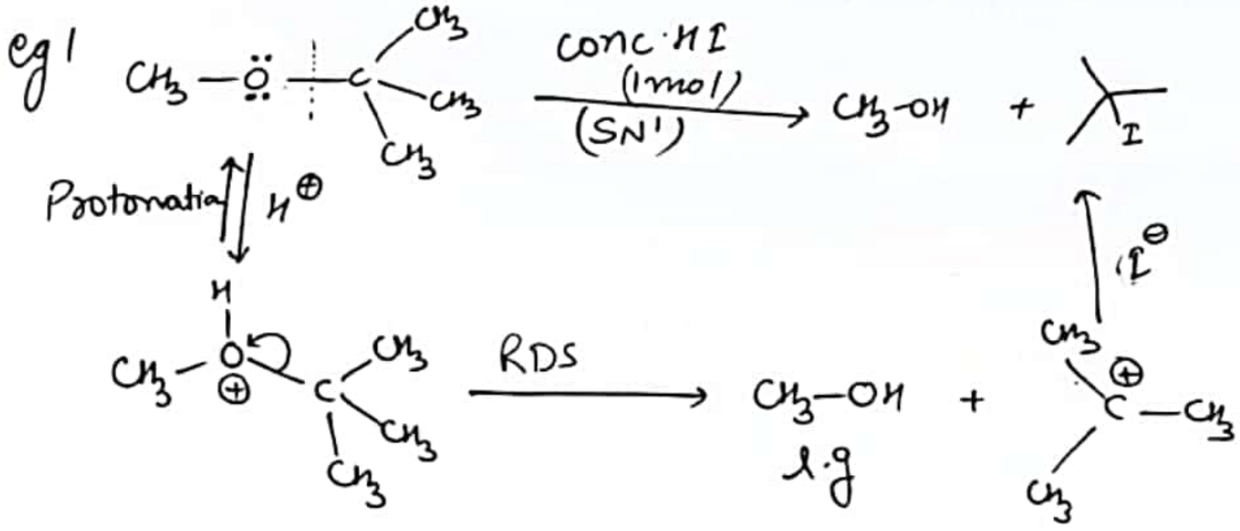
⇒ anhydrous HI give S_N2 reaction with ethers. since there is no water for solvation of ions

ORGANIC REACTION MECHANISM – III

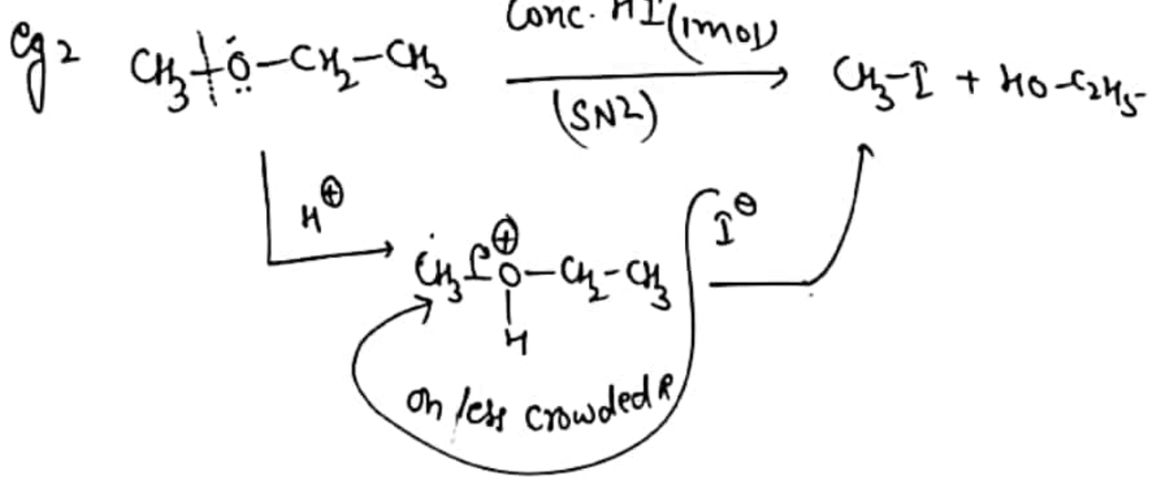
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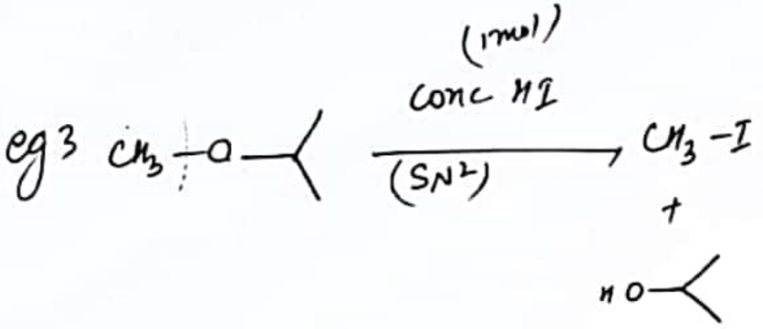
eg 1



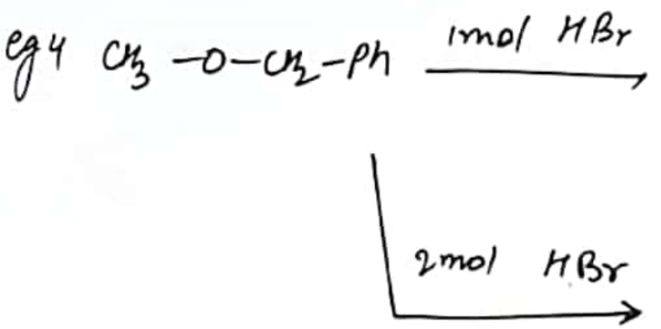
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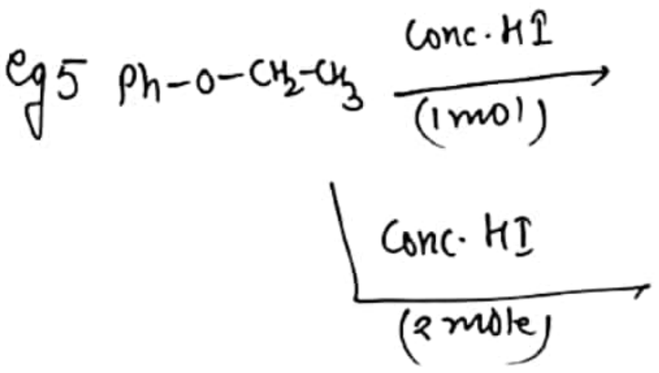
eg 3



eg 4



eg 5

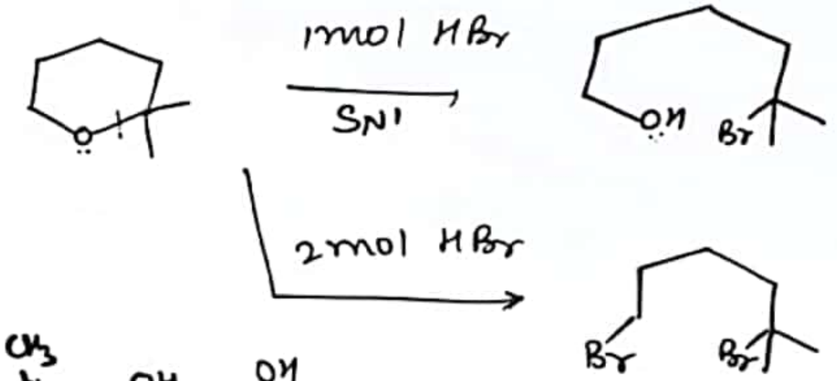


ORGANIC REACTION MECHANISM – III

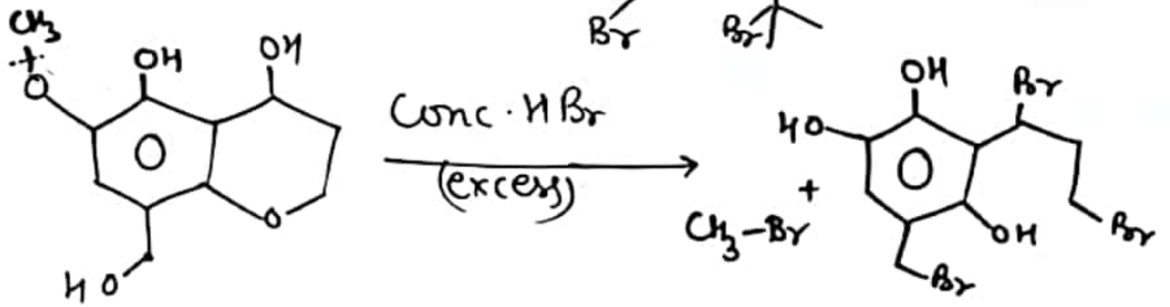
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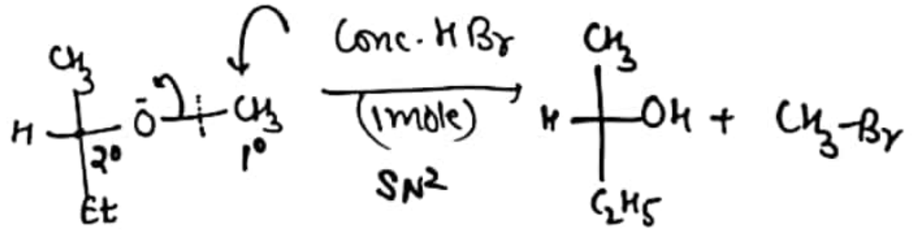
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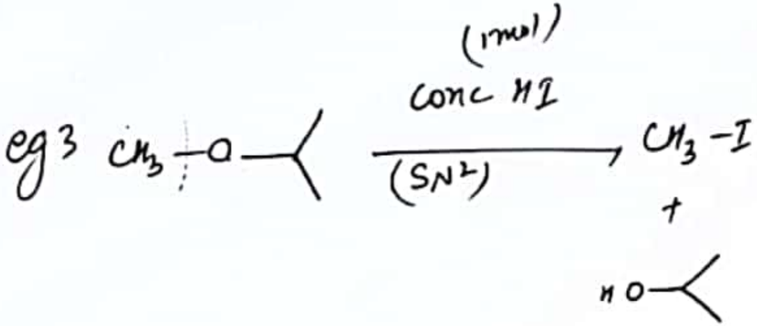
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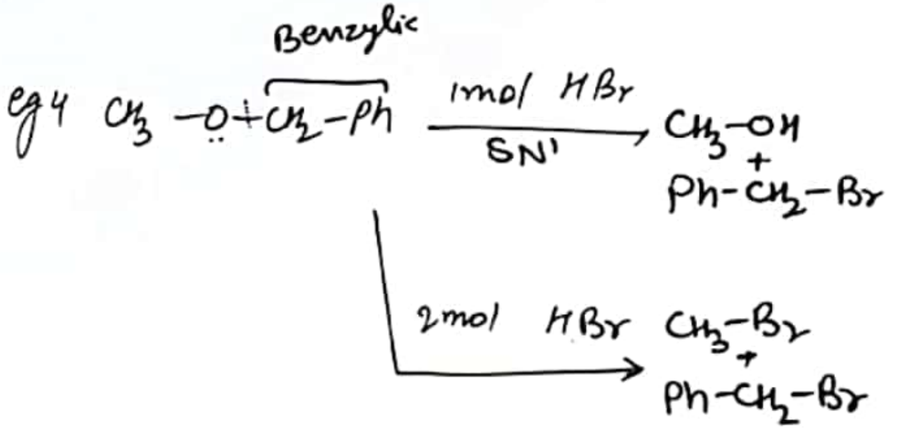
eg 8



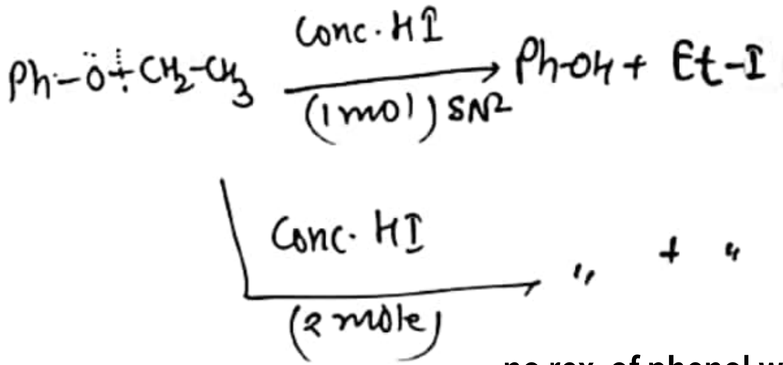
eg 3



eg 4



eg 5



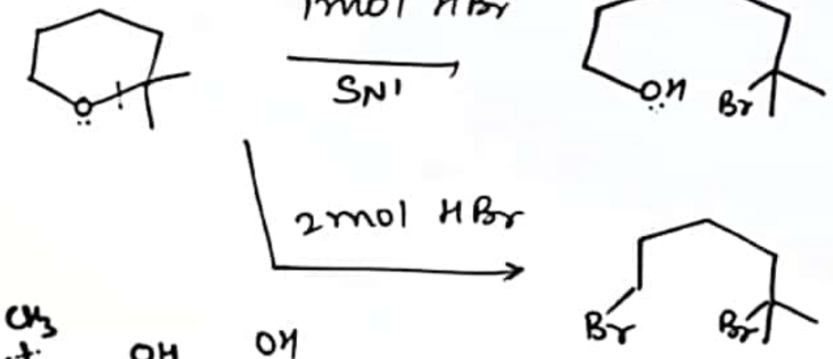
no rex. of phenol with HI

ORGANIC REACTION MECHANISM – III

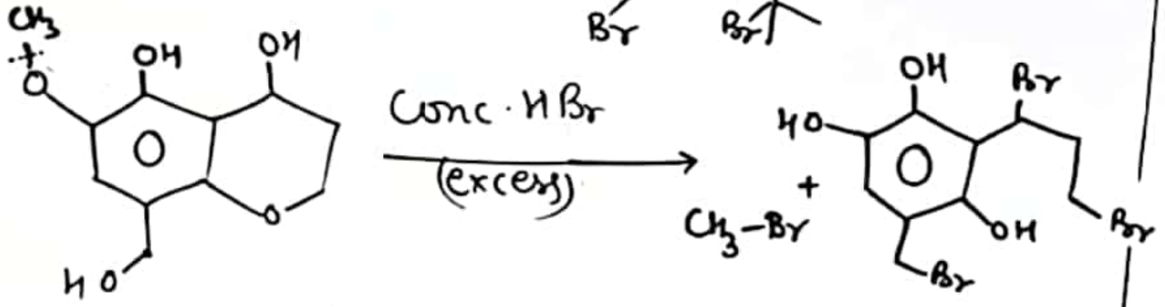
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[XI A] Reaction of ether with HX:–

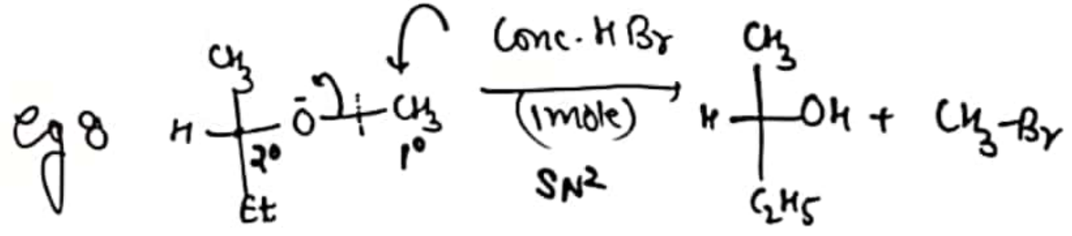
eg 6



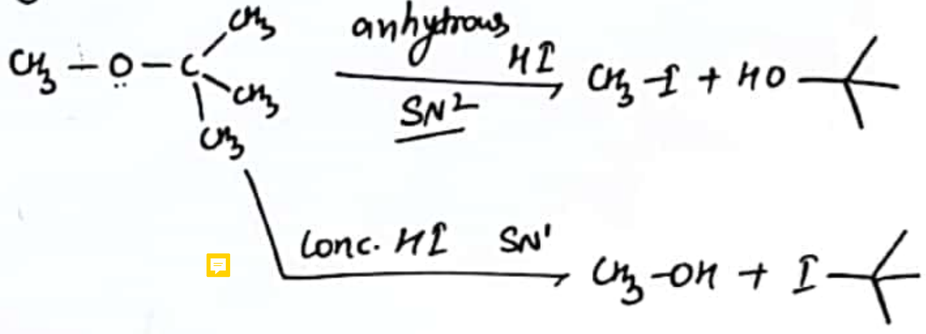
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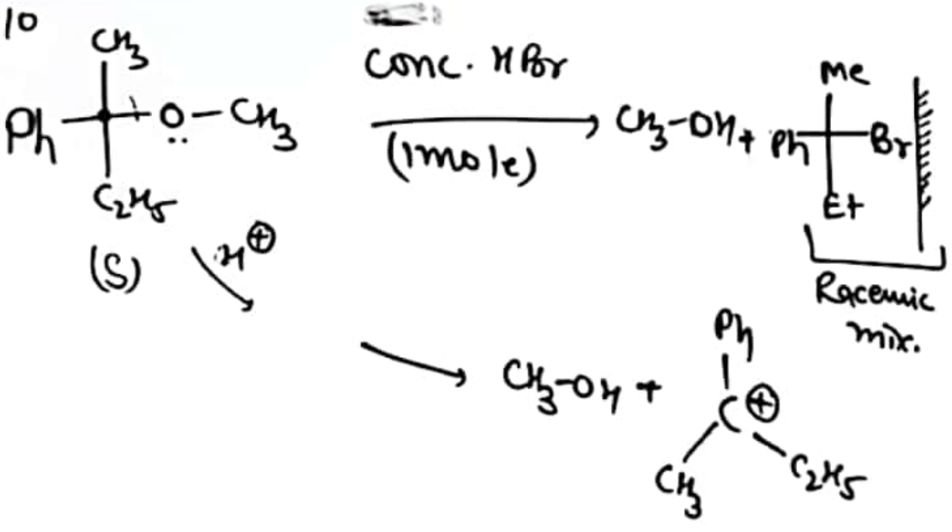
eg 8



eg 9



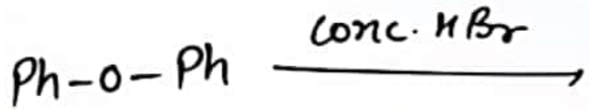
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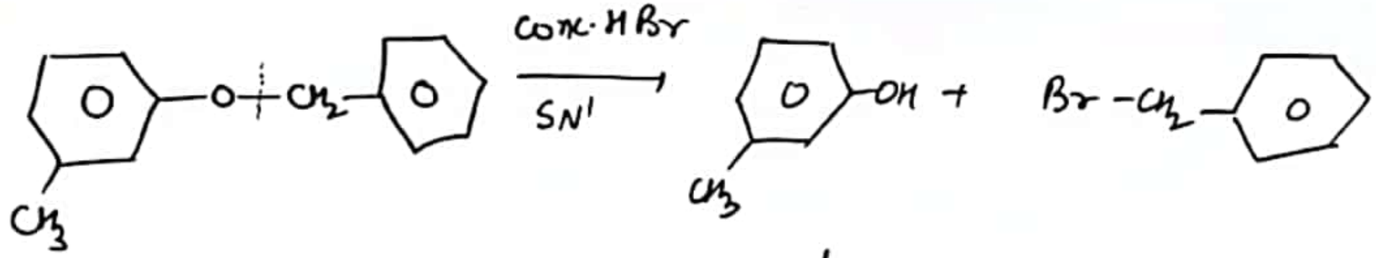
(XI) Nucleophilic Sub. Rxn of Ethers:-

[XI.A] Reaction of ether with HX:-

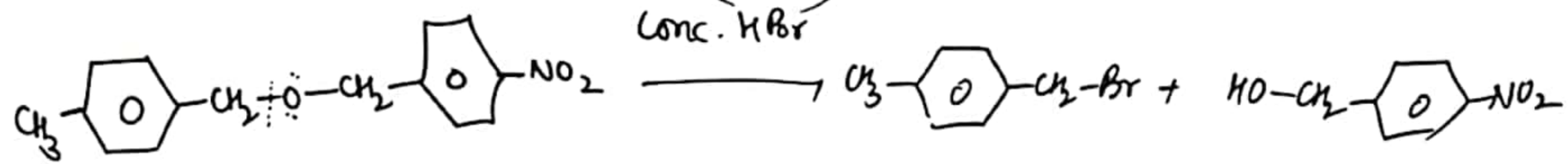
eg 11



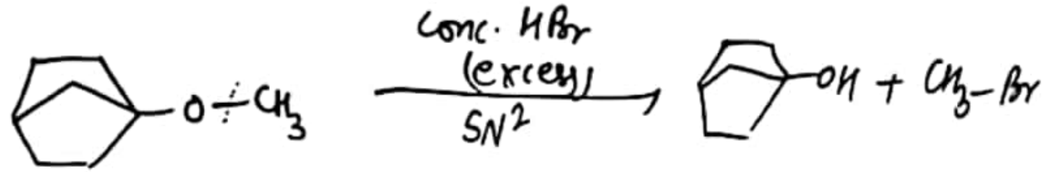
eg 12



eg 13



eg 14

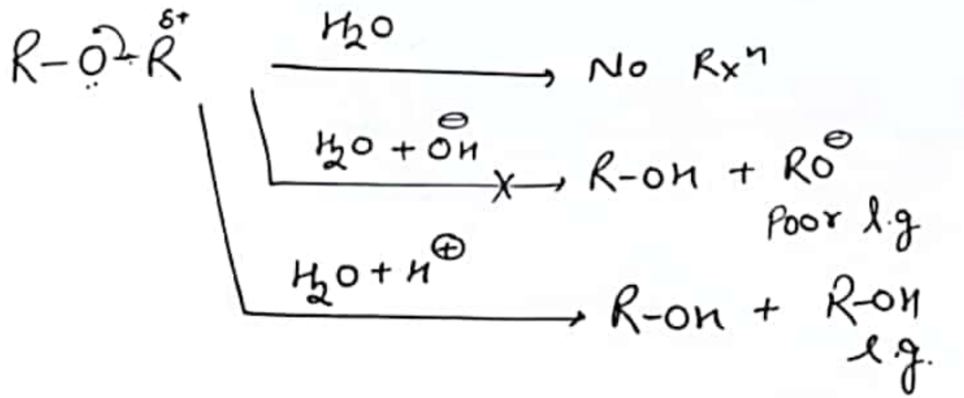


Bredt's rule

ORGANIC REACTION MECHANISM - III (ORM-III)

(XI) Nucleophilic Substitution Reactions of Ethers :-

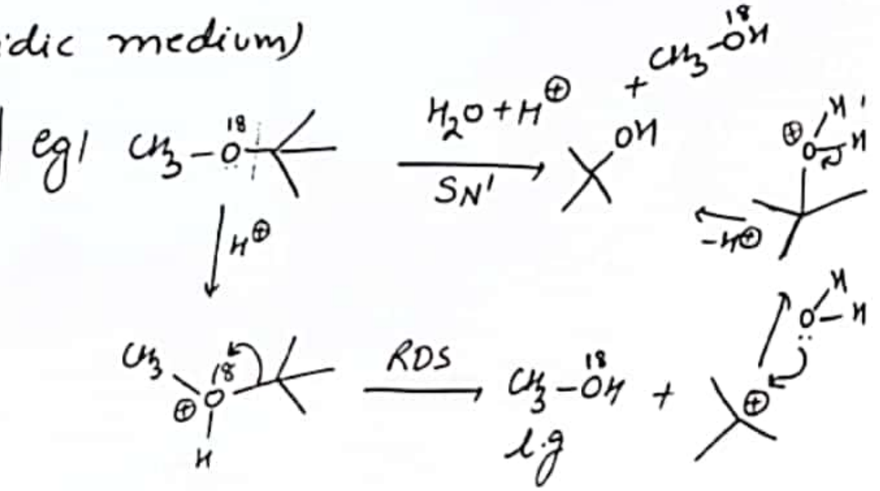
(XI.B) Hydrolysis of ether :- (possible only in acidic medium)



⇒ any alkyl group of ether is 3°, 2°, Benzylic } $\xrightarrow[\text{Hydrolysis}]{H_3O^{\oplus}}$ S_N1

⇒ When both alkyl groups of ether is 1° } $\xrightarrow{\text{Hydrolysis}}$ S_N2

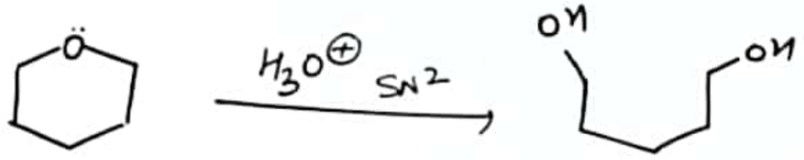
eg!

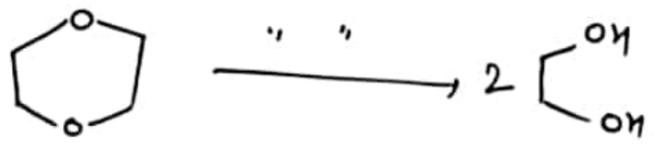


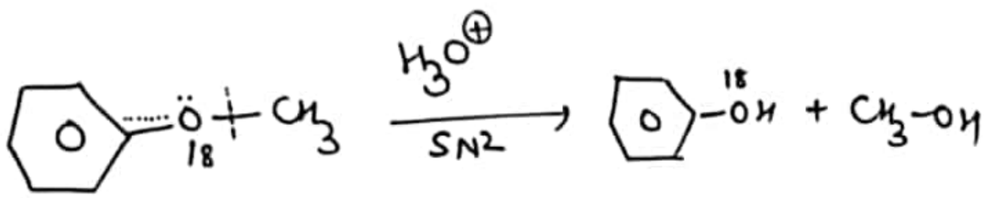
ORGANIC REACTION MECHANISM - III (ORM-III)

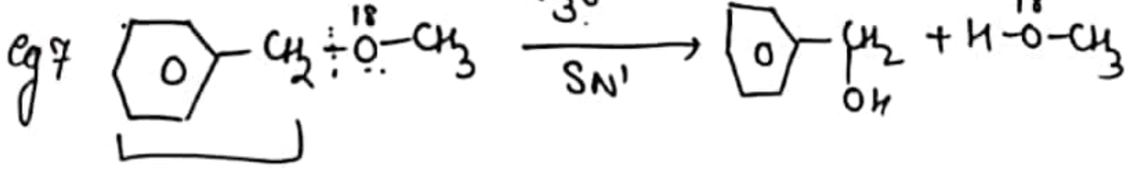
(XI) Nucleophilic Substitution Reactions of Ethers :-

(XI.B) Hydrolysis of ether :- (possible only in acidic medium)

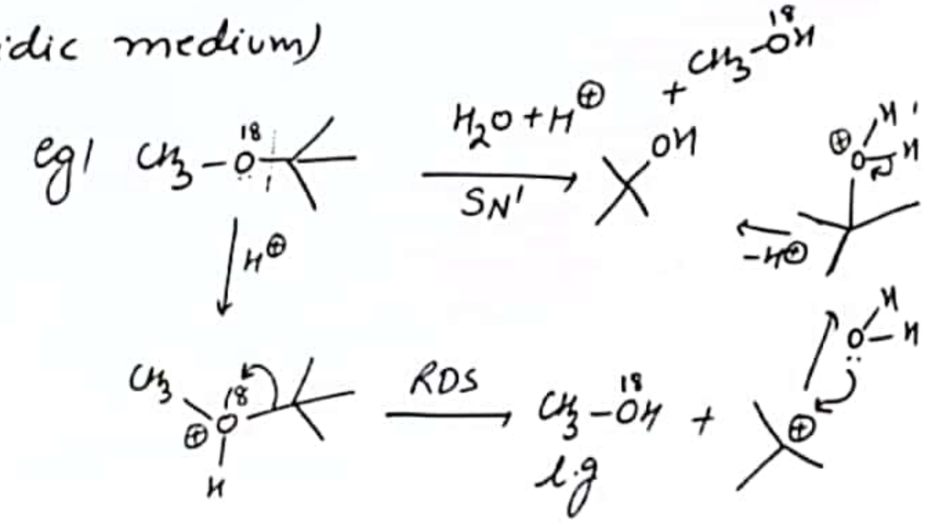
eg 4 

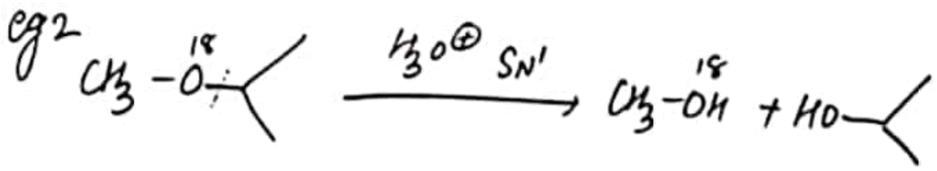
eg 5 

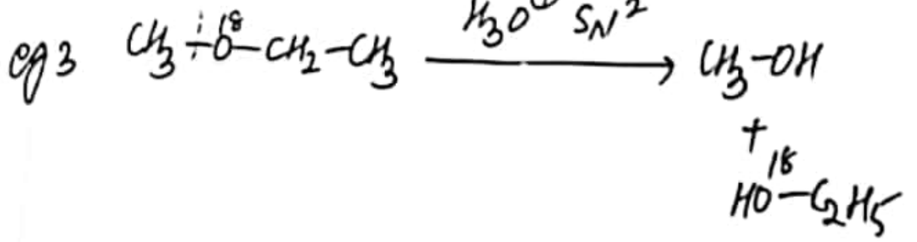
eg 6 

eg 7 

Benzylic

eg 1 

eg 2 

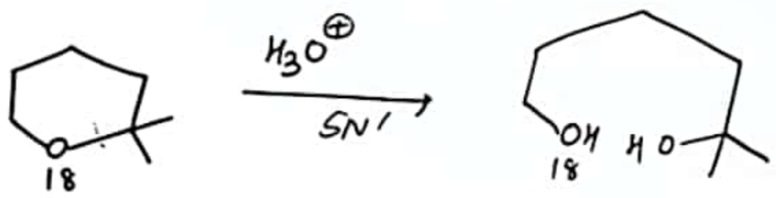
eg 3 

ORGANIC REACTION MECHANISM - III (ORM-III)

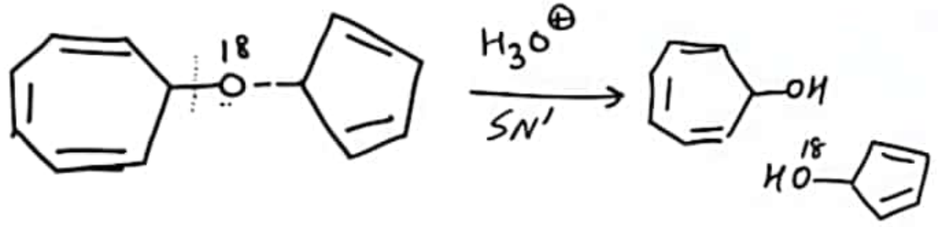
(XI) Nucleophilic Substitution Reactions of Ethers :-

(XI.B) Hydrolysis of ether :- (possible only in acidic medium)

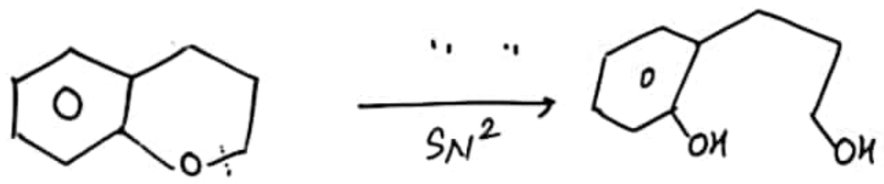
Eg 8



Eg 9

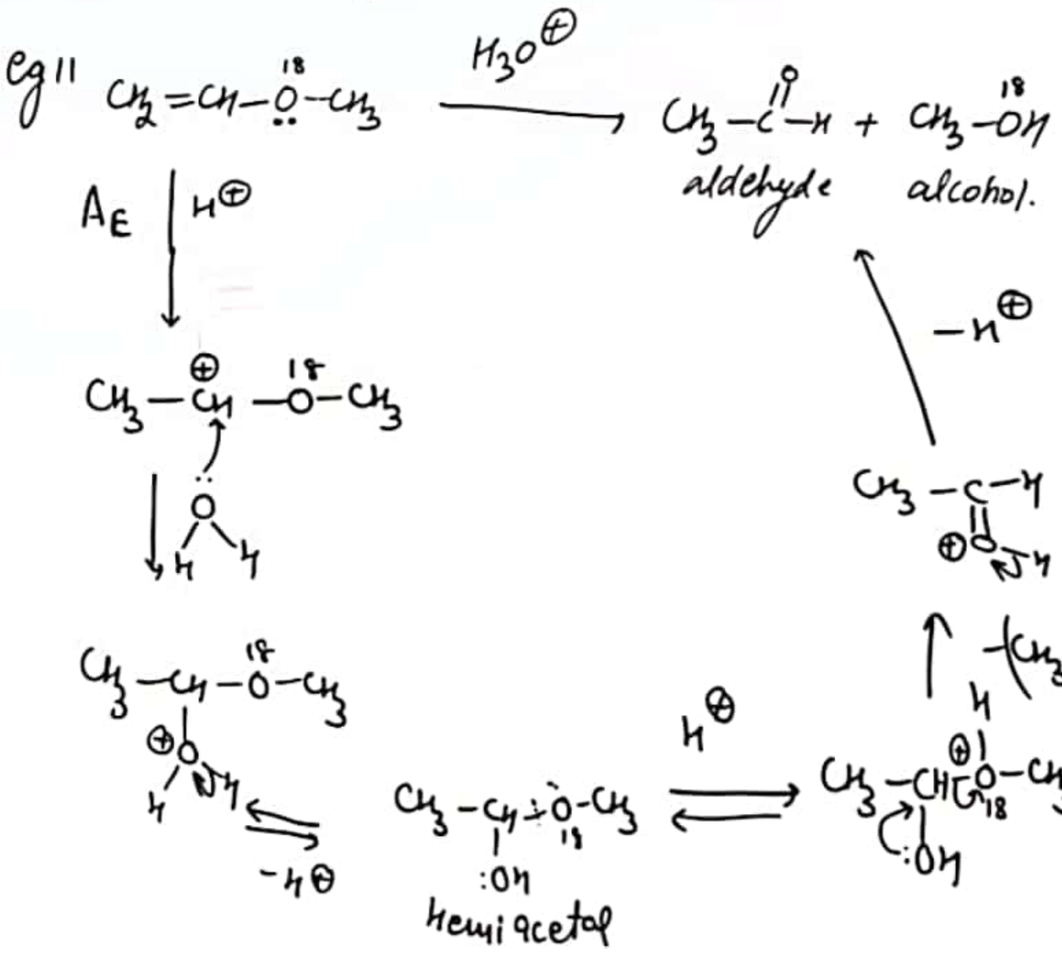


Eg 10



→ Hydrolysis vinylic esters.

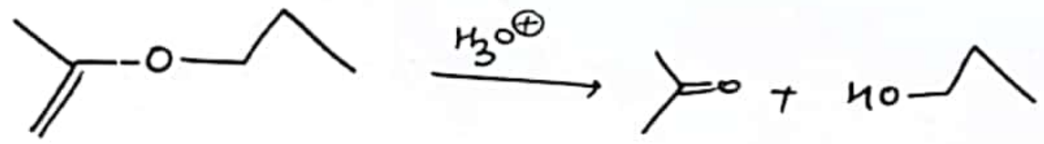
Eg 11

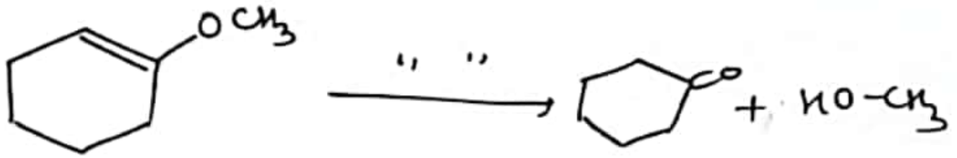


ORGANIC REACTION MECHANISM - III (ORM-III)

(XI) Nucleophilic Substitution Reactions of Ethers :-

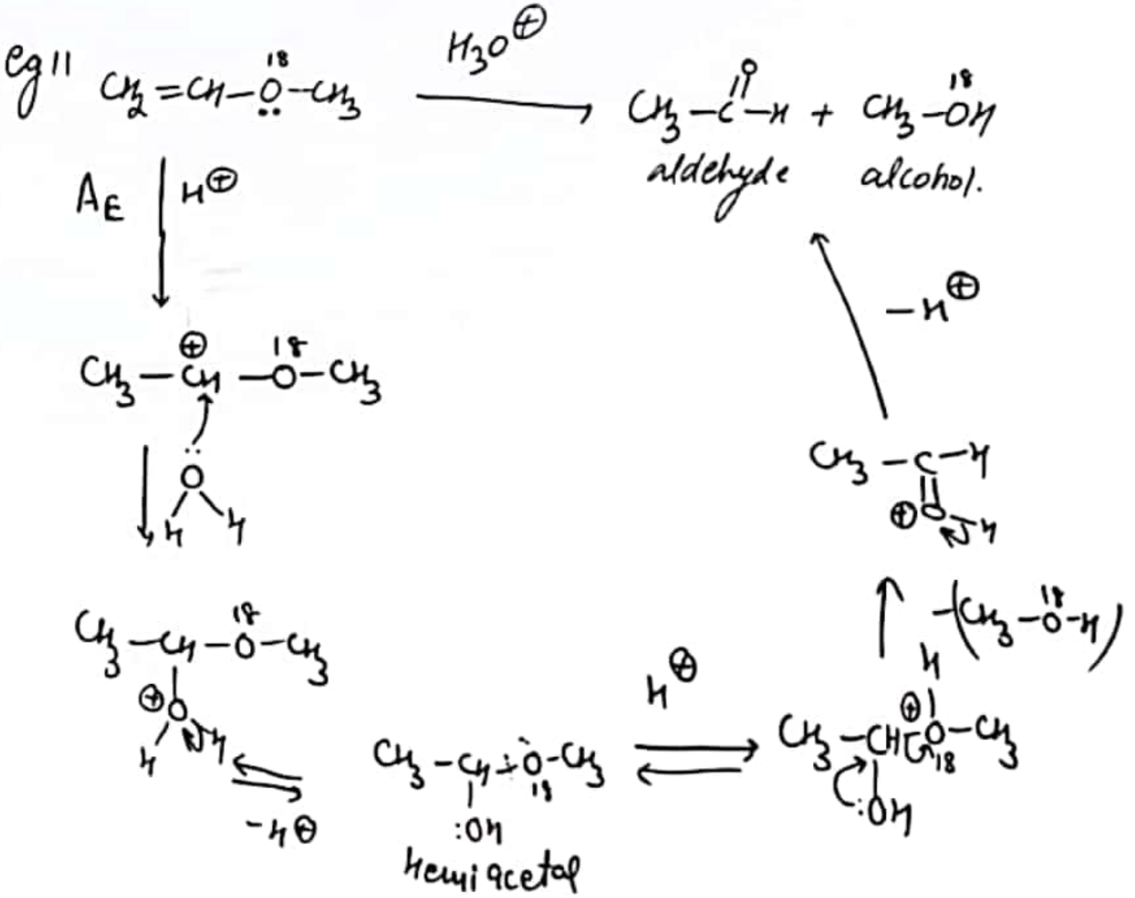
(XI.B) Hydrolysis of ether :- (possible only in acidic medium)

eg 12 

eg 13 

eg 14 $Ph-O-Ph \xrightarrow{\text{..}} \text{No Rx}^{n}$.

→ Hydrolysis vinylic esters.

eg 11 

ORGANIC REACTION MECHANISM – III (ORM-III)

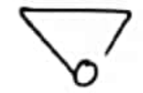
(XI) Nucleophilic Substitution Reactions of Ethers :-

XI.c Reaction of ether with PCl_5 :-

$$R-O-R \xrightarrow[S_N2]{PCl_5} 2\,R-Cl + POCl_3$$

ORGANIC REACTION MECHANISM - III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides

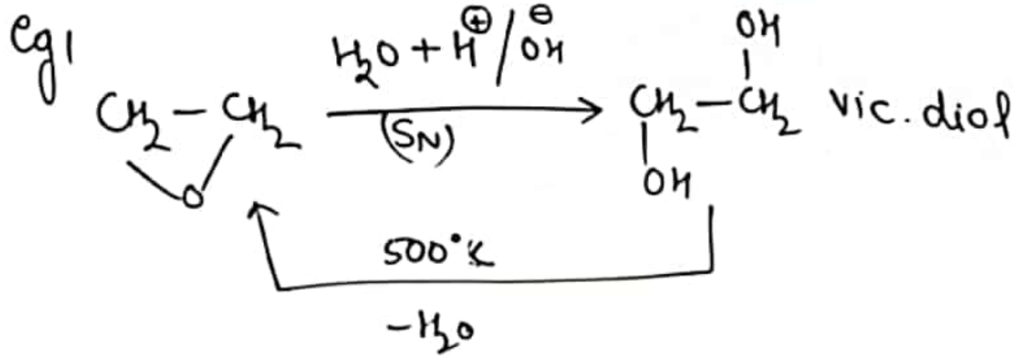


epoxide

⇒ High Ring Strain

⇒ higly reactive

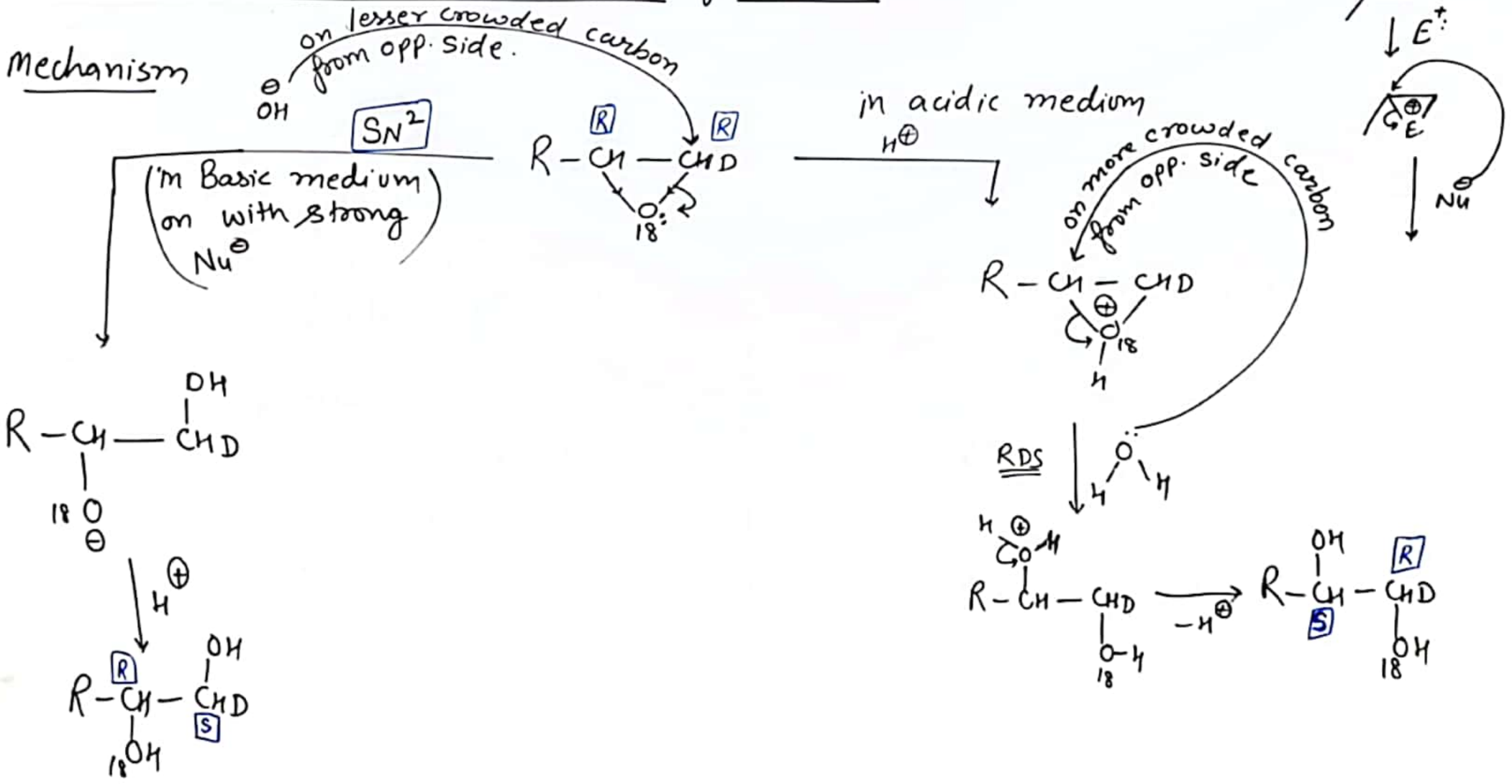
⇒ Can give S_N Reactions in acidic medium as well as basic medium.



ORGANIC REACTION MECHANISM – III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides

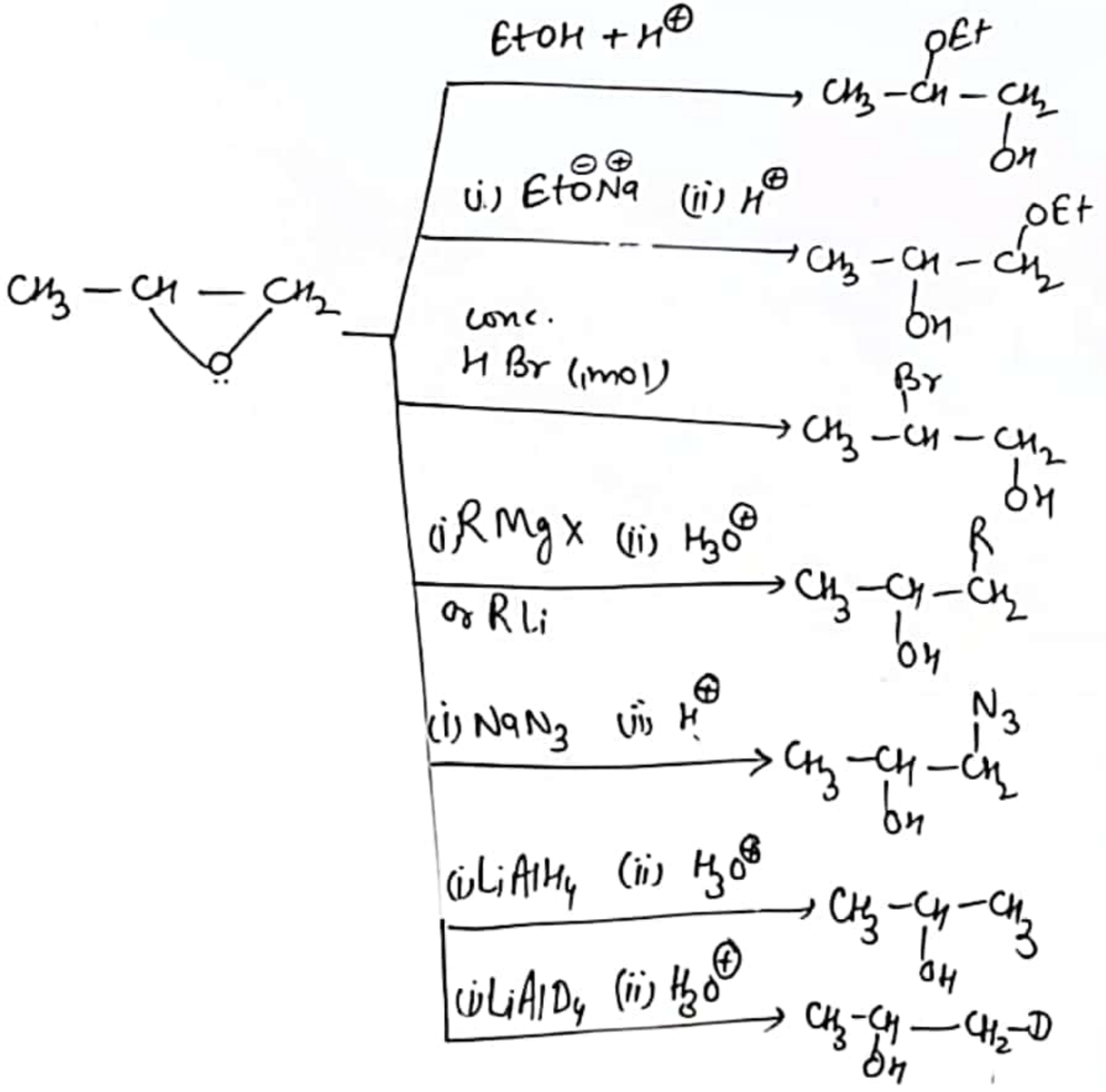
Mechanism



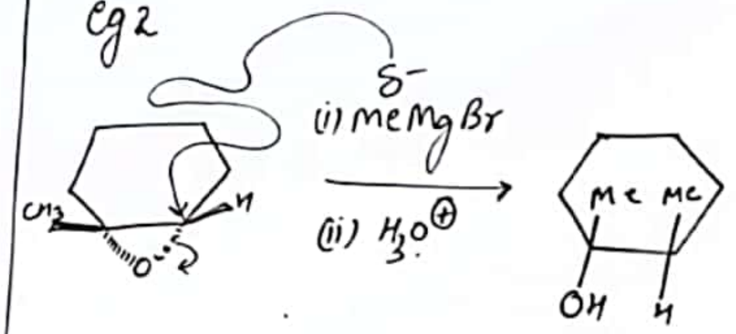
ORGANIC REACTION MECHANISM - III (ORM-III)

(XII) Nucleophilic Substitution Reactions of Epoxides

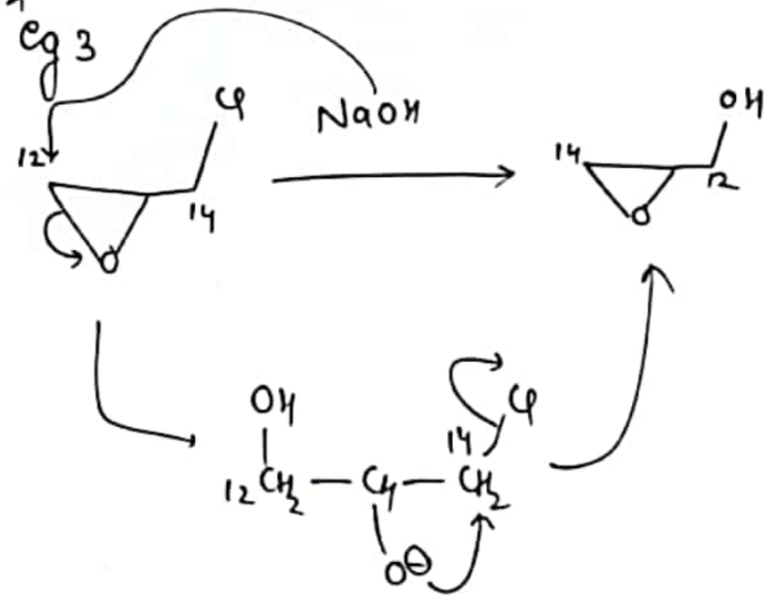
eg 1



eg 2



* eg 3



ORGANIC REACTION MECHANISM – III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(A) ArS_N1

(only in diazonium salt)

in aromatic compound

(B) $Ar S_N2$

(When strong EWG is/are present on ortho / Para position of leaving group.)

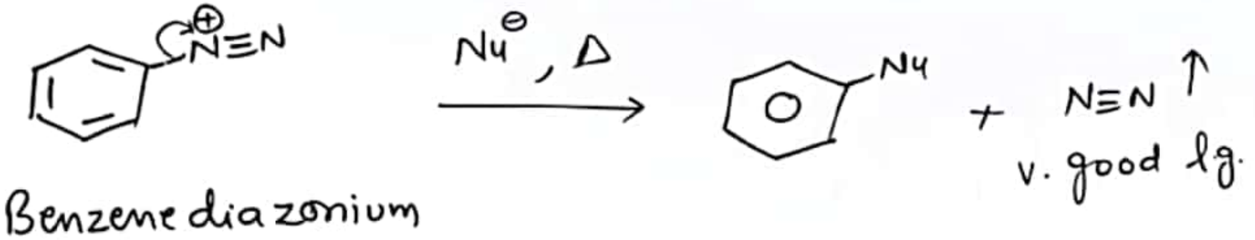
(c) Benzyne mechanism

(In difficult condition with strong base/$Nu^{\ominus}$)

ORGANIC REACTION MECHANISM – III (ORM-III)

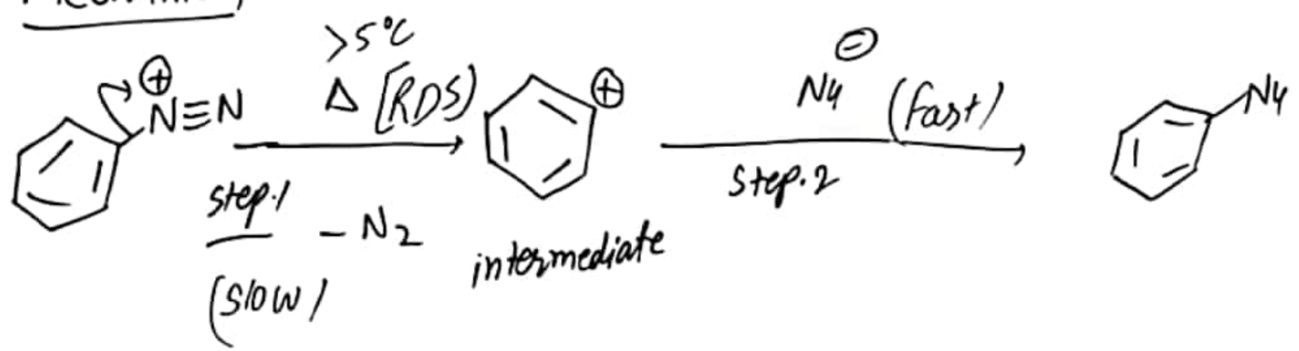
(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(XIII.A) $Ar S_N1$ Reaction (only introduction)



Benzene diazonium Chloride.
(Stable upto 5°C)

mechanism



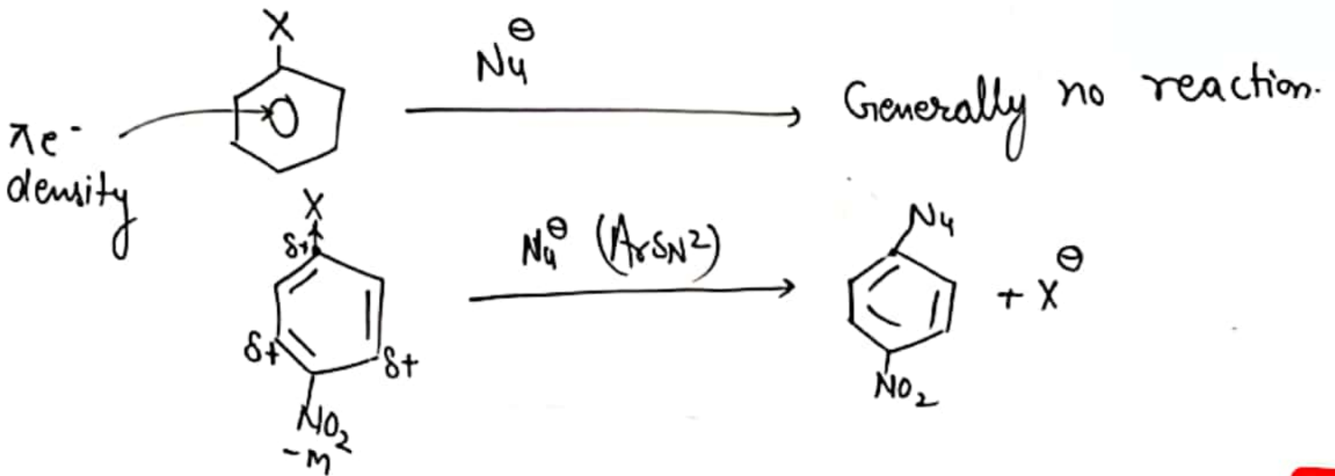
ORGANIC REACTION MECHANISM – III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

(XIII.B) $Ar S_N 2$

Generally Nucleophilic attack on aromatic ring do not possible due to high e^- density on aromatic ring.

But in presence of strong e^- withdrowing group on ortho or para position. of leaving group Nucleophilic attack possible.



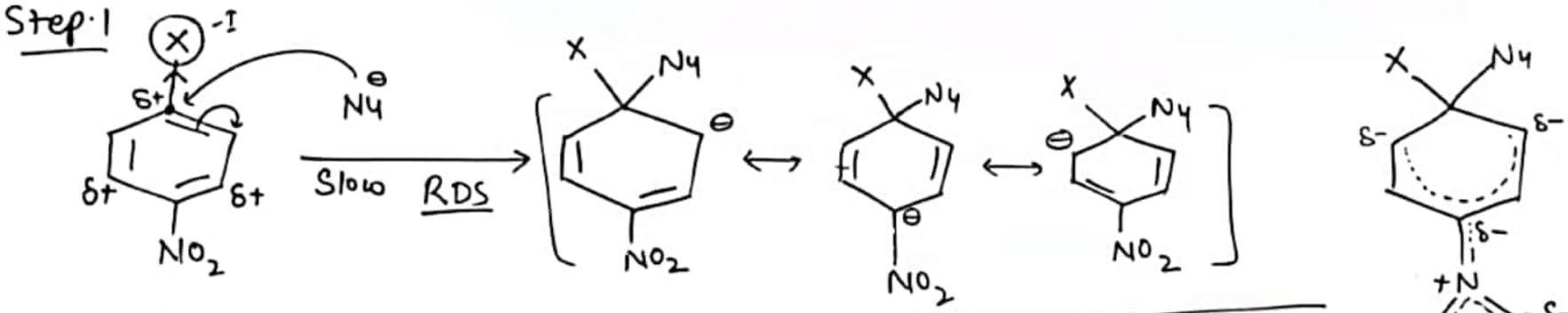
ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

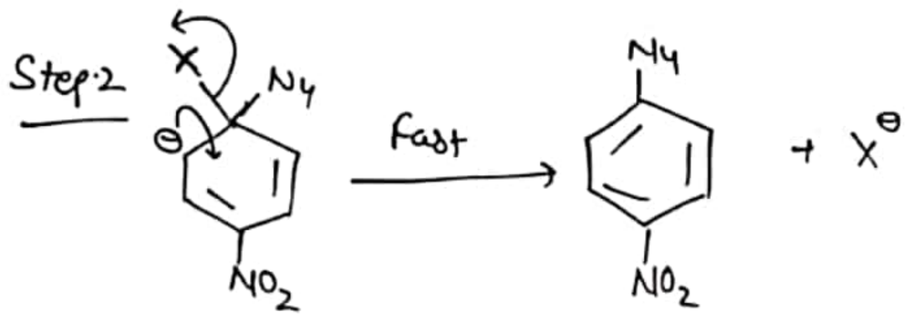
(XIII.B) $ArSN^2$

Mechanism

Step·1



Step·2



⇒ Rate of $ArSN^2 \propto [Ar\text{-}X]\,[N\bar{u}]$

⇒ 2nd order

⇒ Reactivity of $ArSN^2$ ↑ in presence of EWG.

⇒ ∴ ∴ ∴ ∝ Stability of intermediate

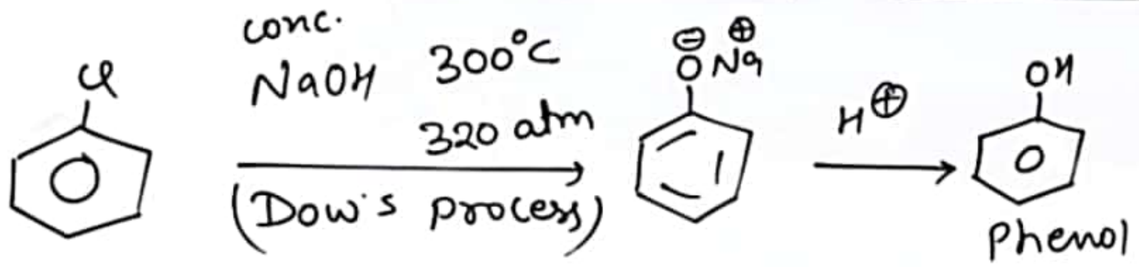
⇒ $X^{\ominus} > N_4^{\ominus}$ stability

Imp. ⇒ Reactivity w.r.t. halogen i.g ⇒ Ar-F > Ar-Cl > Ar-Br > Ar-I

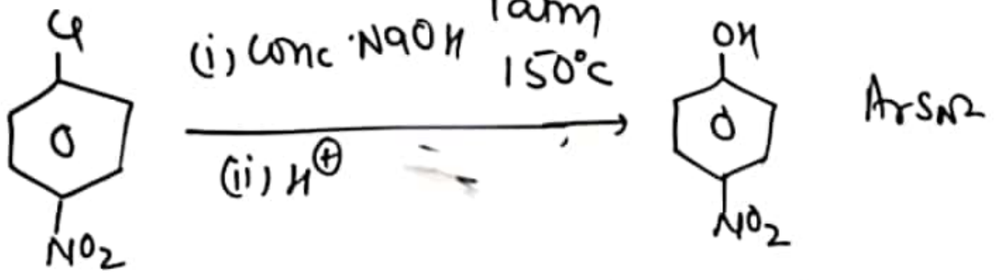
ORGANIC REACTION MECHANISM - III (ORM-III)

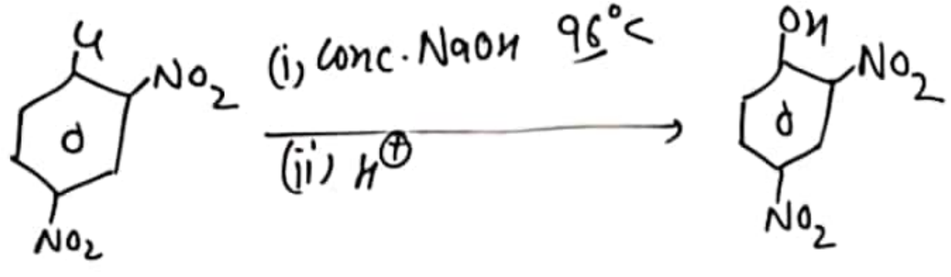
(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rxn on aromatic ring)

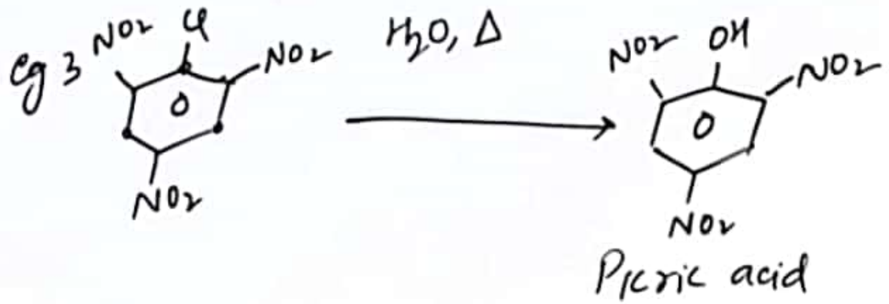
(XIII.B) $Ar S_N 2$

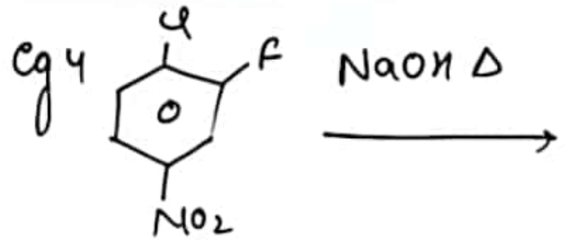


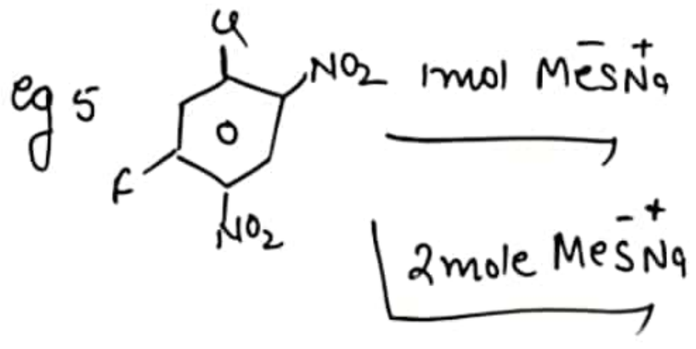
Benzyne mechanism (not $Ar S_N 2$)

eg 1 

eg 2 

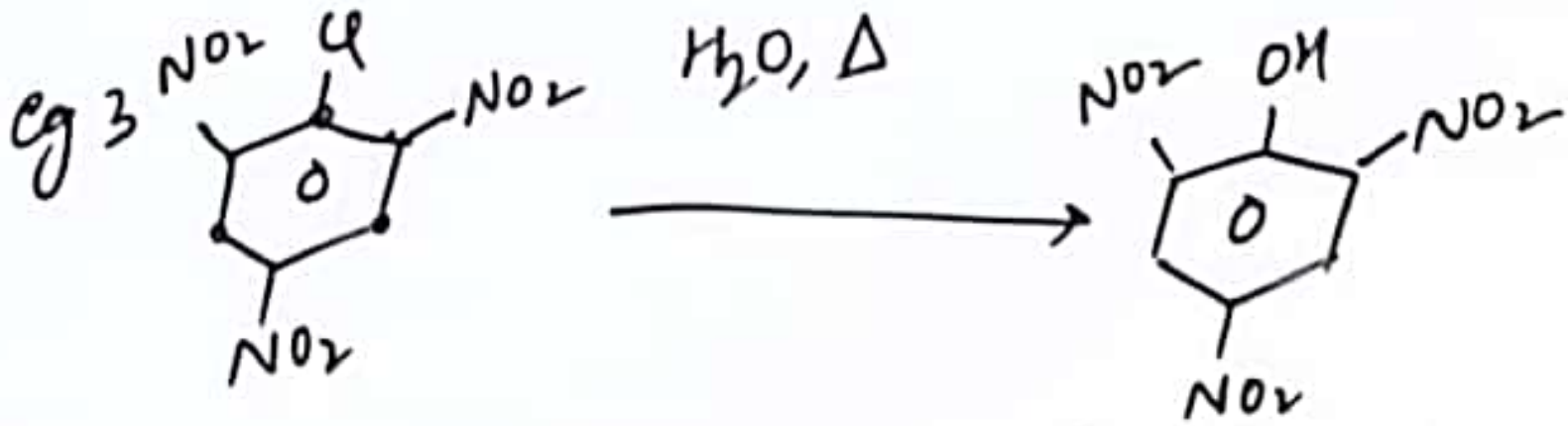
eg 3 

eg 4 

eg 5 

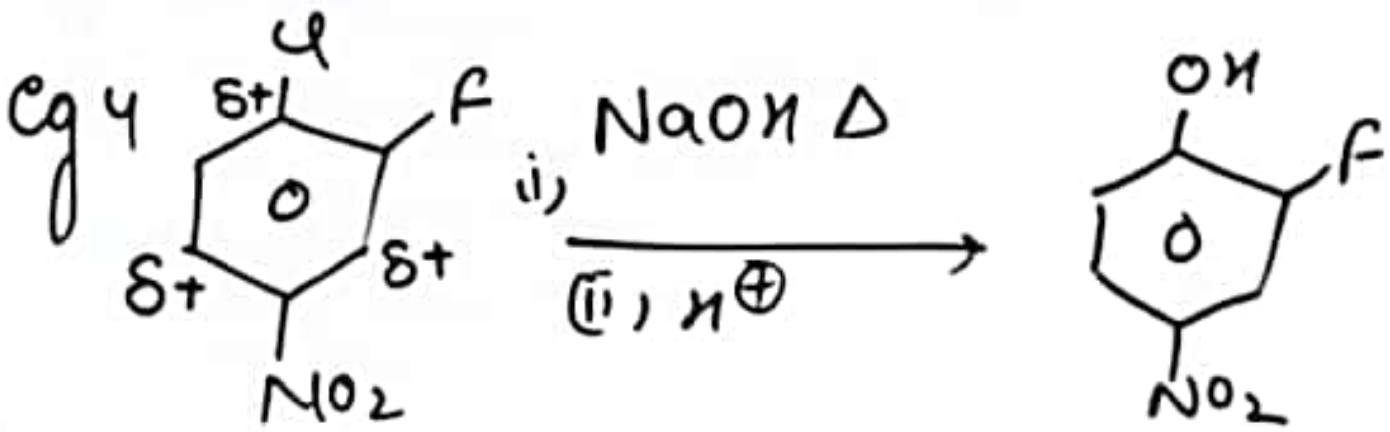
aromatic ring)

eg 3

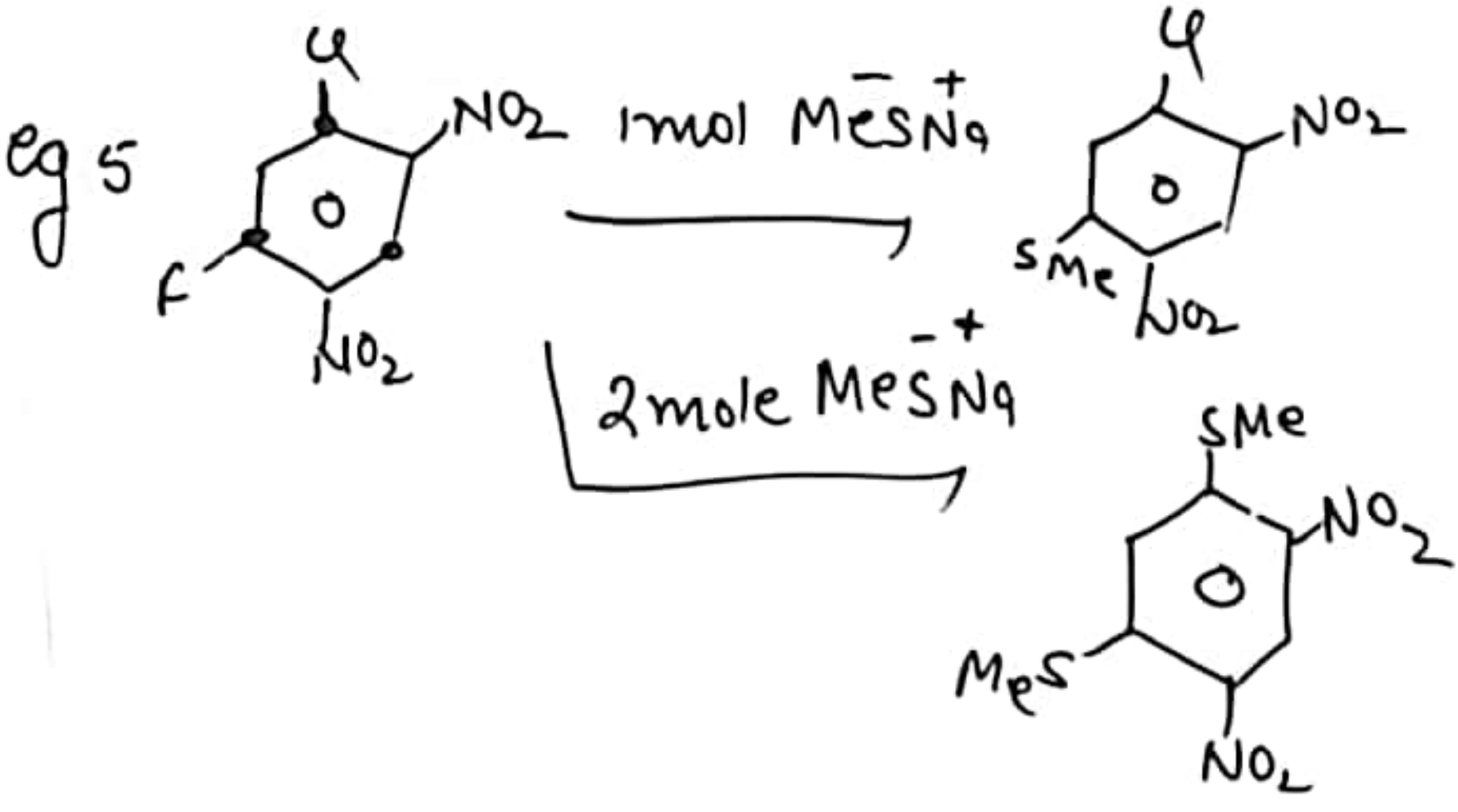


Picric acid

eg 4



eg 5

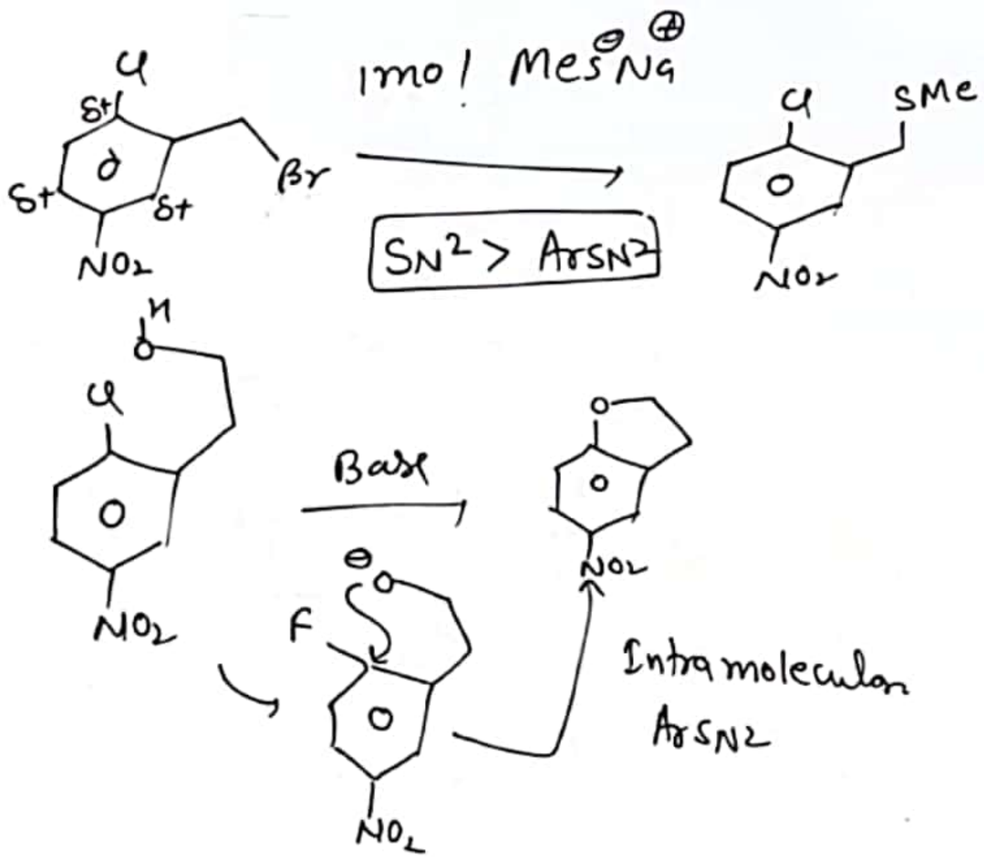


ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) $Ar S_N$ Reaction (Nucleophilic sub. Rx^n on aromatic ring)

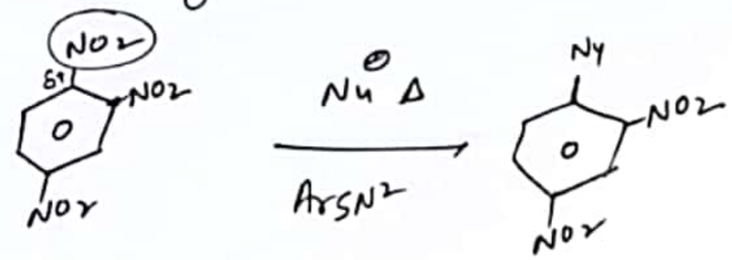
(XIII.B) $ArSN^2$

eg6

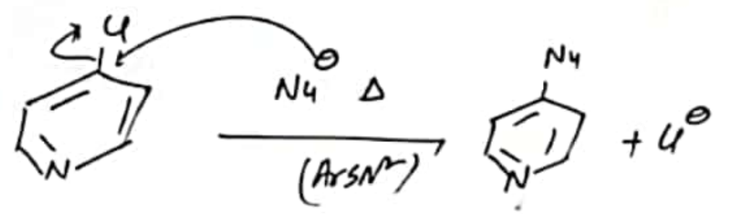


eg7

eg8



Chichibabin Rx^n

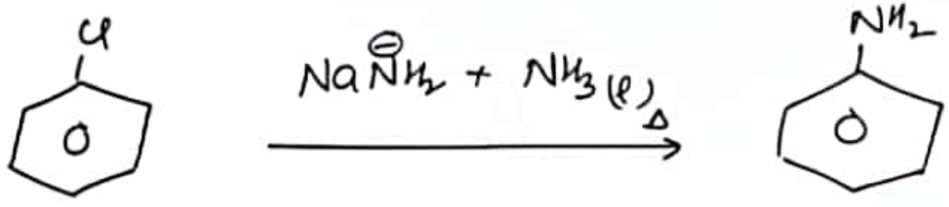


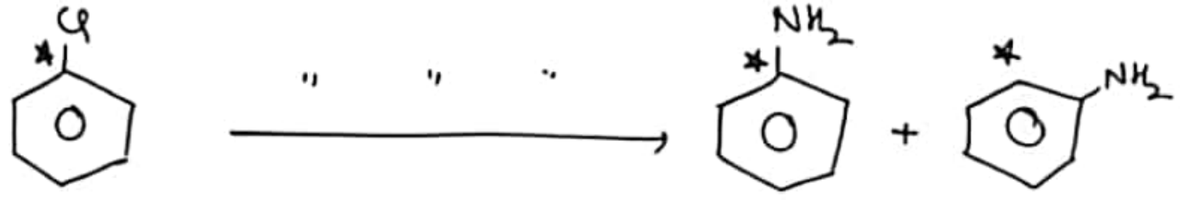
ORGANIC REACTION MECHANISM - III (ORM-III)

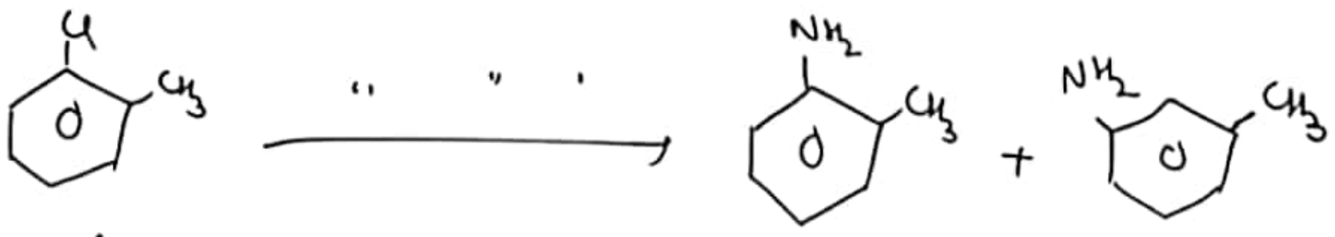
(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

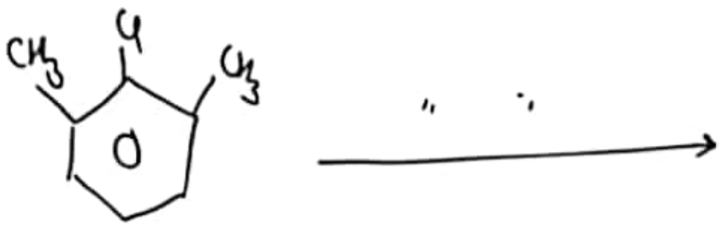
(XIII.C) Benzyne mechanism:-

⇒ It is possible in difficult conditions (Strong base + high temp)

eg1 

eg2 

eg3 

eg4 

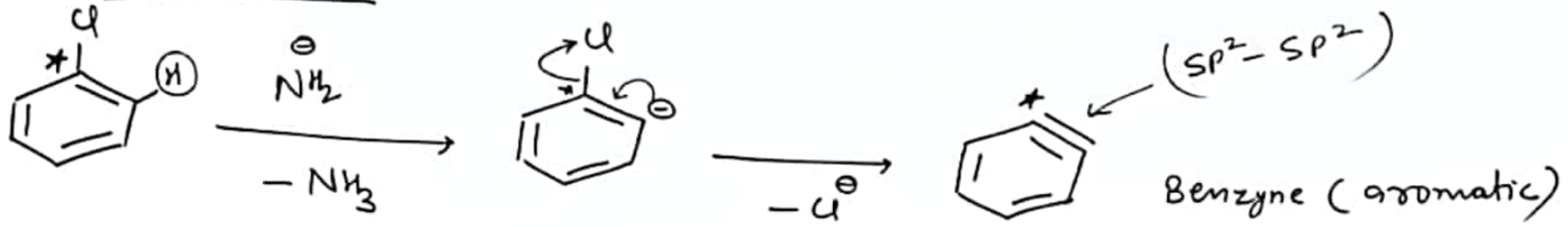
ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

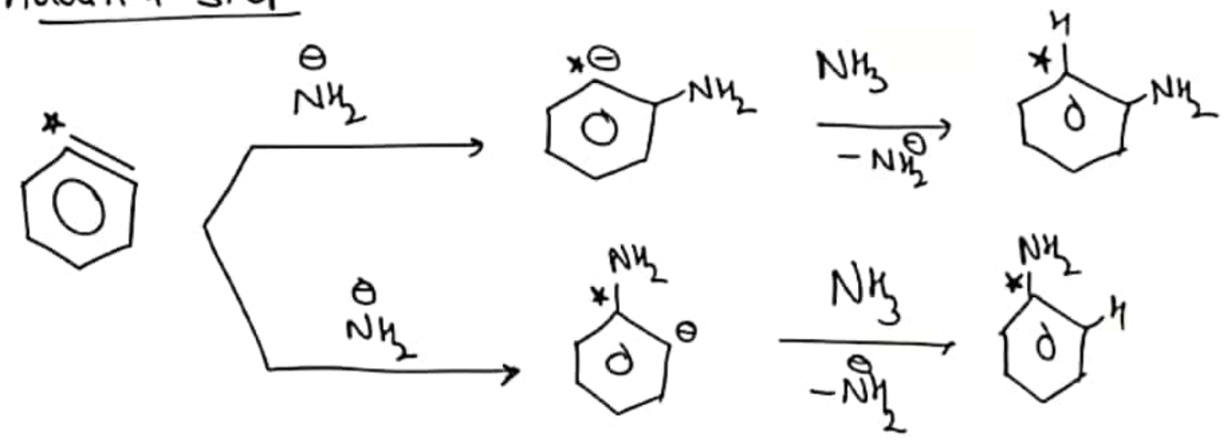
(XIII.C) Benzyne mechanism:- (Elimination - Addition mechanism)



⇒ Elimination step.



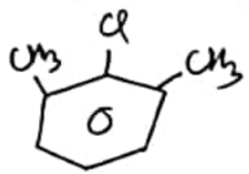
⇒ Addition step



ORGANIC REACTION MECHANISM - III (ORM-III)

(XIII) Nucleophilic Substitution Reactions of Aromatic compounds.

(XIII.C) Benzyne mechanism:- (Elimination - Addition mechanism)

eg 4 

$\xrightarrow{NaNH_2 + NH_3(l)}$ No R_x^n

eg 5 